

Peace or Push? Understanding Global Patterns of Refugee Return

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Aug 17, 2026

Abstract

When do refugees return to their countries of origin? A core premise of the international refugee regime is that refugees remain protected abroad while conflict continues at home and return once conditions improve. Using a new global dataset covering 293 origin–asylum dyads from 1980 to 2023, I show that most returns instead occur during ongoing conflict, while many refugees remain abroad long after war ends. I explain these patterns by conceptualizing return as a constrained choice. Whereas existing research treats return as reflecting refugees’ preferences between origin and asylum, I argue that constraints on remaining in asylum can force return during conflict, while constraints on moving home can inhibit return after peace. Most refugees live in neighboring countries, where they may face conflict, government pressure to leave, and insufficient resources to return and rebuild their lives. Empirically, realized return is substantially higher under coercive pressure to leave asylum, including while conflict continues at home, while post-war return is lower when asylum is more attractive and aid for repatriation is limited. The findings recast refugee return as a politically and materially constrained process rather than mostly a response to improved security at home.

*I am grateful to James D. Fearon, Jeremy Weinstein, Jens Hainmueller, Michael Allen, and Judith Goldstein for their invaluable advising and feedback. I thank Idean Salehyan, Lisa Blaydes, Andrew Shaver, Claire Adida, Perisa Davutoglu, and participants in the Online Peace Science Colloquium, the Stanford-Berkeley IPE Colloquium, and the Stanford IPE Workshop for helpful comments. I thank the research assistants of the Political Violence Lab for their help in constructing the Global Refugee Return dataset: Samantha Terriquez, Franklin Huang, David Hsueh, Joe Stout, Evan Richardson, Caroline Robbins, Kianna Victor, Citlali Hernandez Pulido, Jade Gomez-Castro, Thomas Lobaton, Zex Liu, Jennifer Lara, Kev Young, Maria Fernanda Bello, Marium Fatima Agha, Ryder Stoddard, Stephanie Udeze, Toros Pinar, Abigail McMurphy, Anvi Gaikwad, Ceyda Akkus, Damara Gomez, and Ulysses Trejo Vasquez. I also thank Georgia Sasso, Suha Choi, and Jelena Hertzler for their research assistance. Finally, I thank the UNHCR archives team for hosting me in their archives.

1 Introduction

The international refugee regime rests on the premise that people who face persecution or war at home should be protected abroad and return once the conditions that drove their displacement have ended. The principle of *non-refoulement* prohibits states from returning refugees to places where they face persecution or serious harm,¹ and voluntary repatriation is UNHCR’s preferred “permanent solution” to refugee crises.² Together, these principles imply that refugees should remain in asylum while conditions at home are unsafe and return once those conditions improve. Existing studies largely support this account, finding that refugees’ intentions to return depend primarily on origin-country security conditions (e.g., Alrababah et al., 2023; Beaman, Onder and Onder, 2022; Adema et al., 2025; Al Husein and Wagner, 2023).

Observed return patterns do not align with these expectations. I introduce the Global Refugee Return (GRR) dataset, which reconstructs annual refugee stocks and returns for 293 origin–asylum dyads from 1980 to 2023 and corrects major inconsistencies in existing global displacement statistics. The data reveal two striking patterns. First, more than three-quarters of recorded returns occurred while conflict was ongoing in the country of origin. Second, peace often produces little repatriation: among wars that ended and remained ended for at least five years, the median case saw fewer than a fifth of the refugee population return. Refugees thus often return when the refugee regime and security-based theories predict that they should remain in asylum, and remain abroad when those accounts predict that they should return.

Why do refugees return during war, yet often remain in asylum after peace? I argue that answering these questions requires distinguishing refugees’ preferences over where to live from their ability to act on those preferences. Recent research has substantially advanced our understanding of the first question, showing that refugees compare conditions in origin and asylum and become more willing to return as security improves at home (e.g., Alrababah et al., 2023; Beaman, Onder and Onder, 2022; Adema et al., 2025; Al Husein and Wagner, 2023). But realized return also depends on whether the alternatives refugees prefer are actually available to them.

I therefore conceptualize refugee return as a *constrained choice*. Refugees may prefer to remain in asylum but be unable to do so, or prefer to return but lack the means to move. These constraints are especially important in the settings where most refugees live: often in neighboring low- and middle-income countries, where asylum can itself be precarious and refugees may have limited opportunities to accumulate resources.

¹The principle is codified in Article 33(1) of the 1951 Convention Relating to the Status of Refugees and its 1967 Protocol and is also recognized as a rule of customary international law, binding states regardless of whether they are parties to the Convention.

²The other two durable solutions are local integration and third-country resettlement. UNHCR has treated voluntary repatriation as its preferred durable solution since the mid-1980s (Barnett and Finnemore, 2004; Loescher, 2001).

Conflict can spread or break out in host countries, governments can pressure refugees to leave, and years spent in camps or under restrictions on employment can leave refugees unable to finance transportation, housing, and livelihoods upon return.

This framework explains both empirical puzzles. Refugees may return during war when they can no longer remain in asylum, while peace may produce little return when refugees prefer to remain abroad or lack the resources to move. The argument thus extends accounts centered on origin-country security by showing that improvements at home translate into return only under particular asylum and resource conditions. The theoretical innovation is that realized return depends not only on the preference to return, but also on the possibility of remaining abroad and the capacity to move.

I evaluate this argument in three ways: using within-dyad changes in realized return over time, decomposing the mechanisms associated with aggregate global return, and tracing major return episodes with archival case studies. The analysis yields three main findings. First, return rises sharply when refugees' ability to remain in asylum deteriorates. Severe forced-return policies are associated with nearly three times the return rate observed in years without such policies, including while conflict continues at home. In a model-based decomposition, setting severe forced-return policy to zero wherever it is observed reduces predicted return by 24.7%, compared with 14.3% for the post-war window. Much global return thus occurs not after peace, but in settings where refugees' ability to remain abroad is constrained.

Second, refugees' preferences over locations continue to matter when asylum remains available. Return is substantially higher following war endings, but this post-war association is much weaker for refugees living in wealthier asylum countries and returning to more repressive origin countries. Most strikingly, a war ending has no detectable association with return among refugees living in relatively wealthy asylum countries. Third, realized return is higher where more UNHCR repatriation assistance is available, consistent with the argument that resources can constrain refugees' capacity to move.

This research makes three contributions. First, the paper develops a theory of realized refugee return as a constrained choice. Recent quantitative research has advanced our understanding of when refugees *want* to return, emphasizing security at home and the relative economic attractiveness of origin and asylum. A separate qualitative and legal literature has documented coercive and involuntary repatriation. I bring these insights into a common framework by distinguishing refugees' preferences from their ability to act on them and by treating security and political conditions in asylum as central to the return decision. This perspective also highlights the importance of studying realized return globally: research focused on return intentions in a small number of cases can identify how refugees evaluate origin and asylum, but cannot examine the forces that account for aggregate patterns of return across refugee populations.

Second, the findings speak to research on international cooperation, law, and institutions. *Non-refoulement* is the core legal principle of the international refugee regime, yet states use coercive policies to induce return, including to countries where conflict is ongoing. UNHCR sometimes facilitates repatriation under these conditions, in part because refusing to do so may leave refugees to return without assistance or jeopardize the organization’s access to refugees. The findings therefore illustrate the limits international law and organizations face in constraining state behavior. At the same time, they raise a broader puzzle about cooperation with the refugee regime. If host governments can push refugees out, and if many refugees remain long after wars end, why do so many states continue to provide asylum rather than force refugees to leave? The paper thus documents a consequential failure of international cooperation while raising a broader question about why states continue to sustain the refugee regime. This broader puzzle motivates my book project.

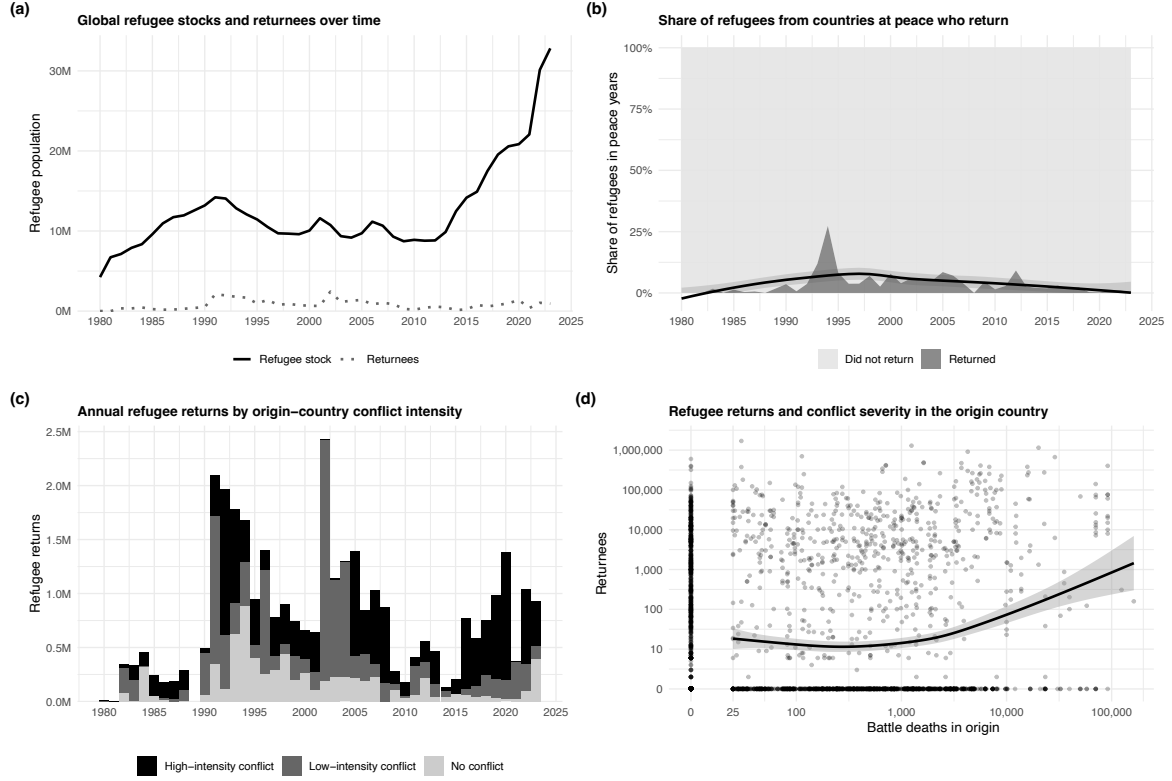
Third, the paper makes a new measurement and data contribution. The Global Refugee Return dataset provides annual measures of refugee stocks and realized return across 293 origin–asylum dyads from 1980 to 2023. It identifies substantial inaccuracies in existing UNHCR statistics, including the omission of forced and spontaneous returns and inconsistencies in reported refugee stocks, and provides corrected measures based on archival evidence. GRR also introduces several new measures constructed from archival records, including repatriation assistance, tripartite repatriation agreements, UNHCR involvement in repatriation, forced-return policies, encampment rates, and UNHCR financing. These measures capture dimensions of refugee return and the institutional context surrounding it that have not previously been collected systematically at the global level.

The article proceeds by documenting the two empirical puzzles, return during conflict and continued displacement after peace. I then develop the theory, introduce the GRR dataset, and test its implications using within-dyad panel models and counterfactual estimates of attributable return. I employ a case study of Ethiopian and Eritrean refugees to show the mechanisms underlying push-driven return and non-return after peace. The article concludes with implications for refugee solutions and directions for future research.

2 Puzzling Patterns of Refugee Return

Existing research on refugee return emphasizes security in the country of origin. Micro-level studies of return intentions consistently find that refugees become more willing to go home as conditions improve (Alrababah et al., 2023; Al Husein and Wagner, 2023; Adema et al., 2025; Beaman, Onder and Onder, 2022). For example, the “threshold model” developed by Alrababah et al. (2023) argues that refugees consider return only once a minimum level of safety in the origin country has been achieved. Below this threshold, deteriorating conditions in asylum should not induce return. These

Figure 1: Return during conflict and non-return during peace



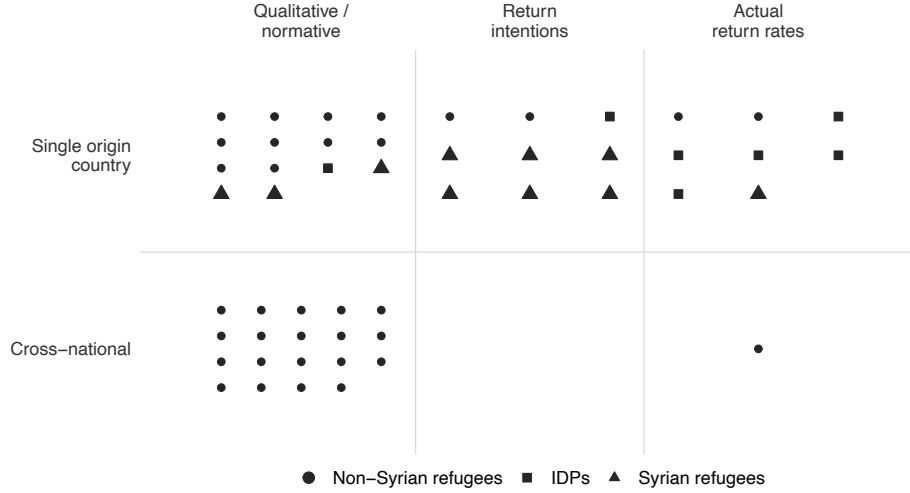
Notes: (a) Global refugee stock (solid) and recorded returns (dotted). (b) Share returning, among refugees who fled war, in dyad-years when the origin is at peace. (c) Annual returns by origin-country conflict intensity (UCDP: ≥ 25 and $\geq 1,000$ battle deaths). (d) Log battle deaths against log returnees. Lines are LOESS smoothers; 1989 omitted.
Sources: GRR; UCDP/PRIO.

accounts therefore imply two broad expectations for realized return: return should predominantly occur when there is peace, while large-scale return should be rare while conflict continues.

The Global Refugee Return dataset, a new dataset described in Section 4, shows that these expectations often do not hold in practice. Panel (a) in Figure 1 plots global refugee stocks and return flows since 1980, illustrating that return is rare relative to the size of displaced populations. Across dyad-years with more than 1,000 refugees, the median return rate is zero and the mean is 5.34 percent; in 89 percent of dyad-years, fewer than 5 percent of refugees return. One possibility is that low return simply reflects the persistence of conflict. Yet Panel (b) shows that even when there is no conflict in the country of origin, the average return rate is only 3.42 percent.³ Peace alone, in other words, rarely produces large-scale repatriation.

³Restricted to origin-year dyads where refugees generally fled war rather than broader non-conflict persecution.

Figure 2: Scope and outcomes studied in the refugee return literature



Notes: Each point is a study from a systematic search (2006–2026) of nineteen political science, migration, development, and area studies journals, supplemented by citation tracing. Triangles denote studies of Syrian refugees, squares studies of IDPs, and circles studies of other populations. The full list of studies is reported in the Appendix.

Panel (c) reveals the opposite puzzle: most returns occur during war, not after it. Between 1980 and 2023, more than three-quarters of all recorded returns took place while conflict was ongoing. Roughly two-fifths occurred during high-intensity conflict, more than a third during low-intensity conflict, and only one-fifth during peace. Panel (d) reinforces this pattern, showing that higher battle deaths are associated with more, rather than fewer, returns.

Why have these patterns received relatively little attention? One reason is that existing quantitative research has primarily studied a different outcome at a different level of analysis. Figure 2 summarizes the scope and outcomes examined in studies of refugee return. Research is concentrated in single-origin studies of individual return intentions, particularly among Syrian refugees, while cross-national studies of realized return remain rare.

Studying intentions in specific cases, however, may not tell us when refugees actually return. Refugees who prefer to remain abroad may nevertheless return when conditions in asylum unexpectedly deteriorate. Moreover, studies of individual variation within particular refugee populations have limited leverage on population-wide shocks that may drive major return episodes. The main cross-national exception, Zakirova and Buzurukov (2021), shows that political and economic conditions in origin and asylum also predict realized return. Yet this work, like the intentions literature on which it builds, focuses on the conditions shaping refugees’ preferences rather than constraints on their ability to act on them, including whether they can remain in asylum or afford to move.

3 Theory

Why do refugees return during war, yet often remain abroad after peace? I conceptualize refugee return as a *constrained choice* organized around three questions. First, *do refugees want to return?* Refugees' preferences depend on the relative attractiveness of origin and asylum. Second, *can refugees stay?* Conflict or government coercion can make continued asylum costly or impossible, inducing return even when refugees prefer to remain abroad. Third, *can refugees move?* Refugees who prefer to return may lack the resources to relocate and rebuild. Realized return therefore need not reveal a preference for home, just as non-return need not reveal a preference for asylum; preferences and realized return can diverge where choices are constrained.

3.1 Three Components of Constrained Choice

Do refugees want to return? When staying and moving are feasible, refugees choose between origin and asylum according to their relative expected welfare. An end to conflict makes the origin country safer and should increase return, as existing theories predict. However, even after war ends, asylum may still offer better economic opportunities, while years abroad can generate jobs, businesses, social networks, and children with few ties to the origin country (Borjas and Bratsberg, 1996; Van Hook and Zhang, 2011). These attachments can make remaining abroad preferable even after the original cause of displacement disappears.

Political insecurity may also persist after war. Civil wars often end under authoritarian governments, and repression can continue after conflict is over (Karreth, Sullivan and Dezfuli, 2020; Carey and González, 2021). Refugees may therefore remain reluctant to return to the government they fled even in the absence of active conflict (Hathaway and Foster, 2014; Bradley, 2013; Long, 2013).⁴

Thus, when refugees can stay and move, return should depend on the attractiveness of origin relative to asylum.

H1 (Preferences). When staying and moving are feasible, return becomes more likely as conditions in the origin country improve relative to conditions in asylum.

Can refugees stay? Refugees who prefer to remain in asylum may nevertheless be unable to stay. Host governments may pressure them to leave before conditions at home have improved. Despite the prohibition on forced return under the 1951 Refugee Convention,⁵ states may circumvent or ignore *non-refoulement* (Schwartz, 2025; Chimni,

⁴For studies incorporating broader political conditions into return decisions, see Zakirova and Buzurukov (2021) and Beber, Roessler and Scacco (2021).

⁵*Non-refoulement* is generally recognized as binding all states as customary international law, while states parties to the 1951 Convention or its 1967 Protocol are additionally bound by their treaty

2004; Harrell-Bond, 1989; Long, 2013; Stein and Cuny, 1994). Governments may withdraw residence permits, close schools and markets, terminate assistance, or threaten expulsion to induce ostensibly “voluntary” repatriation. UNHCR may support repatriation under these conditions both to reduce the harm refugees would face if expelled without assistance and because it relies on host-state cooperation to maintain access (Barnett, 2001*a,b*; Barnett and Finnemore, 2004; Whitaker, 2003).

Conflict in the asylum country can similarly make continued residence untenable. Most refugees live in neighboring countries, where conflict risks are correlated across borders (UNHCR, 2025; Fearon and Laitin, 2003; Collier and Hoeffler, 2004; Gleditsch, 2007; Lischer, 2005). Refugees may therefore be victims of a new conflict in asylum, which forces them to leave. Consider, for example, Zaire (DRC) in 1996:

“In October, an outbreak of civil war in Zaire, which engulfed Rwandan refugee camps there, produced what UNHCR described as the ‘largest and swiftest’ repatriation in memory. About a half-million Rwandan refugees returned home during a four-day period in mid-November. Most returnees repatriated on foot, jamming the main Rwandan highway from the border. The sea of humanity stretched nearly 100 miles during the height of the return.” (*WRS, Zaire, 1997*)

When refugees are pushed out, return may be their only feasible option rather than onward movement to a third country. Many live near origin-country borders because of established migration routes, cross-border ethnic ties, and the strategic placement of camps (Camarena, 2024).⁶ Authorities may restrict internal travel or entry into neighboring states (Whitaker, 2003), conflict may make onward travel dangerous, and refugees may lack the resources to relocate elsewhere. Deteriorating conditions in asylum can therefore produce return even when conditions at home have not improved.

H2 (Stay Constraint). Constraints on refugees’ ability to remain in asylum increase return independently of improvements in origin-country conditions. Coercive host policies and conflict in asylum should therefore increase return, including during ongoing conflict at home.

Can refugees move? Refugees may prefer to return but lack the resources to do so. Repatriation requires transportation, housing, replacement of lost property, and the re-establishment of livelihoods, often in economies damaged by war. Refugees who have faced employment restrictions or spent years in camps may have few savings to cover these costs.

International assistance can relax this constraint. UNHCR repatriation programs provide transportation, cash, food, shelter, and reintegration assistance that can make an

obligations.

⁶For example, 79 percent of UNHCR camps in Africa lie within 50 kilometers of a border (Fein and Van Den Hoek, 2022).

otherwise unaffordable move feasible. The repatriation of Sierra Leonean refugees from Guinea in the early 2000s illustrates the scale of support that a repatriation program can involve:

“About half of the returning refugees received assistance from UNHCR and other agencies in the form of transportation, cooked meals, and medical care during the journey home. . . Returnees received blankets, cooking utensils, shelter materials, and a two-month food supply. UNHCR appealed for a nine-fold increase in its budget, to \$18 million, to support reintegration programs including schools, health clinics, and livelihood programs.” (*Guinea & Sierra Leone, WRS, 2002–2004*)

This constraint is distinct from refugees’ preferences over where to live. Two refugees may both prefer to return, but only one may have the resources to make the move and begin their life again in their home country. Repatriation assistance can therefore increase realized return without changing the underlying preference between origin and asylum.

H3 (Mobility Constraint). Return becomes more likely as refugees gain access to resources that reduce the financial constraint on relocation. In particular, repatriation assistance should increase realized return.

3.2 Formalization

Let U_{oi} and U_{ai} denote refugee i ’s expected utility from living in the origin country, o , and the asylum country, a , respectively:

$$U_{oi} = v_o(s_o, e_o, \rho_o), \quad U_{ai} = v_a(s_a, e_a).$$

Here, s denotes security, e economic conditions, and ρ_o political repression. The functions $v_o(\cdot)$ and $v_a(\cdot)$ map these conditions into the expected utility associated with each location.

Let P_i indicate whether refugee i prefers the origin country to the asylum country:

$$P_i = \mathbb{1}\{U_{oi} > U_{ai}\}.$$

Thus, $P_i = 1$ when the refugee prefers to return and $P_i = 0$ when the refugee prefers to remain in asylum. Better conditions at home raise U_{oi} , while better conditions in asylum raise U_{ai} . A war ending therefore makes return more attractive without necessarily making the origin preferable.

The constrained-choice framework adds two conditions. Let $A_i = 1$ if continued asylum

Table 1: Preferences, constraints, and realized return

Case	Conditions	Implication
Unconstrained choice	$A_i = 1, M_i = 1$	Return if and only if the origin is preferred, $P_i = 1$.
Constrained return	$A_i = 0$	Return can occur even when asylum is preferred.
Constrained non-return	$A_i = 1, M_i = 0$	Refugees can remain abroad even when the origin is preferred.

Notes: A_i indicates whether continued asylum is feasible; M_i indicates whether voluntary return is financially feasible; P_i indicates whether the origin is preferred to asylum.

is available and $A_i = 0$ if conflict or coercion makes staying infeasible. Let c_i denote the resources required to return, r_i the refugee's available resources, and z_i external repatriation assistance. Let M_i indicate whether return is materially feasible:

$$M_i = \mathbb{1}\{r_i + z_i \geq c_i\}.$$

Thus, $M_i = 1$ when the refugee has sufficient resources to cover the costs of return, and $M_i = 0$ when those resources are insufficient. Let R_i indicate whether refugee i actually returns to the origin country, with $R_i = 1$ for return and $R_i = 0$ for remaining in asylum. Realized return satisfies

$$R_i = 1 \iff A_i = 0 \text{ or } (A_i = 1, P_i = 1, M_i = 1).$$

This expression captures two routes to return. If continued asylum is unavailable, $A_i = 0$, the refugee returns regardless of whether the origin country is preferred. If asylum remains available, $A_i = 1$, return occurs only when the refugee both prefers the origin, $P_i = 1$, and has sufficient resources to move, $M_i = 1$.

The conventional location-choice model is the special case in which both staying and moving are feasible, $A_i = M_i = 1$. Under these conditions, realized return reflects preferences alone: refugees return if and only if they prefer the origin country. Once either constraint is allowed to bind, however, realized behavior can diverge from preferences. When asylum becomes unavailable, refugees may return even if they would prefer to remain abroad, producing constrained return. Conversely, when asylum remains available but return is not financially feasible, refugees may remain abroad even if they would prefer to return, producing constrained non-return.

4 The Global Refugee Return (GRR) Dataset

To test the theory, I introduce the Global Refugee Return (GRR) dataset, which covers all protracted refugee situations at the origin–asylum dyad–year level from 1980 to 2023. The dataset includes origin countries with at least 25,000 refugees in exile in any year, which is the UNHCR’s definition of a protracted refugee situation. For each origin country, I include asylum countries accounting for at least 75 percent of the total refugee population.⁷ The resulting sample comprises 293 dyads across 87 origin populations and 93 asylum countries.

GRR makes two contributions. First, it corrects systematic measurement error in UNHCR data on refugee stocks and returns. UNHCR records only voluntary repatriations, excluding forced returns,⁸ and captures spontaneous returns incompletely, since refugees who return without UNHCR assistance may not be recorded. Stock figures often reflect “planning estimates” negotiated with host governments rather than actual counts, omit large unregistered populations, and are not revised retroactively when more accurate censuses become available. These problems are compounded by political incentives: host governments may overstate stocks to sustain aid flows, while origin governments may overstate returns to signal post-conflict stability. UNHCR, which depends on the cooperation of these governments, may adopt government figures rather than risk losing access.

Rather than relying on official statistics, GRR reconstructs stocks and returns using qualitative assessments from experts in each context. The primary source is the *World Refugee Survey* (WRS), published annually from 1980 to 2008, which reports on stocks, forced and spontaneous returns, and unregistered populations, as well as providing a qualitative assessment of the reliability of official data. The WRS draws on consultations with governments, international organizations, and field-based sources, and is widely regarded as an authoritative account of refugee conditions (USCRI, 2008). Since reports are organized by host country rather than by return event, they capture smaller return movements as well as large episodes. For 2009–2023, I draw on the U.S. Department of State’s *Country Reports on Human Rights Practices*. Because these reports less frequently include stock and return statistics, for these years I use UNHCR data as a baseline and supplement with information on unregistered populations and forced returns.

GRR improves on existing data not only by coding reports that are less vulnerable to institutional bias, but also by prioritizing internal consistency within origin–asylum dyads. Official reports often provide a range for the refugee population, but may alter-

⁷The 75 percent threshold is calculated using UNHCR data, supplemented with Palestinian refugees under UNRWA and Mozambican refugees in South Africa, listed as “Unknown” in UNHCR data.

⁸Note that such repatriations often include returns that are coerced in practice but classified as voluntary.

nate between the upper and lower bounds across years even when there is no evidence that the underlying stock changed. GRR applies a consistent coding rule: when the reported range remains unchanged, it retains the same estimate unless the archival evidence indicates a genuine change in the refugee population. Because each series is hand-coded and checked multiple times, year-to-year changes are more likely to reflect actual movements in refugee stocks rather than revisions in reporting conventions. This makes GRR particularly well suited to estimating within-dyad changes over time.

Second, GRR complements existing *de jure* datasets such as DWRAP by introducing two measures of refugees' *de facto* conditions in asylum: forced return practices and the share of each refugee population living in camps (Blair, Grossman and Weinstein, 2022).⁹ Additionally, GRR introduces new archival measures of the repatriation process, including UNHCR-assisted repatriation programmes and tripartite repatriation agreements, as well as new data on UNHCR financing, measuring both overall funding per refugee within each asylum country and funding allocated specifically to repatriation.

The dataset was constructed through detailed coding of archival reports. I hand-coded the 25 largest refugee-producing countries and their primary asylum destinations (83 dyads), and the remaining dyads were coded by a team of 26 trained research assistants using a standardized protocol, which I then reviewed. I cross-validated major discrepancies between GRR and UNHCR figures and assessed internal consistency through desk research. See Section 6 for a discussion of the dataset's limitations and the Appendix for additional details.

The value of these corrections is illustrated by the example of Ethiopian and Eritrean refugees in the Horn of Africa. UNHCR data record 2 million Ethiopian refugees in Somalia in 1980, falling to 700,000 in 1981 and remaining around that figure until 1988; GRR records 700,000 throughout the early 1980s, declining to 350,000 by 1988. The WRS indicates that the decline in official statistics reflected "more accurate counting" rather than large-scale return (Somalia, 1982), and that the Somali government's later figure of 840,000 was "arbitrarily agreed upon by the government and UNHCR" and "always considered far too high" by others (WRS, Somalia, 1989). UNHCR data also miss the pushed return that followed: as the Somali civil war engulfed refugee-hosting areas in 1990–1991, large numbers of Ethiopian refugees fled back to Ethiopia, yet UNHCR records no repatriations because its personnel had been evacuated. The reverse problem appears in Sudan, where UNHCR data show a sharp drop in the Eritrean refugee stock after the fall of the Mengistu regime in 1991, but GRR records no comparable decline because large-scale repatriation was delayed by funding disputes;

⁹Existing data on camp residence are insufficient for this purpose: UNHCR accommodation data tend to undercount camp populations, while Zhou and Shaver's (2021) refugee site data do not identify which refugee groups occupy each site or what share of each group lives in camps, limiting their use for studying variation across groups facing different conditions within the same country.

only “some 10,000 Eritreans returned by their own means” in the year of independence (WRS, Sudan, 1992). This case is discussed in more detail in Section 7.

5 Empirical Analysis

5.1 Global Descriptive Patterns

I begin by documenting where and when refugee return occurs. Figure 3 summarizes the conditions under which global return takes place, while Figure 4 shows the timing of origin-country conflict, war endings, asylum-country conflict, and forced-return policies across the refugee dyads accounting for the most return.

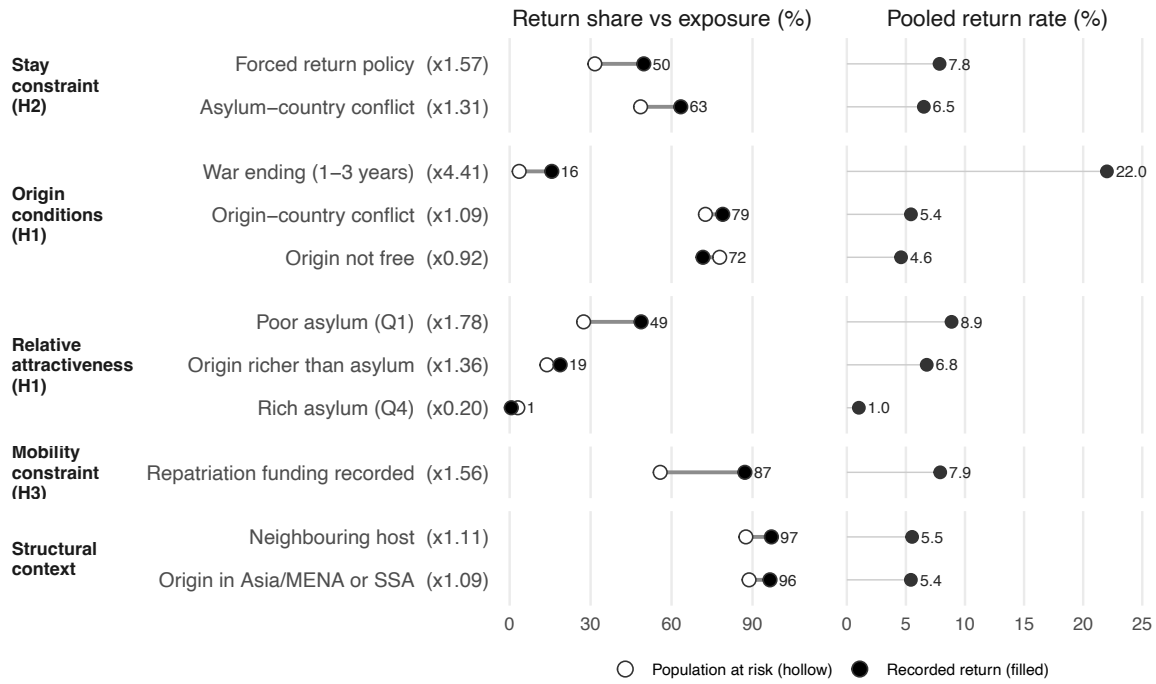
Return is highly concentrated. The top 5% of dyad-years account for 93.8% of all recorded return, while 53.8% of dyads never record any return. Even among dyads that do experience return, the median dyad experiences 49.9% of its lifetime return in a single year. Realized return is therefore better characterized as an episodic movement affecting substantial portions of refugee populations than as a steady annual flow. This concentration motivates a focus on population-wide shocks that can change refugees’ options abruptly, rather than on slow-moving differences between origin and asylum, such as economic conditions.

The descriptive patterns provide initial evidence on each of the three components. First, conditions associated with refugees’ ability to remain in asylum are disproportionately present when return occurs. Asylum-country conflict occurs in dyad-years accounting for 63% of all return, while some form of forced-return policy occurs in dyad-years accounting for 50%. Severe forced-return policies are particularly concentrated among major return episodes. Only 150 dyad-years are coded as severe, yet they account for 35.3% of all recorded return. Figure 4 further shows that these policies frequently occur while conflict in the country of origin remains ongoing.

Second, the descriptive evidence also shows that improvements in origin-country security matter, but do not determine return on their own. Seventy-nine percent of recorded return occurs while the origin country is at war, and 17 of the 20 largest return episodes take place during ongoing conflict. The first three peaceful years following a war ending account for only 16% of global return. Return rates can nevertheless be high when these post-war periods occur, averaging 22.0% annually, which suggests that ending a war can substantially increase the attractiveness of return for affected populations.

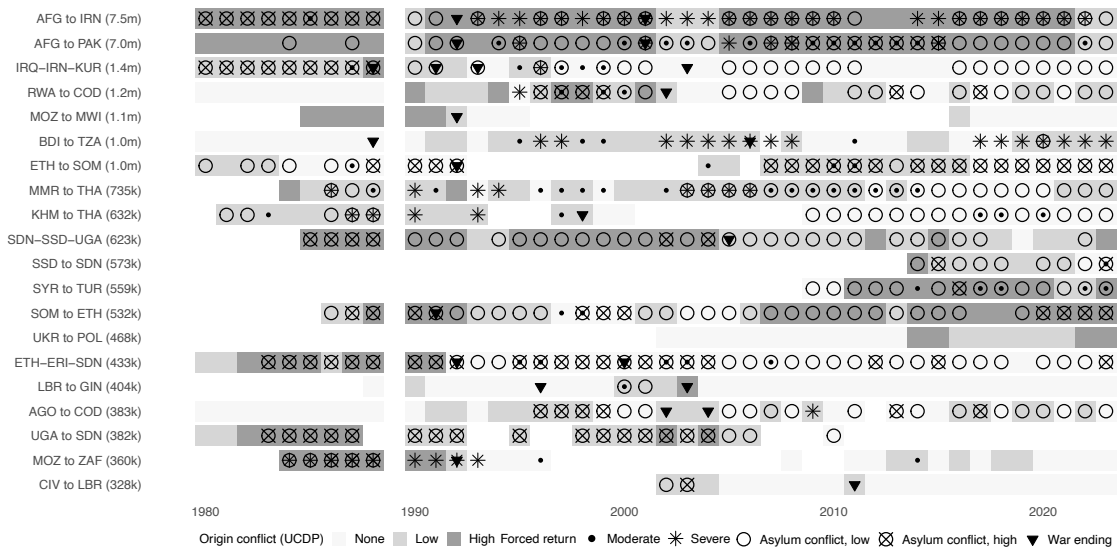
Other descriptive patterns are consistent with the preference component. Return from wealthy asylum countries is relatively rare, including when refugees from the same origin countries return from poorer hosts. This is not simply a distinction between high-income countries and the rest of the world. Excluding Europe and North America, average GDP per capita is \$2,752 in asylum countries compared with \$1,040 in countries

Figure 3: Correlates of Global Refugee Return



Notes: Each row compares a condition's share of all recorded return (filled) with its share of the population at risk (hollow); the multiplier in the row label is the ratio of the two. The right panel gives the pooled return rate. Conditions overlap. Descriptive comparisons have no controls or fixed effects.

Figure 4: Conflict, war endings, and forced return in the twenty dyads with the most return



Notes: Twenty dyads with the highest total recorded return, 1980–2023, together accounting for 78% of all return. Row labels report total return. Shading indicates origin-country conflict intensity; circles indicate asylum-country conflict; dots and stars indicate moderate and severe forced-return policies; triangles mark war endings. Open circles denote low-intensity conflict and crossed circles high-intensity conflict. Policy marks are plotted over conflict circles when both occur. Full GRR panel.

of origin. Among peaceful origin-years without asylum-country push, dyads in which the origin country is richer contain only 24% of refugees but account for 42% of return. Similarly, origins classified as “Not Free” account for less return than their share of the population at risk, and repression frequently persists after fighting ends.¹⁰ Ending a war therefore changes an important component of the relative attractiveness of origin and asylum without necessarily reversing the overall comparison.

Finally, return must be feasible. Among the dyad-years for which UNHCR repatriation funding is coded, repatriation programmes account for 87% of recorded return while covering 56% of the refugee population at risk. This comparison is descriptive, but highlights how frequently large repatriation movements coincide with external resources for transportation and reintegration.

These descriptive patterns are consistent with the three components, but they do not show that these conditions themselves drive return; they may simply reflect differences across refugee populations and host countries. I therefore turn next to whether changes in these conditions predict changes in return within the same origin–asylum dyads over time.

5.2 Empirical Strategy

The theory concerns refugees’ preferences and constraints, which are not directly observed in the dyad-year data. I therefore test observable implications of the three components by examining whether changes in the conditions associated with each predict changes in the rate of realized return within the same origin–asylum dyads over time. Dyad fixed effects absorb time-invariant differences across refugee populations and host-country relationships, while year fixed effects account for shocks common to all dyads. Because the analysis is observational, I interpret the coefficients as conditional within-dyad associations rather than as causal effects. The tests ask whether changes in realized return occur in the directions implied by the theory after accounting for time-invariant dyad characteristics and common year shocks.

5.2.1 Independent Variables

Preferences (H1). I measure changes in the relative attractiveness of origin and asylum using security, economic, and political conditions in both locations. *War Ending* is an indicator for the first three peaceful years following the end of a war in the origin country. War endings come from an extension of Fearon’s (2017) civil war list through 2023, merged with inter-state wars from the Correlates of War project. A year enters the post-war window only if the origin country is also at peace according to the same war list. This means that if one war ends but another war in the same country remains

¹⁰Among origins observed during both conflict and peace, 81% of refugees in peace years still come from states classified as “Not Free,” compared with 88% during high-intensity conflict.

ongoing or begins immediately afterward, those years are not coded as post-war.

Conflict in COO is coded from UCDP as 0 for no conflict, 1 for low-intensity conflict with 25–999 battle deaths, and 2 for high-intensity conflict with at least 1,000 battle deaths (UCDP, 2024). The categories enter as indicators with no conflict as the reference category.

Economic conditions are measured using GDP per capita in the country of asylum, *GDP (COA)*, and country of origin, *GDP (COO)*, from the World Bank. *Civil Liberties Restrictions (COO)* is the Freedom House civil liberties index, ranging from 1 for the greatest protection of civil liberties to 7 for the least. Higher values therefore indicate greater non-conflict political insecurity in the country of origin.

Stay Constraint (H2). I use two measures of conditions that can constrain refugees' ability to remain in asylum. *Conflict in COA* is coded analogously to conflict in the origin country as no, low-intensity, or high-intensity conflict.

Forced Return is an ordinal measure of coercive return policy at the asylum–origin dyad level, hand-coded from the WRS and USHR archival reports. It takes values 0 for no policy, 1 for moderate coercion, and 2 for severe coercion, and enters the models as indicator variables with no forced-return policy as the reference category. Moderate policies include threats or encouragements to return. Severe policies are implemented measures intended to bring about the widespread return of the refugee population, such as the workplace raids and the revoking of residence permits. Coding is based on the policy imposed rather than on the amount of return subsequently observed, and follows rules detailed in the Appendix.

Mobility Constraint (H3). Because refugees' own resources are not observed systematically, I use UNHCR repatriation assistance to measure external resources that can reduce the costs of return. *Repatriation Assistance* is the log of one plus UNHCR repatriation funding per member of the origin country's global refugee population, hand-coded from UNHCR Executive Committee documents and Yearbooks for 1980–1999 and 2011–2020. Because historical budgets cannot consistently be allocated across asylum countries, this measure varies at the origin-country-year level.

I also construct two dyad-level measures: *UNHCR Repatriation Programme*, indicating whether a UNHCR-assisted programme was operating, coded from the *World Refugee Survey* for 1980–2008, and *Tripartite Repatriation Agreement*, indicating an active agreement among UNHCR and the origin and asylum governments, coded from UNHCR archives and supplemented with Camarena (2022). Because these measures cover different periods and do not fully overlap with the funding measure, I estimate them in separate models rather than include them in the main regression table; their estimated return-rate ratios are reported in Figure 5. To my knowledge, these archival measures have not previously been assembled systematically across refugee populations. Because

UNHCR assistance may respond to expected return and other political conditions, I interpret the estimates as associations consistent with H3 rather than causal effects of assistance.

Controls. *Refugee Rights* is the DWRAP index of legal rights afforded to refugees in the country of asylum (Blair, Grossman and Weinstein, 2022). *Conflict Duration* is the log of cumulative years of conflict in the country of origin. Section 6 assesses robustness to alternative measures of conflict, broader political persecution, welfare, and international assistance.

5.2.2 Specification

I estimate Poisson pseudo-maximum likelihood (PPML) models with dyad and year fixed effects.¹¹ The unit of analysis is the directed origin–asylum dyad-year. The dependent variable is the count of refugees in dyad i , from origin o and hosted in asylum country a , who return during year t .

I include the log of the population at risk as an offset, so the coefficients describe proportional changes in the return rate rather than changes driven mechanically by refugee population size. The population at risk is the previous year’s refugee stock.¹² I restrict the estimation sample to dyad-years with more than 1,000 refugees in the previous year, reducing the influence of small denominators that can generate unstable return rates.

The baseline specification is:

$$E[Y_{it} \mid \mathbf{X}_{it}] = \exp(\beta' \mathbf{X}_{it} + \alpha_i + \gamma_t + \ln S_{it-1}), \quad (1)$$

where Y_{it} is the number of returnees from dyad i in year t , $\mathbf{X}_{it}$ is a vector of time-varying covariates, α_i and γ_t denote dyad and year fixed effects, and $\ln S_{it-1}$ is the offset for the population at risk. Standard errors are clustered at the dyad level.

The hypotheses concern different components of this model. H1 predicts that return should increase following improvements in origin-country conditions and decrease as asylum becomes more attractive relative to origin. I therefore examine the post-war coefficient, the economic and political covariates, and whether the association between a war ending and return varies with these comparative conditions. H2 predicts positive

¹¹The PPML estimator is well suited to this setting because refugee return counts are non-negative, highly right-skewed, and contain many zeros. PPML accommodates these features while remaining consistent under general forms of heteroskedasticity, unlike log-linearized OLS specifications that require transforming or dropping zero-valued observations.

¹²The exception is 21 dyad-years in which a large refugee inflow arrived and returned within the same year. For these observations, the denominator is the estimated within-year peak population.

coefficients for forced-return policies and asylum-country conflict, including when conflict continues in the country of origin. H3 predicts a positive coefficient on repatriation assistance.

Finally, I distinguish each mechanism’s within-dyad association from its aggregate importance. A condition may strongly predict return but account for little global movement if few refugees are exposed to it. I therefore use a pooled PPML model to estimate the reduction in predicted return when severe coercion, the post-war window, or high-intensity asylum-country conflict is set to zero wherever observed. I use a pooled specification because dyad fixed effects drop zero-return dyads and absorb between-dyad differences in exposure. These estimates are descriptive model-based decompositions, not causal effects.

5.3 Results

Table 2 presents the main estimates. Model 1 is the primary specification. Model 2 adds UNHCR repatriation assistance and therefore uses the smaller subset of origin-years for which funding is recorded. Because the refugee population at risk enters as an offset, exponentiated coefficients describe proportional differences in the return rate. The mean return rate in the Model 1 sample is 6.6%, compared with 5.0% in the broader estimation frame, because fixed-effects PPML drops dyads with no variation in return.¹³

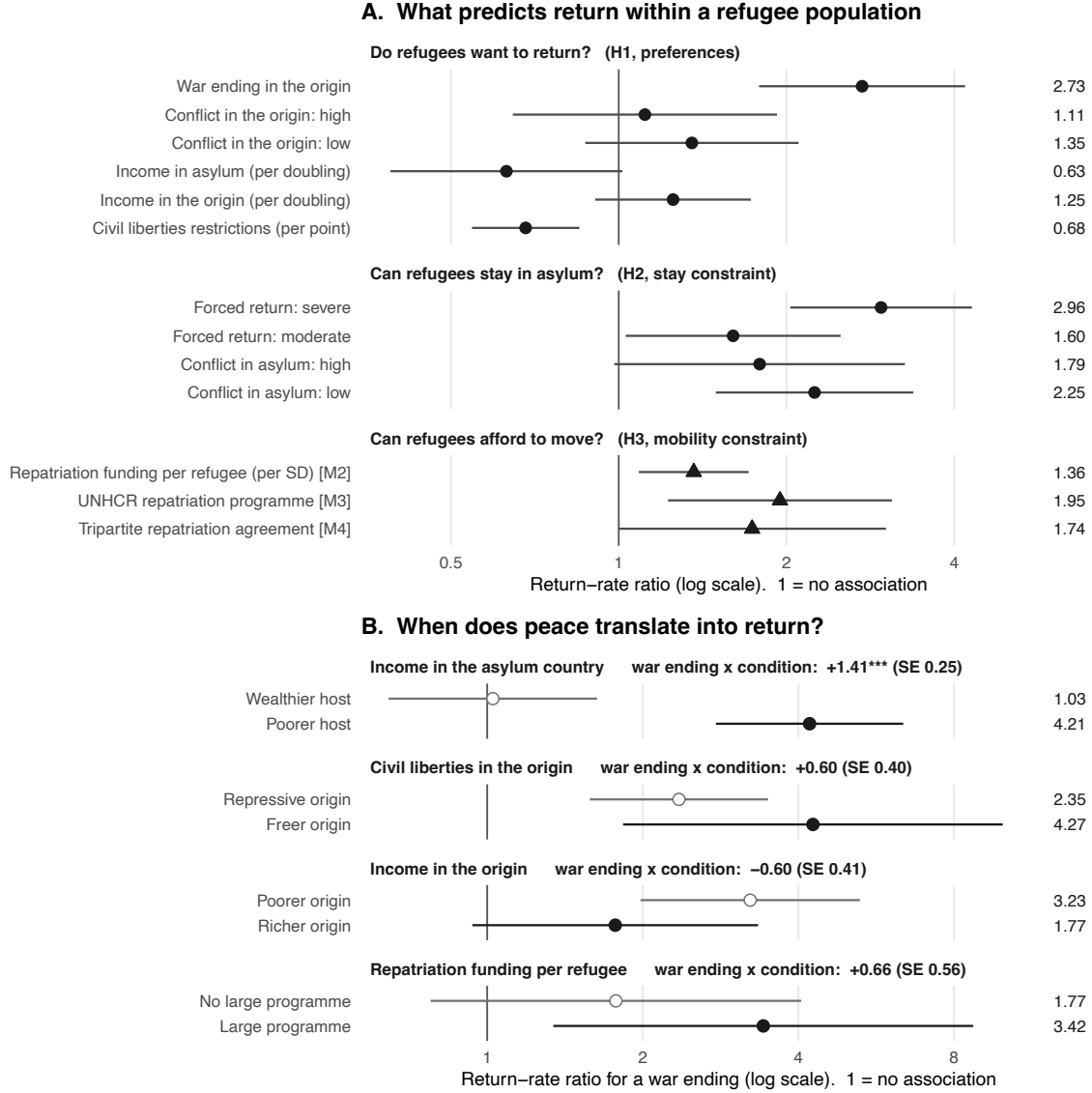
Because this restriction excludes many refugee populations that never experience return, I also estimate pooled PPML models with year fixed effects but without dyad fixed effects. These retain 255 of the 285 dyads in the estimation frame, subject to covariate missingness, and produce very similar results (Appendix).

Preferences (H1). H1 predicts that, conditional on the constraints on staying and moving, return should respond to the relative attractiveness of origin and asylum. The baseline results first confirm the importance of origin-country security. The coefficient on the post-war window is 1.01 (SE = 0.22), implying a return rate approximately 2.7 times as large during the first three peaceful years following a war ending (Panel A of Figure 5). By contrast, neither low- nor high-intensity origin-country conflict is statistically distinguishable from peace once the remaining covariates and fixed effects are included. This suggests that changes in conflict intensity have relatively little association with return. What matters is the discrete transition from war to peace.

Model-predicted rates indicate that this increase is concentrated immediately after con-

¹³Percentage-point translations below are anchored to the regression-sample mean of 6.6%. The estimation sample contains 2,327 dyad-years across 110 dyads, from an estimation frame of 5,131 dyad-years in 285 dyads. Of the excluded dyads, 156 record no return in any year and 19 are lost to covariate missingness. The fitted sample contains 75.6% of all return recorded in GRR and 58.7% of refugee stock-years in the estimation frame. The Appendix documents sample attrition in detail.

Figure 5: What predicts refugee return, and when peace translates into return



Notes: Return-rate ratios from PPML models with dyad and year fixed effects, log refugee stock at risk as an offset, and dyad-clustered standard errors. Bars show 95% confidence intervals; the vertical line at one indicates no association. Circles report the baseline specification in Table 2 (M1). Triangles report auxiliary models adding UNHCR repatriation funding (M2), a UNHCR repatriation programme indicator (M3, 1980–2008), or a tripartite repatriation agreement indicator (M4, 1980–2002), each estimated on the available sample. Income is scaled per doubling and civil liberties per Freedom House point. In Panel B, estimates show the return-rate ratio associated with a war ending at the indicated value of each condition, based on the corresponding interaction model.

Table 2: Poisson pseudo-maximum likelihood estimates of refugee return

Term	(1) Main	(2) With assistance
Preferences (H1)		
War Ending	1.01*** (0.22)	1.05** (0.38)
Conflict in COO: high	0.11 (0.28)	0.38 (0.35)
Conflict in COO: low	0.30 (0.22)	0.22 (0.25)
GDP (COA)	-1.12 [†] (0.59)	-1.40 (0.86)
GDP (COO)	0.39 (0.28)	0.80 (0.50)
Civil Liberties Restrictions (COO)	-0.53*** (0.16)	-0.10 (0.26)
Stay constraint (H2)		
Forced Return: severe	1.08*** (0.19)	1.29*** (0.31)
Forced Return: moderate	0.47* (0.23)	0.13 (0.28)
Conflict in COA: high	0.58 [†] (0.31)	0.79 [†] (0.43)
Conflict in COA: low	0.81*** (0.21)	0.95** (0.36)
Mobility constraint (H3)		
Repatriation Assistance	–	0.31** (0.12)
Controls		
Refugee Rights	-0.01 (0.17)	-0.19 (0.22)
Conflict Duration	-0.03 (0.09)	-0.06 (0.12)
Fixed effects and fit		
Dyad and year fixed effects	Yes	Yes
Sample	Full	Funding coded
Dyad-years	2,327	1,378
Dyads	110	100
Pseudo R2	0.799	0.782

Notes: COO = country of origin; COA = country of asylum. Column 2 adds UNHCR repatriation assistance on the smaller sample of origin-years for which funding is recorded. PPML with dyad and year fixed effects and the log lagged refugee stock as an offset. Continuous variables are standardized separately, so their coefficients are not directly comparable. Dyad-clustered standard errors in parentheses.

flict terminates. One year after a war ending, predicted return is 27.3% of the refugee stock at risk, compared with 14.5% when the post-war indicator is counterfactually set to zero; it remains elevated in the second year before declining in the third (Appendix). Return is therefore substantially higher immediately following war endings, as conventional security-based accounts and H1 predict.

The remaining covariates provide evidence for the comparative component of H1. Return falls as political repression in the origin country increases: the coefficient on civil-liberties restrictions is -0.53 ($SE = 0.16$). Return is also lower from wealthier asylum countries, with a coefficient of -1.12 ($SE = 0.59$), although this estimate is less precise. Origin-country income is not statistically distinguishable from zero in the baseline model.

The more demanding implication of H1 is that the association between peace and

return should vary with the attractiveness of asylum. Panel B of Figure 5 reports the return-rate ratio associated with a war ending at each value of the four conditions, together with the interaction that tests whether the condition changes this association rather than the level of return. The evidence supports this implication most clearly for asylum-country income. A war ending has no detectable association with return in wealthier asylum countries, where the ratio is 1.03 (coefficient 0.03), but a large and precisely estimated association in poorer ones, at 4.21 (1.44); the interaction between the two, 1.41, is statistically significant. Post-war return is therefore substantially higher when remaining abroad is relatively unattractive.

The other three interactions are weaker. Post-war return is higher for less repressive origins, but that interaction is not statistically significant, and origin-country income does not significantly condition the post-war association. The clearest conditional evidence for H1 thus comes from the asylum side of the comparison. The Appendix reports the four interaction models in full.

H1 therefore receives its clearest support from the post-war pattern and the asylum-income interaction: return rises following war endings, but this association is much weaker where refugees have an attractive option in asylum. Continued displacement after war is thus consistent with refugees responding to improved security at home while still sometimes preferring to remain abroad.

Stay constraint (H2). H2 predicts that return rises with push factors in asylum, even if conditions at home have not improved. The clearest evidence comes from coercive host-state policies. Relative to years without such a policy, a moderate forced-return policy is associated with a 60% higher return rate ($e^{0.47} = 1.60$). A severe policy is associated with nearly three times the return rate ($e^{1.08} = 2.96$), the largest association in Panel A of Figure 5. In the year a severe forced-return policy begins, predicted return is 9.2%, compared with 4.0% when the policy is counterfactually set to zero. One year later, the corresponding rates are 19.1% and 5.5% (Appendix).

Conflict in the asylum country points in the same direction, but the evidence is weaker. Low-intensity asylum-country conflict is associated with a return rate approximately 2.2 times that observed under peace ($e^{0.81}$), while the point estimate for high-intensity conflict implies a rate approximately 1.8 times as large ($e^{0.58}$). Only the low-intensity coefficient is precisely estimated, and the two coefficients are not statistically distinguishable from one another. I therefore treat asylum-country conflict as suggestive evidence for H2 rather than as a result comparable in strength to coercive policy.

Most importantly, the forced-return result does not depend on peace at home. Of the 129 severe-policy dyad-years in the estimation sample, only 8 occur inside the post-war window, while 82 occur during ongoing origin-country conflict. Interacting severe forced return with ongoing conflict leaves the severe-policy coefficient at 1.19

(SE = 0.36), while the interaction is small and statistically insignificant, -0.15 (SE = 0.32, $p = 0.64$). The return associated with severe coercion is therefore statistically indistinguishable depending on whether conflict at home is ongoing.

This is the distinctive observable implication of H2. If realized return simply tracked improving conditions at home, coercive policy should be associated with little additional return while those conditions remained unsafe. Instead, severe host-state pressure is associated with substantially greater return even while the insecurity that originally generated displacement persists. The evidence for conflict in asylum is more qualified. More broadly, these patterns show why observed return cannot always be interpreted as revealed preference for the origin country.

Mobility constraint (H3). H3 predicts that refugees who want to return may nevertheless remain abroad if they lack the resources to move, and that assistance reducing these costs should increase realized return. The new archival measures make it possible to examine systematically, for the first time across refugee populations, whether return is higher when resources that reduce the costs of relocation are available.

All three measures of UNHCR repatriation assistance are positively associated with return. In Model 2, a one-standard-deviation increase in logged repatriation funding per refugee is associated with a 36% higher return rate (coefficient 0.31, SE = 0.12). A UNHCR repatriation programme is associated with 1.95 times the rate of return (0.67, SE = 0.24), while an active tripartite repatriation agreement is associated with 1.74 times the rate (0.55, SE = 0.28), although the latter estimate is less precise. Two years after a war ending, predicted return is 10.0% with no recorded repatriation funding and 19.5% at the seventy-fifth percentile of funding.

Panel B of Figure 5 shows a similar pattern. A war ending is associated with 1.77 times the return rate without substantial assistance, compared with 3.42 times the rate when substantial assistance is present (coefficients 0.57 and 1.23). Although the difference is not statistically significant, the pattern is consistent with H3: improved conditions at home may increase refugees' willingness to return without producing movement when they lack the resources to do so.

A concern with these results is that UNHCR may provide assistance when it expects refugees to return, in which case assistance would mark anticipated return rather than relax a constraint on movement. Several patterns weigh against this interpretation. First, within the funding data, the positive association is concentrated in the *amount* of assistance rather than simply whether any funding is present. Among origin-years receiving some funding, the coefficient on funding per refugee is 0.82 (SE = 0.21), while an indicator for any funding is small and statistically indistinguishable from zero (0.14, SE = 0.30). When both enter together, the amount remains strongly positive (0.64, SE = 0.16). This pattern is more consistent with a resource mechanism, since the amount

available determines how much of the cost of return can be covered.

Second, peace and repatriation assistance often occur at different times. Of the 152 post-war dyad-years for which the programme measure is observed, only 41% have a programme operating. When programmes begin after a war ending, the median lag is one year, 45% begin at least two years later, and one-quarter begin five or more years later.

These delayed cases provide additional leverage on whether refugees move when conditions improve or when resources become available. Across the fourteen post-war episodes in which assistance began at least three years after the war ended, return had already subsided by the time assistance arrived. The return rate averaged 0.8% in the two years before assistance and rose to 9.9% over the three years beginning with its arrival. Post-war situations that never received assistance show no comparable increase, averaging 1.4% during years four through eight after a war ending.¹⁴ For these cases, assistance brings about a renewed increase in returns, suggesting that it can have an effect on return independent of war ending.

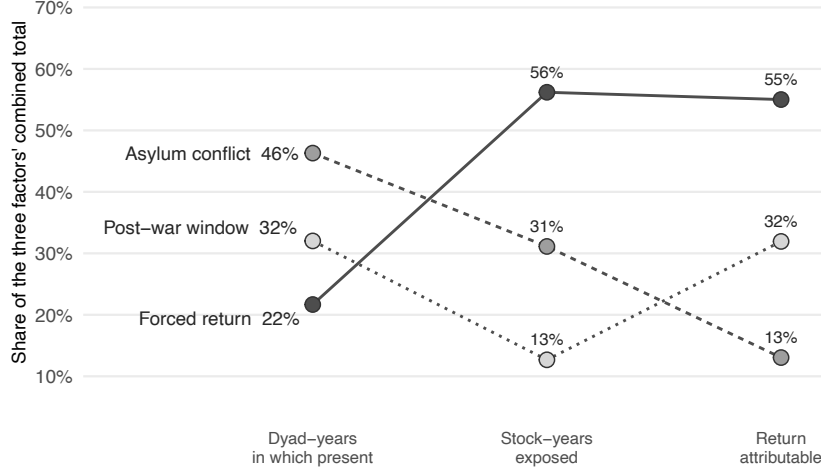
The timing of tripartite agreements points in the same direction. Across agreements signed in dyads in the estimation sample, the pooled return rate averages 3.7% over the four years preceding signature, then rises to 8.1% in the year of signature, 18.8% in the following year, and 18.2% in the second year after signature. Return therefore tends to rise after an agreement is concluded rather than while it is being negotiated.

These results do not identify a causal effect of repatriation assistance: funding, programmes, and tripartite agreements may respond to expected return and broader political conditions, the timing comparisons rely on relatively few episodes, and the measures cover only part of the panel. Their contribution is instead to provide the first systematic cross-national evidence on a previously difficult-to-measure dimension of repatriation. Across three independently constructed archival measures, return is consistently higher when repatriation resources and institutional support are available, and the timing evidence further supports H3's implication that limited resources can constrain refugees' capacity to return.

Global patterns of return. The three hypotheses concern different components of an individual refugee population's return decision. They do not, by themselves, establish which mechanism matters most for the global pattern documented at the beginning of the paper. Aggregate importance depends on exposure as well as the size of the conditional association: a factor that strongly predicts return can account for

¹⁴These fourteen episodes occur across nine dyads, with six or seven dyad-years contributing to each relative-year cell, so the pattern is suggestive rather than precisely estimated. Return after assistance begins also accounts for only 15% of total recorded return in these episodes. For these delayed programs, assistance therefore appears to restart return after an earlier wave has subsided rather than account for most post-war movement.

Figure 6: Model-implied contributions to predicted global return



Notes: Each line compares a factor's frequency, refugee stock-years exposed, and model-implied share of predicted return. Estimates use pooled PPML with the log stock at risk as an offset, retaining 255 of 285 dyads and 73% of refugee stock-years. Contributions are calculated by setting each factor to zero wherever observed. These are descriptive decompositions, not causal effects; conditions overlap.

relatively little global movement if few refugees experience it.

Figure 6 separates these two components. For each factor, I compare fitted return under the observed data with a counterfactual prediction in which that factor is switched off wherever it occurs. I use a pooled specification for this decomposition because the dyad fixed-effects PPML drops all dyads without within-dyad variation in return. The pooled model retains 255 of 285 dyads and 73% of refugee stock-years, making it better suited to a claim about aggregate global return.¹⁵

In this model-based decomposition, setting severe forced-return policy to zero wherever it is observed reduces predicted return by 24.7%, compared with 14.3% for the post-war window and 5.9% for high-intensity asylum-country conflict. Setting moderate forced-return policy to zero reduces predicted return by an additional 4.8%. Added together, dyad-years with any forced-return policy are associated with 29.5% of predicted return.¹⁶

¹⁵Distance, contiguity, and population size are included directly to account for important between-dyad differences. The two specifications yield similar attributable shares. Severe forced return accounts for 24.7% of predicted return in the pooled specification and 25.8% under dyad fixed effects; the corresponding figures are 14.3% and 12.6% for the post-war window and 5.9% and 6.3% for high-intensity asylum-country conflict.

¹⁶Because these conditions overlap, their attributable shares cannot be added mechanically. Twenty-four dyad-years contain two of the three factors, and switching all three off simultaneously removes 41.1% of predicted return rather than the 44.9% obtained by summing their individual shares. These quantities are also imprecisely estimated. A dyad-cluster bootstrap gives a ratio of the severe-coercion

The difference between coercion and peace is driven primarily by exposure. Severe forced-return policies occur in only 132 dyad-years, compared with 195 post-war dyad-years, but they affect much larger refugee populations. Severe policies cover 99.9 million refugee stock-years, compared with 22.5 million in post-war windows. The average severe-policy dyad-year contains approximately 757,000 refugees at risk, compared with 116,000 in the average post-war dyad-year.

This distinction reconciles the results across the three components. H1 shows that refugees respond strongly to improvements in origin-country security, while H2 helps explain why most global return nevertheless occurs during conflict: severe coercive policies are imposed on much larger refugee populations.

When do governments force refugees to return? A question that follows from the analysis is why host governments impose forced-return policies. Although this is not a formal test of the theory, I briefly consider what the results suggest. Plausible explanations include host repression and weak refugee protections, hostility toward socially distant or militarized groups, worsening violence in the origin, and pressure from hosting an increasingly large refugee population. I examine these alternatives in the Appendix using pooled comparisons, comparisons across refugee groups within the same host, across hosts receiving refugees from the same origin, and within the same origin-asylum dyad over time.

The clearest result is the importance of scale. A ten-fold increase in refugee stock is associated with roughly five times the odds of severe coercion in a model containing origin conflict and population size. Severe-policy episodes begin at a median population of roughly 160,000 refugees, and 61% begin after the population exceeds 100,000. This does not appear to reflect either general host burden or simple “host fatigue.” Once the size of the particular origin group is included, the host’s total refugee population contributes little. Among populations already above 100,000, policy onset is essentially unrelated to how long the refugee situation has lasted.

The resulting pattern is one of coercion concentrated in large, unresolved displacement crises. Severe policy is more common when hosts are accommodating a large refugee population while the conflict producing that population remains unresolved, leaving little reason to expect the population to decline and the possibility of further displacement. Regime type, social and cultural ties, origin-country violence, and refugee militarization also predict coercion in some raw comparisons, but much less consistently in the within-host, same-origin, and within-dyad analyses. The Appendix analyses do not identify why governments adopt coercion, but they show that refugee-population size is the most consistent correlate across pooled, same-host, same-origin,

share to the post-war share of 1.72, with a 95% interval from 0.44 to 9.89; the coercion share exceeds the post-war share in 72% of resamples. The ordering is therefore more securely established by the descriptive shares of recorded return than by the ratio of attributable shares.

and within-dyad comparisons.

6 Alternative Explanations and Robustness

I address the main threats to inference here and report the full set of robustness checks in the Appendix. The most important concern is measurement, since GRR departs from UNHCR, the only other global source of refugee stocks and returns. I compare the two directly in the Appendix. The datasets agree closely on refugee stocks (correlation 0.96) but substantially less on returns, particularly in poorer asylum countries where most return occurs. UNHCR records roughly three-quarters as much return as GRR, and the twelve largest episodes missing from UNHCR alone account for 18% of all return in GRR. Re-estimating the models with UNHCR data produces much noisier and substantively implausible results, including a positive association between origin-country conflict and return while income and civil liberties matter little. Rather than suggesting that conflict itself encourages return, this pattern is more consistent with differential measurement: UNHCR is better able to observe and record returns in conflict settings where it maintains an operational presence. Neither source provides ground truth, but these results raise concerns about using UNHCR return data to estimate the relationship between origin-country conditions and realized return.

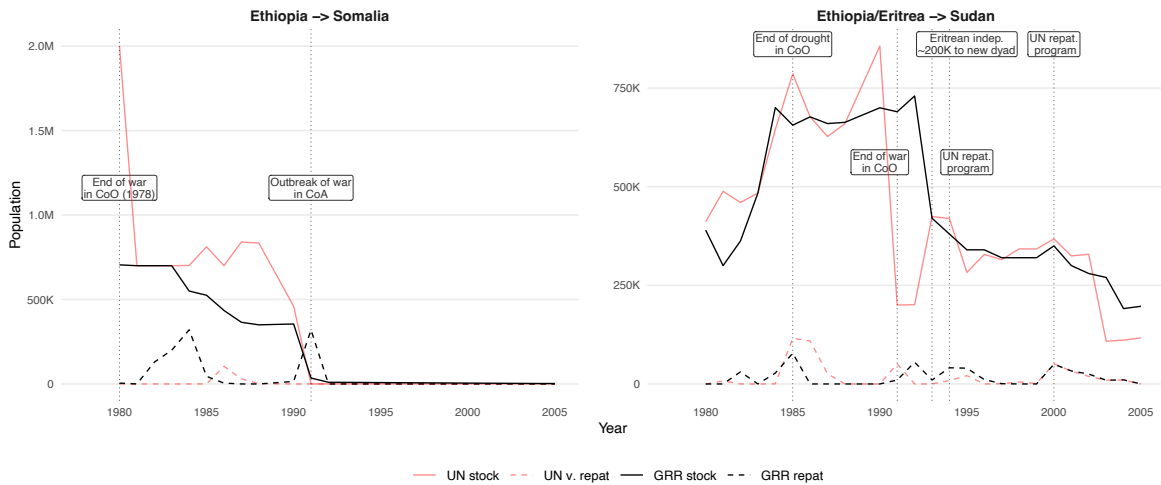
A second concern is the forced-return measure itself. Because it is hand-coded from archival reports, coding necessarily involves judgment. I therefore retain the source text underlying every coding decision and apply explicit thresholds distinguishing moderate and severe policies. The measure also does not mechanically reproduce the outcome: 36% of moderate-policy and 8% of severe-policy dyad-years record no return, and severe forced return remains predictive when the dependent variable is restricted to returns coded as voluntary. As an external validity check, I compare the measure with DWRAP and find that forced-return policies are less common where domestic law provides stronger protections against refoulement.

The main conclusions are robust to alternative measures, specifications, and samples. Severe forced return policies remain strongly associated with return when using alternative definitions of war endings and conflict, alternative measures of repression and refugee stocks, alternative estimators and fixed effects, and samples excluding Europe and North America or populations not initially displaced by war. Severe forced return also remains significant when all regressors are lagged, when refugee-warrior cases are excluded, and when the largest dyads, return episodes, origins, asylum countries, or regions are removed in turn. Alternative measures of the dependent variable, including discrete return episodes and a three-year moving average, produce the same substantive conclusion. Evidence for the preference component is somewhat more sensitive: the post-war increase in return survives alternative definitions of conflict termination but weakens when major return episodes are excluded and in the post-2009 sample.

I also consider two alternative explanations. First, social and ethnic ties may determine which refugee groups return. In the Appendix, I show that shared language, transnational kinship, and refugees' political status at home predict some cross-sectional variation in return, but including these factors leaves the severe forced-return result essentially unchanged. Second, one might argue that return during war is not especially puzzling if refugees are returning to relatively safe areas within a country that remains at war. Ideally, wartime return does occur to safer areas, but this does not resolve the timing puzzle. Such areas often existed before large-scale return began, yet refugees had left and generally remained abroad. Their existence therefore cannot explain why return occurs at a particular moment. Moreover, the refugee regime treats ongoing conflict nationwide as grounds for continued protection abroad, so large-scale return before conflict ends still departs from the conventional expectation that refugees remain in asylum until conditions at home have sufficiently improved.

7 Case Study: Ethiopian Refugees in Somalia and Sudan

Figure 7: Returns of Ethiopian/Eritrean refugees from Somalia and Sudan (1980–2005)



Notes: The 1993 drop in the Ethiopia/Eritrea → Sudan panel reflects the split of the dyad at Eritrean independence, when roughly 200,000 refugees were reassigned. In the Ethiopia → Somalia panel, civil war broke out in 1991 but UCDP codes the conflict as severe from 1988. *Sources:* GRR; WRS archives.

The statistical analysis above identifies which factors predict return within dyads and presents descriptive patterns to understand global trends across all protracted refugee situations. To trace the theory's mechanisms in a specific context, I examine two dyads from the same origin country that together illustrate the core mechanisms. Ethiopian displacement produced two major refugee populations during the Cold War: ethnic

Somalis who fled the Ogaden War into Somalia, and Tigrayans and Eritreans who fled the civil war into Sudan. Figure 7 plots refugee stocks and returns for both dyads and overlays UNHCR data on the GRR estimates, demonstrating the substantial data discrepancies described for this case in Section 4.

Ethiopia → Somalia: Push-Driven Return

An estimated 700,000 Ethiopians were displaced into Somalia by the Ogaden War in 1977–78. In the early 1980s, the end of active fighting in the Ogaden, combined with a UNHCR repatriation program providing cash grants and six months of food rations, generated substantial return among refugees. The transition costs for these refugees were comparatively low: most were living in camps in poor conditions rather than integrated into Somali communities, and had been in asylum for only a few years.¹⁷ Yet more than half chose to stay. As conditions in the camps improved and a local settlement program allowed refugees to cultivate land, expected quality of life in Somalia came to exceed what many anticipated at home, even after insecurity in the origin country had fallen.¹⁸

By 1990–1991, a new wave of Ethiopian refugees, largely Oromos displaced by forced villagization, had joined the remaining Ogaden refugees. Ethiopia remained insecure. Yet mass return occurred, effectively ending the protracted situation. The driver was not improving conditions at home but the collapse of conditions in asylum:

“Beginning in December 1990, as fighting in the Somali civil war intensified in areas where the refugees lived, UNHCR personnel were evacuated from Somalia, and assistance programs for the refugees stopped. Large numbers of Ethiopian refugees began fleeing Somalia and returning to Ethiopia... Only some 35,000 Ethiopians who were trapped at Qoryoley refugee camp or who fled from various camps to Mogadishu expecting to find help there stayed in Somalia.” (*WRS, Somalia, 1991–1992*)

This episode illustrates the stay constraint (H2): when insecurity in asylum rises sufficiently, refugees return even when the origin country remains insecure. It corresponds to the constrained-return case in Table 1, in which continued asylum becomes infeasible and return can occur without an improvement in conditions at home.

Ethiopia/Eritrea → Sudan: Non-Return Despite Peace

The second dyad tells a contrasting story. An estimated 700,000 Tigrayans and Eritreans fled Ethiopia’s civil war into Sudan beginning in the 1960s. This case illustrates

¹⁷Initial camp conditions were severe: “Problems of water supply, firewood stripped from an almost bare landscape, disease and starvation face refugees and relief officials daily” (*WRS, Somalia, 1981*).

¹⁸By the mid-1980s, conditions had improved to the point where local Somalis obtained food in the camps, “where conditions are often better than in the surrounding desert” (*WRS, Somalia, 1985*).

that the end of conflict is not sufficient to prompt return when transition costs are high and quality of life in asylum exceeds that at home, reflecting both the preference component (H1) and the mobility constraint (H3).

In May 1991, the Mengistu regime fell and the EPLF took control of Eritrea, ending the conflict. Under conventional accounts, this should have triggered large-scale repatriation. Instead, only about 10,000 returned in the year of independence. A dispute between the Eritrean government and UNHCR over repatriation funding prevented the launch of a return program; Eritrea demanded more rehabilitation assistance than donors were willing to provide. Without adequate support to reduce transition costs, the mobility constraint remained binding for many refugees even though the origin country was at peace.

A repatriation program began in 1994 and generated some return, approximately 81,000 over two years, but roughly 320,000 remained. The stalled return was attributed to “financial disputes over the scope of the repatriation program, and deteriorating relations between the governments of Sudan and Eritrea” (WRS, Sudan, 1997), alongside insecurity at the border. When a second program launched around 2000, providing transportation, reintegration grants, and housing, return accelerated: approximately 130,000 repatriated over five years, consistent with H3’s prediction that reducing transition costs increases return. Yet even then, approximately 260,000 refugees chose to remain. In 2002, UNHCR invoked the cessation clause, revoking Eritreans’ refugee status and requiring them to “repatriate, file individualized asylum claims to remain in Sudan as refugees, or take steps to become permanent legal residents of Sudan” (WRS, Sudan, 2003). Only 25,000 repatriated in 2002 and 9,500 in 2003; the vast majority stayed.

The archival evidence points to quality-of-life considerations as the primary explanation. On the origin side, a survey found that roughly 85 percent of refugee households indicated that assistance levels in Eritrea would be a major determinant of their return decision (WRS, Sudan, 1997). On the asylum side, refugees had built substantial lives over decades. By the early 1990s, half lived in camps with access to food aid, health and education programs, and technical training, often among relatives or co-ethnics and renting agricultural land, while the other half lived in urban areas.¹⁹ When encouraged to return in 2001, for example, many chose to remain longer to harvest their crops.²⁰ For this population, the quality of life built over decades of exile exceeded

¹⁹ “Eritrean refugees who had lived for decades in settlements in the east among relatives or members of similar ethnic groups were able to farm and graze their livestock” (WRS, Sudan, 2005). Camp residents “received food aid, health and education programs, literacy and technical training, and special aid for women and children from UNHCR and other assistance groups. About 12,000 children attended 30 refugee schools in the camps” (WRS, Sudan, 1999).

²⁰ “Many other would-be returnees chose to remain longer in Sudan to harvest their crops” (WRS, Sudan, 2002). This is supported by the fact that the largest annual returns of this population was not after conflict but during it: in 1985, 78,000 refugees returned following the resumption of rains in

what they expected in Eritrea, and they opted to stay even after conflict had ended and transition costs had been reduced, consistent with the preference component (H1).

8 Conclusion

This paper revisits a foundational premise of the international refugee regime: that refugees receive protection while conditions at home remain unsafe and return once those conditions improve. Using the Global Refugee Return dataset, which covers 293 origin–asylum dyads from 1980 to 2023, I show that realized return often follows a different pattern. Most recorded returns occur while conflict is still ongoing in the country of origin, while many refugees remain abroad long after war has ended.

I argue that these patterns reflect refugee return as a constrained choice with three components: whether refugees want to return, whether they can remain in asylum, and whether they have the resources to move. The results provide evidence for each. Return is substantially higher after war endings, but this post-war association is much weaker when refugees have better alternatives in asylum. Severe forced-return policies are associated with nearly three times the rate of return observed in years without such policies, including while conflict continues in the country of origin. The evidence for conflict in the asylum country is more qualified. Finally, repatriation assistance is positively associated with realized return, consistent with the argument that refugees who would prefer to return may nevertheless lack the resources required to do so. In the model-based global decomposition, setting severe forced-return policy to zero reduces predicted return by 24.7%, compared with 14.3% for the post-war window, reflecting in large part the much larger refugee populations exposed to severe coercion.

These findings challenge the common interpretation of return as evidence that conditions at home have improved sufficiently for refugees to go back. Realized return is an imperfect indicator of either safety or preference: refugees may return under constraint, just as continued displacement may persist after the conditions that initially caused flight have changed.

This distinction also matters for how repatriation is understood within the refugee regime. Voluntary return after peace can provide a durable solution when refugees want to return and can do so safely, but observed return is not necessarily evidence that these conditions have been met. Some of the largest return movements instead occur under coercive pressure and while conflict continues at home. This raises questions about the welfare and reintegration of refugees who return under such conditions (Fransen, Ruiz and Vargas-Silva, 2017), as well as the consequences of mass return for origin communities and post-conflict politics (Schwartz, 2019; Blair, Van Dijke and Wright, 2026). The conditions driving return may therefore shape both returnees’

Ethiopia (WRS, Sudan, 1986), despite ongoing conflict.

prospects and the consequences of repatriation after they arrive home, and warrant further investigation.

The findings also speak to broader questions about international cooperation, law, and institutions. *Non-refoulement* is the core legal principle of the refugee regime, yet states use policies that induce refugees to return to countries where conflict is ongoing. UNHCR sometimes facilitates repatriation under these conditions because refusing to participate may leave refugees to return without assistance or jeopardize the organization's access to refugee populations. This illustrates the constraints international organizations face when implementation depends on state cooperation.

Finally, the analysis underscores the difficulty of measuring refugee return. Existing UNHCR statistics omit many forced and spontaneous returns, including precisely the coercive movements central to the argument here. GRR provides corrected measures of refugee stocks and realized return, together with new data on forced-return policies, encampment, and UNHCR financing, but it too remains an imperfect measure of movements that are often undocumented and politically contested. Future research should continue to triangulate administrative records and archival sources with new forms of measurement, including satellite-based measures of population movement and border-crossing data.

Several open questions follow from the analysis. First, this paper focuses on refugees who have already entered an asylum country and subsequently return. It therefore does not capture related forms of coercion that occur before or at the point of entry, including maritime pushbacks, border closures, and other policies designed to prevent asylum seekers from reaching protection in the first place (FitzGerald, 2019; Goodwin-Gill, 2011; Schwartz, 2025). Second, the analysis treats return as movement from asylum back to the country of origin, but refugee mobility is often more complex. Refugees may move repeatedly between origin and asylum countries, return temporarily, or leave again after an unsuccessful return, patterns that fall outside a simple refugee–returnee binary (Masterson and Vidarte, 2026). Better data on these movements would help establish when return represents a durable end to displacement and when it is one stage in a longer process of repeated mobility.

Taken together, these findings change how we should think about both refugee return and the refugee regime more broadly. Repatriation is not simply the endpoint of displacement that follows the restoration of peace. It may instead reflect the erosion of protection in asylum, while many refugees who are no longer fleeing active war remain abroad for years or decades, either because they prefer to remain or because returning is difficult. This creates a broader puzzle for the study of international cooperation. Host governments often have the practical ability to push refugees out despite their international legal obligations, and accepting refugees may entail a far longer commitment than the regime's temporary-protection logic suggests. Why, then, do so many states continue to provide asylum rather than force refugees to leave? The

findings in this paper document an important failure of the refugee regime, but they also make its persistence more puzzling. Explaining why states continue to sustain this system despite weak enforcement and potentially enduring costs is the broader question that motivates my book project.

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Peace or Push? Understanding Global Patterns of Refugee Return

Mae MacDonald

Supplementary Materials

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1 Global Refugee Return (GRR) Dataset

The Global Refugee Return (GRR) dataset is an original global dataset on refugee stocks, returns, and the conditions shaping repatriation. It covers origin–asylum dyad-years from 1980 to 2023. The dataset makes three core contributions: (1) improved stock and repatriation data that corrects systematic errors in UNHCR figures and incorporates forced returns; (2) new indicators of *de facto* coercive return policies as actually implemented; and (3) UNHCR financing data for asylum-country operations and repatriation programs.

Unit of analysis. Each observation records conditions for refugees from a specific country of origin residing in a specific country of asylum in a given year.

Geographic coverage. The dataset includes all **origin countries** with at least 25,000 refugees in exile in any given year, consistent with UNHCR’s threshold for a protracted refugee situation. For each origin country, the dataset includes **asylum countries** that together account for at least 75% of the origin’s total refugee population, calculated using UNHCR refugee stock data. Dyad-years in which no refugee population is present are excluded, as they are uninformative for the study of repatriation and incompatible with the Poisson offset specification used in the main models.

The dataset contains 7,526 dyad-year observations across 293 dyads (87 origin countries and 93 asylum countries).

Temporal coverage. The dataset spans 1980–2023. The year 1989 is missing for all variables coded from the World Refugee Survey (WRS), because the 1990 WRS report was not published.

Primary sources. GRR draws on four primary sources: (1) the World Refugee Survey (WRS), 1980–2008, providing annual country reports on refugee stocks and durable solutions; (2) the U.S. Department of State Country Reports on Human Rights Practices (USHR), 2009–2023, with data extracted from Section 2.d on freedom of movement; (3) UNHCR Executive Committee (ExCom) reports and archival documents, 1980–1997; and (4) UNHCR statistical yearbooks and reports, 1998–2023. Variables derived from these sources were hand-coded by the author and research assistants, with each case coded by one assistant and verified by the author. Additional variables are drawn from the UCDP Armed Conflict Dataset, the Ethnic Power Relations (EPR) Dataset, DWRAP (Blair et al., 2022), the World Bank World Development Indicators, and the CEPII GeoDist dataset.

1.1 Codebook

Tables 1–5 describe all variables used in the main analysis. Variables with the prefix **a_** refer to asylum-country characteristics, **o_** to origin-country characteristics, and **d_** to dyad-level variables. Variable suffixes follow a consistent convention: **_z** denotes a standardized (z-scored) variable, **_l** a log-transformed variable, and **_zl** a standardized log-transformed variable. In the regression models, several variables are transformed for estimation. Binary high/low indicators (e.g., **a_conflict_high**, **a_conflict_low**) are derived from the ordinal variables by splitting into separate dummies for the high and low categories, with zero as the reference. Continuous variables used in standardized or logged form in the regressions are noted below; standardization is performed on the estimation sample (dyad-years with more than 1,000 refugees in the previous year). The dependent variable is **trepat** (count of returnees), and **refugees** enters as the offset, lagged one year.

Table 1: Codebook: Identifiers

Variable	Description
id	Unique dyad identifier, formatted as COO–COA.
year	Year of observation.
coo	Country of origin name.
coo_iso	Country of origin ISO 3166-1 alpha-3 code.
coa	Country of asylum name.
coa_iso	Country of asylum ISO 3166-1 alpha-3 code.

Table 2: Codebook: Stock and Return Variables

Variable	Description	Source
refugees	Stock of forcibly displaced persons in the dyad at year-end, including refugees, asylum-seekers, individuals in refugee-like situations, and unregistered refugees. Primary stock variable.	WRS (1980–2008); UNHCR + USHR unregistered (2009–2023)
refugees_formal	Stock of forcibly displaced persons excluding unregistered refugees.	WRS (1980–2008); UNHCR (2009–2023)
trepat	Total refugees repatriated (voluntary + forced).	Derived: vrep + frep
vrep	Voluntary repatriation count.	WRS (1980–2008); UNHCR (2009–2023)
frep	Forced repatriation count.	WRS (1980–2008); USHR (2009–2023)

Table 3: Codebook: Asylum-Country Variables (prefix: **a_**)

Variable	Description	Source
a_conflict	Conflict intensity in the asylum country (UCDP): 0 = none, 1 = low (25–999 battle deaths), 2 = high ($\geq 1,000$). In regressions, split into a_conflict_high (=1 if 2) and a_conflict_low (=1 if 1); no conflict is the reference category.	UCDP Armed Conflict Dataset
a_conflict_bd	Battle deaths in the asylum country (UCDP ‘best’ estimate). Available from 1989.	UCDP Battle-Related Deaths Dataset
a_forced	Coercive forced-return policies in the asylum country at the dyad level: 0 = none; 1 = moderate (threats or encouragement to induce return without policy implementation, or implemented policies targeting <10% of the refugee stock); 2 = severe (implemented policies intended or expected to affect $\geq 10\%$ of the refugee stock). Coding reflects the policy itself, not realized return. In regressions, split into a_forced_high (severe) and a_forced_low (moderate); no forced return is the reference category. See Table 6 for example coding.	Hand-coded from WRS and USHR
a_camp	Ordinal share of the refugee population living in camps: 0 = no camps, 1 = <10% in camps, 2 = 10–50%, 3 = 50–95%, 4 = 95–100%.	Hand-coded from WRS and USHR
a_camp_most	Binary: 1 if most refugees live in camps (a_camp $\in \{3, 4\}$), 0 otherwise.	Derived from a_camp
a_urban_z	Standardized (z-score) measure of settlement type. Underlying ordinal variable: 3 if no camps or <10% in camps (a_camp $\in \{0, 1\}$), 2 if 10–50% in camps (a_camp = 2), 1 if 50–100% in camps (a_camp $\in \{3, 4\}$).	Derived from a_camp
a_funding	UNHCR funding allocated to the asylum country (USD, country-year level).	UNHCR ExCom (1980–2009); UNHCR Yearbooks (2010–2023)
a_funding_pp	UNHCR funding per refugee in the asylum country. In regressions, used in standardized log form (a_funding_pp_zl).	Derived: a_funding / asylum-country refugee stock
a_rights	Refugee rights index (inverse-covariance weighted) from DWRAP. In regressions, standardized (a_rights_z).	Blair et al. (2022)

Table 3: Codebook: Asylum-Country Variables (prefix: **a_**) (*continued*)

Variable	Description	Source
a_gdp_l	Log GDP per capita of the asylum country.	World Bank WDI
a_edu	Mean years of schooling for the 20–24 age cohort in the asylum country (a near-completion measure of recent educational attainment), linearly interpolated from 5-year reporting intervals to annual frequency.	Wittgenstein Centre
a_region	World Bank regional classification of the asylum country.	World Bank

Table 4: Codebook: Origin-Country Variables (prefix: **o_**)

Variable	Description	Source
o_conflict	Conflict intensity in the origin country (UCDP): 0 = none, 1 = low (25–999 battle deaths), 2 = high ($\geq 1,000$). Matched to the specific conflict driving displacement. Missing for non-conflict displacement (e.g., Venezuela). In regressions, split into o_conflict_high and o_conflict_low ; no conflict is the reference category.	UCDP Armed Conflict Dataset
o_conflict_bd	Battle deaths in the origin country (UCDP ‘best’ estimate). Available from 1989. Matched to the specific conflict producing refugees.	UCDP Battle-Related Deaths Dataset
o_conflict_years	Cumulative years of any armed conflict in the origin country. In regressions, used in standardized log form (o_conflict_years_zl), captured as $\log(\text{years} + 0.01)$ and standardized.	Derived from UCDP Armed Conflict Dataset
o_funding_repat_pp	UNHCR repatriation funding allocated to the origin country, divided by the origin’s global refugee population (USD per refugee). In regressions, used in standardized log form (o_funding_repat_pp_zl).	UNHCR ExCom (1980–1999); UNHCR Yearbooks (2011–2023). 2000–2010 missing.
o_eth_name	Name of the refugee group’s ethnic identity (e.g., “Kurds,” “Tutsi,” “Bosniaks/Muslims”).	EPR Refugee Dataset
o_eth_id	EPR group identifier (gwgroupid) for the refugee group.	EPR Refugee Dataset; EPR Dataset
o_eth_status	EPR power-status category of the refugee group in the origin country (categorical: monopoly, dominant, senior partner, junior partner, powerless, discriminated, irrelevant, self-exclusion, state collapse).	EPR Dataset
o_eth_high	Binary: 1 if the refugee group holds high political status (‘monopoly,’ ‘dominant,’ ‘senior partner,’ or ‘junior partner’), 0 otherwise.	EPR Dataset; EPR Refugee Dataset
o_eth_low	Binary: 1 if the refugee group holds low political status (‘discriminated’ or ‘powerless’), 0 otherwise.	EPR Dataset; EPR Refugee Dataset
o_eth_power	Ordinal power score (0–6) derived from o_eth_status : 6 = monopoly, 5 = dominant, 4 = senior partner, 3 = junior partner, 2 = state collapse, 1.5 = discriminated, 1 = powerless, 0.5 = irrelevant, 0 = self-exclusion.	Derived from o_eth_status
o_eth_cat	Three-category collapse of o_eth_status : “In Power” (monopoly, dominant, senior/junior partner), “Excluded” (powerless, discriminated), or “Other” (irrelevant, self-exclusion, state collapse).	Derived from o_eth_status

Table 4: Codebook: Origin-Country Variables (prefix: o_) (*continued*)

Variable	Description	Source
<code>o_eth_inpower</code>	Binary: 1 if the refugee group is “In Power,” 0 otherwise. Equivalent to <code>o_eth_high</code> .	Derived from <code>o_eth_status</code>
<code>o_eth_excluded</code>	Binary: 1 if the refugee group is “Excluded,” 0 if “In Power,” NA for “Other” categories. Differs from <code>o_eth_low</code> in coding “Other” as NA rather than 0.	Derived from <code>o_eth_status</code>
<code>o_eth_claim</code>	Binary indicator for whether the rebel group in the conflict producing refugees has made an exclusive claim to fight on behalf of an ethnic group (ACD2EPR).	ACD2EPR (Wucherpfennig et al.)
<code>o_civlib_z</code>	Freedom House Civil Liberties index for the origin country, standardized.	Freedom House
<code>o_terror_z</code>	Political Terror Scale for the origin country, standardized.	Political Terror Scale (Amnesty International, Human Rights Watch, U.S. State Department)
<code>o_genocide</code>	Binary indicator for genocide or politicide in the origin country.	PITF Genocide/Politicide Dataset
<code>o_pitf_any</code>	Binary indicator for any PITF political violence in the origin country (ethnic war or revolutionary war).	PITF Ethnic/Revolutionary War Datasets
<code>o_pagree</code>	Binary indicator for a peace agreement signed in the origin country in a given year.	UCDP Peace Agreement Dataset
<code>o_gdp_l</code>	Log GDP per capita of the origin country.	World Bank WDI
<code>o_edu</code>	Mean years of schooling for the 20–24 age cohort in the origin country (a near-completion measure of recent educational attainment), linearly interpolated from 5-year reporting intervals to annual frequency.	Wittgenstein Centre
<code>o_region</code>	World Bank regional classification of the origin country.	World Bank

Table 5: Codebook: Dyad-Level Variables (prefix: d_)

Variable	Description	Source
<code>d_distance</code>	Distance between origin and asylum countries (km). In regressions, used in log form (<code>d_distance_l</code>).	CEPII GeoDist
<code>d_contig</code>	Binary: 1 if the countries share a border, 0 otherwise.	CEPII GeoDist

1.2 Forced Return Measurement

The *Forced Return* variable is hand-coded from the qualitative country narratives in the *World Refugee Survey*. Non-zero dyad-years are assigned to one of two categories—*moderate* or *severe*—using two criteria: (i) whether forced return is implemented or only threatened, and (ii) the share of the refugee stock the policy is intended or expected to affect.

A dyad-year is coded **severe** when the asylum government implements a forced return policy intended or expected to affect at least 10% of the refugee stock. Severe policies include large-scale arrests and deportations, non-renewal of residence cards or refugee status, closures of camp schools and marketplaces, and cuts to food aid expressly intended to induce return. A dyad-year is coded **moderate** when the asylum government either (a) threatens or encourages forced return without policy implementation (e.g., political rhetoric, parliamentary debate, public ultimatums), or (b) implements a forced return policy targeting fewer than 10% of the refugee stock. The 10% threshold separates broad coercive policies—which we expect to

Table 6: Examples of Forced Return Policies

Moderate forced return policies	Severe forced return policies
“In a move protested by UNHCR, authorities in Congo-Brazzaville forced 19 asylum seekers back to Congo-Kinshasa in April.” (COD-COG, 2002) “The Tibetan refugee leadership encourages most of those on pilgrimages and many who come to study to return to Tibet, both to counteract the overwhelming Chinese presence in Tibet and because the two dozen Tibetan refugee settlements in India cannot absorb so many people.” (CHN-IND, 1999) “There were scattered reports that the government intended to expel Palestinians who had not arrived in 1948, but who had come to Lebanon from other Arab countries, such as Egypt and Syria. If implemented, this could affect up to 50,000.” (PAL-LEB, 1993) “UNHCR expressed concern that some refugees were being manipulated to repatriate prematurely to Congo-Kinshasa based on misinformation about conditions there. Some refugees reportedly repatriated and had to flee again to Rwanda because conditions in Congo-Kinshasa remained dangerous.” (COD-RWA, 2000) “In western Sudan, 45,000 Chadian refugees remain in UNHCR-assisted reception centers despite government and UNHCR attempts to persuade them to return to Chad.”	“South Africa enacted new legislation in 1991, the Aliens Control Act, giving immigration authorities even more power to seek out and deport Mozambicans... South Africa has forcibly repatriated thousands of Mozambican refugees to a land of deprivation and death in recent years. This practice violates international standards and must cease,” USCR stated. (MOZ-ZAF, WRS 1993) “In April, the Government announced that it would expel all Rwandans whose cases it rejected and the number of applicants declined substantially... Deterred by the low acceptance rates, some 2,000 never registered with the Government and more than 10,500 of those who registered dropped their claims and returned.” (RWA-BDI, WRS 2006) “The events in Zaire produced strains in Congo that resulted in the Congolese authorities forcibly deporting some 10,000 Zaireans in November.” (COD-COG, WRS 1992) “Yemeni authorities forcibly returned 438 Somalis, at least 200 of whom were UNHCR-recognized refugees. They had been part of a group of 1,200 persons who had been rounded up in Aden as part of a sweep of the city for illegal immigrants.” (SOM-YEM, 1996) “UNHCR protested when the government’s roundup of 1,000 undocumented immigrants in January resulted in the detention of several refugees who possessed UNHCR permits. Persons judged to be undocumented were deported without UNHCR screening to determine claims to refugee status.” (COD-ZMB, 1993)

materially shift return rates—from narrower actions targeting specific individuals or groups, which may stoke fear but are less likely to drive aggregate return.

Coding depends only on the policy itself, not on the return outcome. A policy that cuts food aid across all camps to induce return is coded as severe regardless of whether refugees subsequently repatriate. This rules out the most direct form of measurement endogeneity: the variable captures the policy environment refugees face rather than the return rate it is intended to predict.

Table 6 provides illustrative examples of moderate and severe coding decisions. The GRR codebook records a verbatim source quote for every non-zero observation, allowing external verification of all coding decisions.

1.3 Summary Statistics

Table 7 reports summary statistics for all variables included in the main specification. Statistics are computed on the estimation sample (dyad-years with more than 1,000 refugees in the previous year). Because repatriation funding data are available for a subset of years, statistics for the repatriation funding variable are computed on the smaller sample in which it is observed (matching column 2 of Table 2 of the paper, the only column in which it enters).

Figure 1 shows the distribution of the dependent variable. The left panel displays the full range of total repatriation counts with the y-axis capped to show detail among non-zero values. The right panel plots positive values on a log scale, showing a roughly log-normal distribution among dyad-years with nonzero repatriation. The heavy concentration of zeros, extreme right skew, and count nature of the dependent variable motivate the use of Poisson pseudo-maximum likelihood estimation with the log stock at risk as the offset.

1.4 Dyadic Return Patterns

Figures 2 and 3 plot refugee stocks and total returns over time for the 40 largest dyads in the dataset (top 20 origin countries by total displaced population, with the two largest asylum destinations for each).

Table 7: Summary Statistics (filtered sample: dyad-years with lagged refugee stock > 1,000)

Variable	Obs	Mean	Std. Dev.	Min	Max
Return and stock					
Total Repatriation	5109	6,513.724	48,844.540	0.000	1,700,000.000
Refugee Stock (Lagged)	5131	129,495.828	400,251.604	1,002.000	6,013,249.000
Return Rate (% of stock at risk)	5109	2.993	11.027	0.000	100.000
Stay constraint (H2)					
Forced Return: Severe	4415	0.034	0.181	0.000	1.000
Forced Return: Moderate	4415	0.066	0.248	0.000	1.000
Conflict in COA: High	5131	0.070	0.256	0.000	1.000
Conflict in COA: Low	5131	0.175	0.380	0.000	1.000
Preferences (H1)					
War Ending	5131	0.041	0.198	0.000	1.000
Conflict in COO: High	5131	0.119	0.324	0.000	1.000
Conflict in COO: Low	5131	0.224	0.417	0.000	1.000
GDP (COA)	5095	-0.000	1.000	-2.233	1.451
GDP (COO)	4845	-0.000	1.000	-2.983	2.225
Civil Liberties Restrictions (COO)	5086	0.000	1.000	-2.817	1.435
Mobility constraint (H3)					
Repatriation Assistance	3015	0.000	1.000	-0.547	2.395
Controls					
Refugee Rights	4911	-0.000	1.000	-2.162	3.794
Conflict Duration	5131	0.000	1.000	-0.783	1.834
Additional measures, used in the robustness columns only					
Education (COA)	5066	0.000	1.000	-2.474	1.396
Education (COO)	4938	-0.000	1.000	-2.330	1.829
Education Gap (COO – COA)	4873	0.000	1.000	-4.319	2.985
Urban	4477	-0.000	1.000	-1.566	0.761
Majority in Camps	4478	0.256	0.436	0.000	1.000

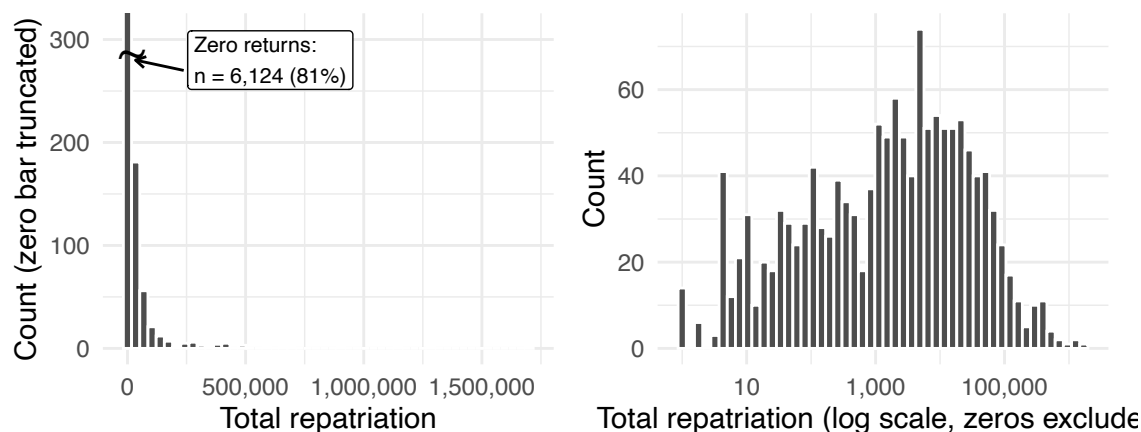


Figure 1: Distribution of the dependent variable (total repatriation). Left panel shows the raw distribution; right panel uses a log scale and excludes zeros.

8

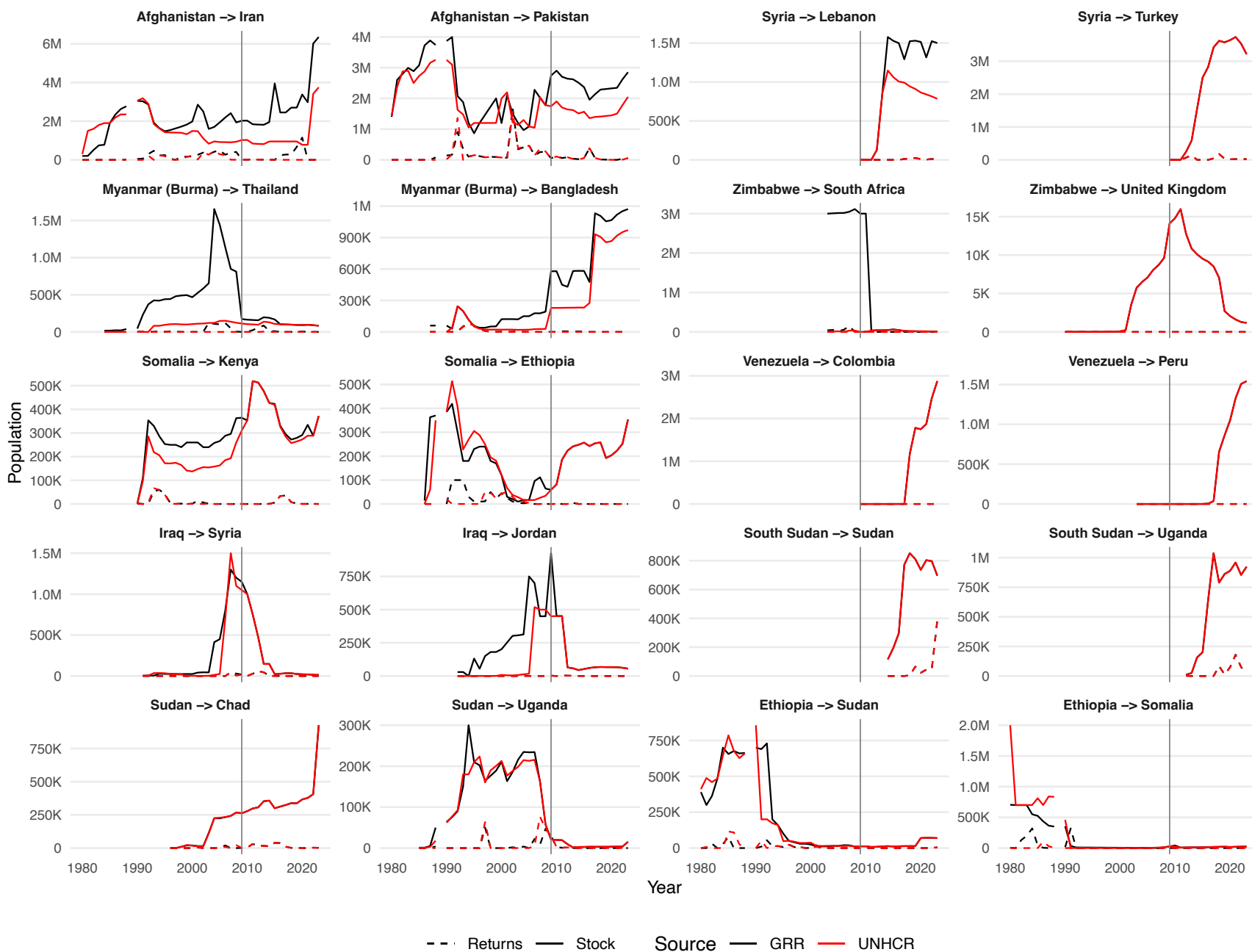


Figure 2: Refugee Stocks and Returns by Dyad (1 of 2)

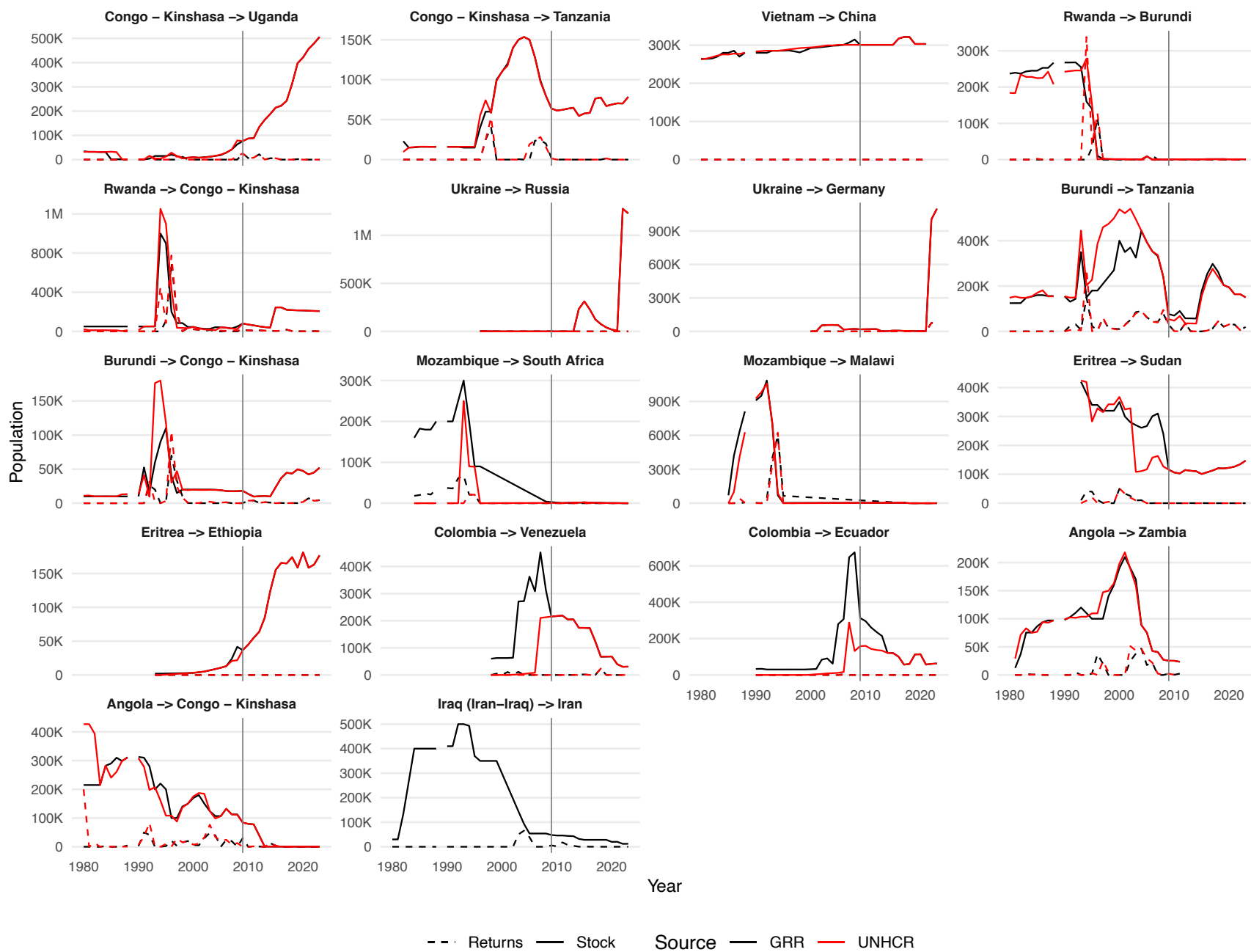


Figure 3: Refugee Stocks and Returns by Dyad (2 of 2)

1.5 Cases

The analysis pools 293 dyads. The tables below provide information on specific cases: the dyads with the most return, the highest return rates, the largest single-year movements, and the largest populations that never returned. They are descriptive and use the full GRR panel rather than the estimation sample, so dyads dropped by the fixed effects still appear.

Return is extraordinarily concentrated. The two Afghan dyads alone account for more recorded return than the next eight combined. 18 of the twenty record a push factor at some point and 9 record a severe forced-return policy specifically. Most of the largest episodes occur while the origin is still at war: 17 of the twenty largest movements of refugees home take place in a year the origin records conflict, and 11 occur alongside a forced-return policy in the host.

177 of the 293 dyads never return 1 per cent of their peak stock, and the twenty largest of them hold 19 per cent of all refugee-years in the panel. In some of these cases return was not expected. The three Palestinian dyads record conflict in the origin in more than eighty per cent of their observed years; Venezuelans were displaced by economic collapse rather than by a war that could end; Tibetans and Bhutanese were expelled by states that continue to exclude them. But 11 of the twenty have origins at peace for at least three-quarters of their observed history, and several — Western Saharans in Algeria, Eritreans in Ethiopia, Guatemalans in the United States — follow wars that ended decades ago.

Table 8: The twenty dyads with the most recorded return, 1980–2023. Return over peak stock can exceed 100 per cent where a population arrived and left more than once over the period.

Dyad	Origin → asylum	Years	Total return	Peak stock	Return/peak (%)	% yrs COO conflict	Push ever
AFG-IRN	Afghanistan → Iran	1980–2023	7,507,327	6,352,317	118	100	Both
AFG-PAK	Afghanistan → Pakistan	1980–2023	7,018,397	3,997,133	176	100	Both
IRQ-IRN-KUR	Iraq (Kurds) → Iran	1980–2023	1,426,658	160,000	892	33	Both
RWA-COD	Rwanda → Congo - Kinshasa	1980–2023	1,190,071	1,000,000	119	49	Both
MOZ-MWI	Mozambique → Malawi	1985–2023	1,126,986	1,085,000	104	44	None
BDI-TZA	Burundi → Tanzania	1980–2023	1,049,147	443,700	236	53	Both
ETH-SOM	Ethiopia → Somalia	1980–2023	1,039,754	705,000	147	52	COA conflict
MMR-THA	Myanmar (Burma) → Thailand	1984–2023	735,024	1,653,500	44	72	Both
KHM-THA	Cambodia → Thailand	1981–2023	631,632	365,000	173	45	Both
SDN-SSD-UGA	Sudan → Uganda	1985–2023	623,004	1,037,898	60	97	COA conflict
SSD-SDN	South Sudan → Sudan	2014–2023	572,695	852,080	67	80	COA conflict
SYR-TUR	Syria → Turkey	2009–2023	559,246	3,737,369	15	87	COA conflict
SOM-ETH	Somalia → Ethiopia	1986–2023	532,300	419,000	127	81	COA conflict
UKR-POL	Ukraine → Poland	2002–2023	468,475	958,519	49	45	None
ETH-ERI-SDN	Ethiopia → Sudan	1980–2023	433,357	730,000	59	26	COA conflict
LBR-GIN	Liberia → Guinea	1988–2023	404,223	420,000	96	14	COA conflict
AGO-COD	Angola → Congo - Kinshasa	1980–2023	383,166	312,700	123	40	Both
UGA-SDN	Uganda → Sudan	1980–2010	382,250	250,000	153	95	COA conflict
MOZ-ZAF	Mozambique → South Africa	1984–2023	359,700	300,000	120	50	Both
CIV-LBR	Côte d'Ivoire → Liberia	2002–2023	328,376	128,067	256	18	COA conflict

Table 9: The twenty highest cumulative return rates, among dyads whose peak stock reaches 25,000. Peak stock is the denominator rather than mean stock because it is the largest population ever plausibly at risk of return and is not deflated by the length of the observation window; the cost is that a dyad which emptied and refilled exceeds 100 per cent.

Dyad	Origin → asylum	Years	Peak stock	Total return	Return/peak (%)	% yrs COO conflict	Push ever
IRQ-IRN-KUR	Iraq (Kurds) → Iran	1980–2023	160,000	1,426,658	892	33	Both
CIV-LBR	Côte d'Ivoire → Liberia	2002–2023	128,067	328,376	256	18	COA conflict
UGA-COD	Uganda → Congo - Kinshasa	1980–2023	90,000	223,191	248	60	COA conflict
BDI-TZA	Burundi → Tanzania	1980–2023	443,700	1,049,147	236	53	Both
COG-COD	Congo - Brazzaville → Congo - Kinshasa	1997–2023	40,000	80,548	201	18	COA conflict
TLS-IDN	Timor-Leste → Indonesia	1999–2012	120,000	220,230	184	12	Both
AFG-PAK	Afghanistan → Pakistan	1980–2023	3,997,133	7,018,397	176	100	Both
BDI-COD	Burundi → Congo - Kinshasa	1980–2023	110,000	191,308	174	53	COA conflict
MLI-BFA	Mali → Burkina Faso	1993–2023	38,776	67,329	174	65	COA conflict
KHM-THA	Cambodia → Thailand	1981–2023	365,000	631,632	173	45	Both
UGA-SDN	Uganda → Sudan	1980–2010	250,000	382,250	153	95	COA conflict
ETH-SOM	Ethiopia → Somalia	1980–2023	705,000	1,039,754	147	52	COA conflict
SOM-ETH	Somalia → Ethiopia	1986–2023	419,000	532,300	127	81	COA conflict
TCD-CMR	Chad → Cameroon	1980–2023	110,000	137,891	125	70	COA conflict
AGO-COD	Angola → Congo - Kinshasa	1980–2023	312,700	383,166	123	40	Both
MOZ-ZAF	Mozambique → South Africa	1984–2023	300,000	359,700	120	50	Both
RWA-COD	Rwanda → Congo - Kinshasa	1980–2023	1,000,000	1,190,071	119	49	Both
AFG-IRN	Afghanistan → Iran	1980–2023	6,352,317	7,507,327	118	100	Both
COD-COG	Congo - Kinshasa → Congo - Brazzaville	1985–2023	133,949	156,666	117	58	Both
SLE-LBR	Sierra Leone → Liberia	1992–2021	120,000	131,805	110	38	COA conflict

Table 10: The twenty largest single-year return episodes. † marks dyad-years in which recorded return exceeds the lagged stock, where the denominator is the year-end stock plus return rather than the lagged stock, so the rate is an upper bound.

Dyad	Year	Origin → asylum	Return	Stock at risk	Rate (%)	COO conflict	COA conflict	Forced return
AFG-PAK	2002	Afghanistan → Pakistan	1,700,000	2,100,000	81.0	Low	Low	Moderate
IRQ-IRN-KUR	1991	Iraq (Kurds) → Iran	1,295,000	1,366,400 [†]	94.8	Low	Low	None
AFG-IRN	2020	Afghanistan → Iran	1,151,789	2,701,142	42.6	High	Low	Severe
AFG-PAK	1992	Afghanistan → Pakistan	910,000	3,997,133	22.8	High	Low	None
RWA-COD	1996	Rwanda → Congo - Kinshasa	700,000	900,000	77.8	Low	High	None
AFG-IRN	2019	Afghanistan → Iran	668,209	2,701,142	24.7	High	Low	Severe
MOZ-MWI	1994	Mozambique → Malawi	600,000	700,000	85.7	None	None	None
AFG-IRN	2005	Afghanistan → Iran	489,600	1,702,800	28.8	High	Low	Severe
AFG-PAK	2005	Afghanistan → Pakistan	475,100	967,200	49.1	High	None	Severe
AFG-IRN	1993	Afghanistan → Iran	471,500	2,862,500	16.5	High	Low	Severe
AFG-IRN	2004	Afghanistan → Iran	419,500	1,600,000	26.2	Low	None	Severe
AFG-PAK	2004	Afghanistan → Pakistan	409,100	1,200,000	34.1	Low	Low	None
AFG-IRN	2008	Afghanistan → Iran	409,000	2,414,700	16.9	High	Low	Severe
UKR-POL	2022	Ukraine → Poland	403,000	1,361,519 [†]	29.6	High	None	None
MOZ-MWI	1993	Mozambique → Malawi	400,000	1,085,000	36.9	None	None	None
SSD-SDN	2023	South Sudan → Sudan	386,793	796,831	48.5	None	High	Moderate
AFG-PAK	2016	Afghanistan → Pakistan	381,275	2,366,950	16.1	High	Low	None
AFG-IRN	2007	Afghanistan → Iran	379,650	2,190,400	17.3	High	Low	Severe
AFG-IRN	2002	Afghanistan → Iran	376,300	2,855,000	13.2	Low	None	Severe
AFG-PAK	1993	Afghanistan → Pakistan	358,800	2,075,000	17.3	High	None	None

Table 11: The non-return cases: the twenty largest dyads whose cumulative return never reaches 1 per cent of peak stock. Ranked by refugee-years, the sum of the annual stock over the dyad’s observed history, which measures how much displacement the dyad represents rather than how long it is observed. The three Palestinian dyads each record the same 233 returns in 1986, which is one reported figure carried across all three rather than three independent observations.

Dyad	Origin → asylum	Years	Refugee-years (m)	Peak stock	Total return	% yrs COO conflict	Push ever
PSE-JOR	Palestinian Territories → Jordan	1981–2008	44.62	2,439,718	233	89	None
VNM-CHN	Vietnam → China	1980–2021	11.97	321,502	0	15	COA conflict
VEN-USA	Venezuela → Colombia	2009–2023	11.93	2,875,743	0	0	COA conflict
PSE-SYR	Palestinian Territories → Syria	1980–2016	10.62	592,200	233	83	COA conflict
PSE-LBN	Palestinian Territories → Lebanon	1981–2023	9.24	500,000	233	81	COA conflict
VEN-PER	Venezuela → Peru	2003–2023	6.98	1,542,004	0	0	COA conflict
ESH-DZA	Western Sahara → Algeria	1980–2023	5.89	165,000	0	21	COA conflict
COL-ECU	Colombia → Ecuador	1990–2023	4.71	674,500	5,102	85	COA conflict
CHN-IND	China → India	1984–2023	4.01	123,000	0	0	COA conflict
UKR-RUS	Ukraine → Russia	1996–2023	3.56	1,275,925	0	42	COA conflict
PHL-MYS	Philippines → Malaysia	1981–2023	2.70	136,075	0	91	COA conflict
BTN-NPL	Bhutan → Nepal	1991–2023	2.33	133,700	0	0	Both
RUS-USA	Russia → United States	1988–2023	2.22	245,238	0	0	None
ERI-ETH	Eritrea → Ethiopia	1993–2023	2.03	181,271	0	8	COA conflict
TUR-DEU	Turkey → Germany	2000–2023	1.98	181,344	0	96	None
IRN-TUR	Iran → Turkey	1982–2023	1.98	257,000	2,208	53	Both
CHN-USA	China → United States	1988–2023	1.96	173,873	0	0	Moderate only
VEN-BRA	Venezuela → Brazil	2011–2023	1.93	510,499	0	0	None
GTM-USA	Guatemala → United States	1988–2023	1.76	192,596	0	20	Moderate only
ARM-AZE	Armenia → Azerbaijan	1992–2013	1.56	229,000	0	38	COA conflict

2 Data Validation

This section addresses concerns about the Global Refugee Return dataset: whether GRR agrees with the UNHCR data on refugee stocks and returns, whether the forced-return measure is coded independently of the returns it is meant to predict, whether the results depend on the annual-count form of the dependent variable, and whether the 2009 change of source introduces a break in measurement.

2.1 GRR and UNHCR Compared

In this section, I compare the GRR dataset of stocks (including unregistered refugees) and returns (including forced returns) with UNHCR’s data.

2.1.1 Summary Statistics and Agreement

Table 12: The six series, over the full GRR panel of 7,526 dyad-years. GRR’s stock includes unregistered populations; UNHCR’s counts registered refugees only.

Series	Non-missing	Coverage	Positive	Total (m)	Median if positive
Refugee stock (GRR)	7,526	100.0%	7,526	700.2	5,752
Refugee stock (UNHCR)	7,305	97.1%	7,139	487.4	4,489
Total return (GRR)	7,498	99.6%	1,374	34.2	2,130
Voluntary return (GRR)	7,475	99.3%	1,217	27.6	2,307
Forced return (GRR)	7,296	96.9%	297	6.6	550
Return (UNHCR)	7,305	97.1%	1,339	25.7	1,219

Table 13: Agreement between the two sources, separately for stocks and for returns.

Comparison	Dyad-years	Correlation	Exact match	Within 10 per cent	One positive, other zero
Stock: GRR vs UNHCR	7,139	0.958	77.4%	85.6%	–
Return: GRR vs UNHCR	1,571	0.425	39.3%	–	31.9%

The two sources largely agree about how many refugees there are and disagree about how many returned. Logged stocks correlate at 0.958 and match exactly in 77% of dyad-years; logged returns correlate at 0.425, and in 32% of the dyad-years where either records a return, one records movement and the other records none.

GRR’s stock exceeds UNHCR’s, since it counts unregistered populations that UNHCR’s registration figures do not. On returns, GRR records 34.2 million returns against UNHCR’s 25.7 million, so UNHCR captures 75% of the total GRR records.

The table below shows examples of return episodes not recorded by UNHCR. The Afghan rows show UNHCR undercounting forced returns, and only counting voluntary returns. The Iraqi Kurdish return of 1991 and the Ethiopian returns from Somalia in 1984 and 1991 UNHCR data records nothing at all, because its personnel were not there to record it. Between them these twelve dyad-years account for 18% of all return GRR records.

Table 14: Agreement between the GRR stock and UNHCR’s, by source era. The final column is the median amount by which GRR exceeds UNHCR where it does.

Era	Dyad-years	GRR = UNHCR	GRR exceeds UNHCR	Median excess
WRS era 1980-2008	3,244	55.9%	24.4%	31%
USHR era 2009-2023	3,895	95.3%	4.7%	76%

Table 15: Largest return episodes recorded by GRR and absent from UNHCR.

Dyad	Year	GRR return	UNHCR return	Unrecorded	Of which forced
IRQ-IRN	1991	1,295,000	0	1,295,000	0
AFG-IRN	2020	1,151,789	947	1,150,842	1,150,842
AFG-IRN	2019	668,209	1,952	666,257	666,257
AFG-IRN	1993	471,500	0	471,500	—
AFG-IRN	2008	409,000	3,656	405,344	406,000
AFG-IRN	2007	379,650	7,462	372,188	363,000
AFG-PAK	1993	358,800	0	358,800	0
ETH-SOM	1984	320,554	0	320,554	0
ETH-SOM	1991	320,000	0	320,000	0
AFG-IRN	2017	274,291	1,202	273,089	273,089
AFG-IRN	2018	275,061	1,972	273,089	273,089
AFG-IRN	2001	263,501	18,000	245,501	120,000

From 2009 the *World Refugee Survey* ends and GRR uses UNHCR’s registration figures as a starting point, adding unregistered populations from State Department reporting and other archival sources on top. The question is how much that addition changes, and whether the series the analysis uses is therefore still informative in those years. The Table shows that this changes little, which is a potential limitation. In the World Refugee Survey years the two stock series are identical in 55.9% of dyad-years and GRR exceeds UNHCR in about a quarter of them. In the State Department years they are identical in 95.3%, and GRR exceeds UNHCR in fewer than one dyad-year in twenty — by a larger margin where it does, a median of 76%, but rarely. The returns series behaves the same way: GRR and UNHCR record an identical figure in 9.7% of dyad-years before 2009 and 84.0% after it.

GRR’s independent contribution is therefore concentrated in the pre-2009 period, and the later years are better described as UNHCR’s series with archival corrections in a minority of cases than as an independent measurement. The comparison in this section is therefore informative mainly for 1980–2008, where the sources genuinely diverge. A future check will code USHR pre-2009 separately and compare with WRS.

2.1.2 Internal Consistency

Table 16: Internal-consistency diagnostics on the stock series. The first four columns are impossibilities or near-impossibilities; the last three are fingerprints of estimation rather than measurement.

Series	Returns exceed prior stock (%)	Unexplained collapse (%)	V-shaped reversal (%)	Spike then collapse (%)	Unchanged from prior year (%)	Exact multiple of 10,000 (%)	Median abs. log change
GRR stock	0.365	3.28	0.848	0.348	8.37	7.49	0.111
UNHCR stock	0.437	3.54	0.941	0.282	4.15	1.82	0.113

Table 17: The same series expressed as rates. A rate above one is not possible.

Series	Dyad-years	Rate above 100%	Rate above 50%	Rate exactly zero	Mean rate
GRR return / GRR stock	4,932	0.36%	1.60%	74.6%	4.93%
GRR voluntary return / GRR stock	4,913	0.35%	1.51%	77.1%	4.67%
UNHCR return / UNHCR stock	4,577	0.44%	1.77%	73.5%	4.84%

Both sources imply sensible rates of internal consistency: pairing GRR’s returns with GRR’s stock puts 0.36% of dyad-years above a rate of one, and UNHCR’s own pairing is comparable. GRR holds a stock unchanged from the previous year more often than UNHCR does, and reports round multiples of ten thousand more often. Both follow from the coding rule because where the archival range does not change, GRR carries the previous estimate forward instead of alternating between bounds, and the sources themselves report round

figures. The rule increases within-dyad stability, which is what a two-way fixed-effects design needs, at the cost of looking less precise than a series that varies because its reporting convention varied.

2.1.3 Replicating the Main Analysis

In the table below, column 1 is the main specification of Table 2 of the paper. Column 2 drops forced returns from the outcome by definition. Columns 3 and 4 then move onto UNHCR’s return measure and onto UNHCR’s stock as well, the second of which is the full replication of the paper on UNHCR data.

Table 18: The paper’s specification — the right-hand side of the baseline table, dyad and year fixed effects, log stock at risk as offset, dyad-clustered standard errors — re-estimated as the dependent variable and the offset are swapped for other measures. Column 1 is the paper’s baseline, reproduced coefficient for coefficient. Samples differ across columns because a rung drops the dyad-years whose own offset is missing or zero. Significance: [†] 0.10, * 0.05, ** 0.01, *** 0.001.

Term	(1) GRR returns GRR stock	(2) GRR voluntary GRR stock	(3) UNHCR returns GRR stock	(4) UNHCR returns UNHCR stock
Forced Return: severe	1.08*** (0.19)	0.51** (0.17)	0.50* (0.22)	0.28 (0.24)
Forced Return: moderate	0.47* (0.23)	-0.20 (0.14)	-0.26 (0.16)	-0.35* (0.16)
Conflict in COA: high	0.58 [†] (0.31)	0.37 (0.30)	0.09 (0.28)	-0.06 (0.28)
Conflict in COA: low	0.81*** (0.21)	0.23 (0.14)	0.15 (0.19)	0.19 (0.21)
War Ending	1.01*** (0.22)	1.20*** (0.20)	1.31*** (0.21)	1.27*** (0.22)
Conflict in COO: high	0.11 (0.28)	0.15 (0.29)	0.64 [†] (0.37)	0.80* (0.37)
Conflict in COO: low	0.30 (0.22)	0.51* (0.23)	0.60* (0.26)	0.64* (0.28)
GDP (COA)	-1.12 [†] (0.59)	-0.29 (0.58)	0.26 (0.66)	-0.76 (0.70)
GDP (COO)	0.39 (0.28)	0.27 (0.28)	-0.02 (0.30)	-0.04 (0.31)
Civil Liberties Restrictions (COO)	-0.53*** (0.16)	-0.47** (0.15)	0.08 (0.16)	0.17 (0.16)
Refugee Rights	-0.01 (0.17)	0.07 (0.17)	0.04 (0.24)	0.04 (0.21)
Conflict Duration	-0.03 (0.09)	-0.04 (0.09)	0.05 (0.11)	-0.00 (0.12)
Dyad-years	2,327	2,281	2,297	2,126
Dyads	110	108	109	107

The severe-policy coefficient is 1.08 on the paper’s own measures and 0.51 once forced returns are removed from the outcome, so about half of it is coerced return counted as return and the other half is voluntary return that a coercive policy brings forward. Moving onto UNHCR’s return measure leaves it at 0.50, and onto UNHCR’s stock as well at 0.28, where it is no longer distinguishable from zero. Re-estimating on UNHCR’s numbers produces results that are hard to reconcile with existing theory: controlling for other factors, war at home appears to *increase* repatriation, while income and civil liberties do little. Since UNHCR records return most reliably in origins where it has an operational presence, which are disproportionately those with active conflict, I read this as evidence about the measure rather than the hypothesis, though no source offers ground truth.

2.2 Validating the Forced-Return Measure

Forced-return policies and forced returns are coded from the same archival sources, which raises the concern that any association between them is mechanical. Three checks bear on this: whether policies and recorded returns are in fact the same variable, whether the result survives when returns coded as forced are removed from the outcome, and whether it survives when the policy measure itself is replaced with one coded independently.

Table 19: Return in the years a policy is coded. If the measure were read off the return, the zero-return column would be near empty.

Policy	Dyad-years	Zero return	Return below 1% of stock	Median return rate
None	3,988	76.3%	84.8%	0.00%
Moderate	290	35.9%	58.3%	0.35%
Severe	150	8.0%	22.0%	10.64%

The table shows that a moderate policy is coded in 290 dyad-years and 36% of them record no return at all; a severe policy is coded in 150 and 8% record none. The coding follows the policy — a threat, an ultimatum, a camp closure, an announced deadline — and not the return, which is why it can be present in a year when nothing happens. The second concern is the reverse: that the returns themselves are an artefact of the same passages. This data shows that they are not. 81% of all return GRR records is coded voluntary, and forced returns appear in only 4.1% of dyad-years.

The strongest version of the concern is that the outcome contains the treatment: if a severe policy is recorded because forced returns were observed, then regressing returns on the policy recovers the coding rule rather than a relationship. Model 2 of Table 18 removes forced returns from the outcome entirely, leaving only returns coded voluntary, and Table 24 does the same on the full estimation sample. The severe-policy coefficient falls by roughly half and remains significant. That is the expected pattern if part of the association is coerced return counted as return and the remainder is voluntary return that a coercive policy encourages. It is not the pattern that a purely mechanical association would produce, which would be a coefficient indistinguishable from zero once the mechanically linked returns are removed.

2.2.1 Amnesty International Alternative Measure

[Placeholder. ANALYSIS UNDERWAY]

2.3 Alternative Outcome: Discrete Return Episodes

Table 20: Seven codings of a discrete return episode

Rule	Definition	Type	Episode-years	Share of dyad-years (%)	Dyads with variation
Rate above 5 per cent	$r_{it} > 0.05$	Level	559	10.9	102
Rate above 10 per cent	$r_{it} > 0.10$	Level	400	7.8	92
Rate above 25 per cent	$r_{it} > 0.25$	Level	199	3.9	75
More than 25,000 returns	$y_{it} > 25,000$	Level	236	4.6	62
More than 50,000 returns	$y_{it} > 50,000$	Level	129	2.5	38
One SD above the dyad mean	$r_{it} > \bar{r}_i + s_i$	Change	304	6.0	122
Three times the dyad median	$r_{it} \geq 3\tilde{r}_i$	Change	274	5.4	75

r_{it} is total return divided by the stock at risk, the same denominator the paper uses as the PPML offset; y_{it} is the return count. $\bar{r}_i$, s_i and $\tilde{r}_i$ are the dyad's own mean, standard deviation and median rate, the median taken over years with positive return. The panel contains 5,131 dyad-years in 285 dyads. The last column counts dyads whose coded outcome is not constant across their observed years; only those dyads enter a conditional logit. It is computed before covariate missingness, so the dyad counts in Tables 21 and 22 are a little lower.

Table 21: Discrete return episodes: seven codings of the dependent variable

Episode rule: Model:	Returns (count) (1) Poisson	Rate > 0.05 (2) Logit	Rate > 0.10 (3) Logit	Rate > 0.25 (4) Logit	> 25,000 (5) Logit	> 50,000 (6) Logit	Mean + 1 SD (7) Logit	3× median (8) Logit
<i>Variables</i>								
<i>Stay constraint (H2)</i>								
Forced Return: Severe	1.084*** (0.1914)	2.907*** (0.3454)	2.746*** (0.3348)	2.156*** (0.5258)	3.421*** (0.4034)	4.722*** (0.7622)	2.164*** (0.3756)	2.080*** (0.4920)
Forced Return: Moderate	0.4727* (0.2265)	0.6887** (0.2505)	0.4602 (0.2837)	0.5052 (0.4221)	0.6783 (0.4293)	0.5177 (0.6327)	0.7264* (0.3101)	0.5584† (0.3155)
Conflict in COA: High	0.5822† (0.3063)	0.2056 (0.4204)	-0.0447 (0.4300)	-0.0164 (0.4546)	1.121 (0.7240)	1.361 (1.341)	0.2330 (0.4323)	0.3159 (0.4661)
Conflict in COA: Low	0.8090*** (0.2081)	0.2508 (0.3076)	-0.1661 (0.3012)	-0.0390 (0.3613)	0.2416 (0.5597)	1.259 (0.8218)	-0.1411 (0.3233)	-0.0044 (0.3386)
<i>Preferences (H1)</i>								
War Ending	1.005*** (0.2171)	1.143*** (0.3452)	1.197** (0.3766)	0.6136† (0.3176)	1.887*** (0.5793)	3.179** (1.179)	1.169*** (0.2657)	0.8843** (0.3237)
Conflict in COO: High	0.1081 (0.2783)	-0.2958 (0.3426)	-0.4352 (0.3733)	-0.4233 (0.4997)	-0.2320 (0.5837)	1.519 (1.232)	-0.0883 (0.3513)	0.0861 (0.3979)
Conflict in COO: Low	0.3022 (0.2248)	-0.0299 (0.2681)	-0.1582 (0.2780)	-0.3726 (0.3638)	-0.0347 (0.5202)	0.9513 (0.6528)	-0.2982 (0.2793)	-0.1745 (0.3277)
GDP (COA)	-1.119† (0.5893)	-0.0949 (0.6024)	-0.6036 (0.6654)	-0.2976 (0.9318)	-0.5044 (0.7902)	-1.273 (2.174)	-0.4393 (0.6443)	-0.1390 (0.6551)
GDP (COO)	0.3890 (0.2842)	0.2132 (0.3439)	0.1808 (0.3173)	0.5141 (0.4445)	0.5034 (0.6751)	1.257 (1.438)	0.9018* (0.3982)	0.4801 (0.4359)
Civil Liberties Restrictions (COO)	-0.5345*** (0.1573)	-0.6866** (0.2129)	-0.7549** (0.2683)	-1.054*** (0.2939)	-0.6430* (0.3198)	-0.9291* (0.4105)	-0.4745* (0.2028)	-0.4341† (0.2228)
<i>Controls</i>								
Refugee Rights	-0.0094 (0.1695)	0.3359* (0.1581)	0.3799* (0.1499)	0.2797 (0.1795)	0.4582† (0.2413)	-0.1519 (0.5895)	0.0992 (0.1533)	0.3340* (0.1619)
Conflict Duration	-0.0281 (0.0855)	0.3261* (0.1371)	0.3752* (0.1673)	0.3283 (0.2022)	0.7578** (0.2443)	0.3891 (0.4192)	0.4158** (0.1590)	0.2296 (0.1848)
Rule type	Count	Level	Level	Level	Level	Level	Change	Change
Dyads	110	93	84	68	55	32	107	72
Episode-years in sample	—	496	367	187	195	98	274	255
<i>Fixed-effects</i>								
Dyad	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>								
Observations	2,327	2,023	1,834	1,530	1,240	732	2,300	1,743
Pseudo R ²	0.79946	0.25170	0.25804	0.25147	0.37856	0.52514	0.17753	0.17404

Clustered (Dyad) standard-errors in parentheses
Signif. Codes: ***, 0.001, **, 0.01, *, 0.05, †, 0.1

Column 1 is the paper's specification: Poisson PML on the return count with the log stock at risk as offset. Columns 2 to 8 are conditional logits on the episode codings of Table 20. All models carry dyad and year fixed effects and cluster standard errors by dyad. Logit coefficients are log odds ratios and are not on the scale of column 1. The conditional logit drops dyads and years without variation in the outcome, which is why the observation and dyad counts fall below column 1.

Table 22: Discrete return episodes: estimators

Dep. Var.:	Logit	Rate > 0.10 LPM	LPM, logit rows	Onset logit	Episode onset Onset hazard	Onset LPM	Returns (count) PPML, logit rows
Model:	(1) Logit	(2) OLS	(3) OLS	(4) Logit	(5) Logit	(6) OLS	(7) Poisson
<i>Variables</i>							
<i>Stay constraint (H2)</i>							
Forced Return: Severe	2.746*** (0.3348)	0.3639*** (0.0475)	0.4367*** (0.0492)	2.690*** (0.5278)	2.191*** (0.3120)	0.2423*** (0.0534)	1.121*** (0.2005)
Forced Return: Moderate	0.4602 (0.2837)	0.0437 [†] (0.0247)	0.0427 (0.0344)	0.8052* (0.3268)	0.6392* (0.2830)	0.0463* (0.0205)	0.5235* (0.2475)
Conflict in COA: High	-0.0447 (0.4300)	0.0079 (0.0374)	-0.0057 (0.0547)	0.4682 (0.5738)	-0.0372 (0.3106)	0.0348 (0.0314)	0.5479 [†] (0.3319)
Conflict in COA: Low	-0.1661 (0.3012)	-0.0127 (0.0263)	-0.0205 (0.0371)	0.0725 (0.3614)	-0.2340 (0.2571)	0.0127 (0.0204)	0.8308*** (0.2059)
<i>Preferences (H1)</i>							
War Ending	1.197** (0.3766)	0.1331** (0.0424)	0.2129** (0.0654)	1.071* (0.4962)	1.210** (0.3964)	0.0551 [†] (0.0328)	0.9851*** (0.2215)
Conflict in COO: High	-0.4352 (0.3733)	-0.0407 (0.0340)	-0.0658 (0.0515)	0.6387 (0.5546)	0.4054 (0.4163)	0.0320 (0.0327)	0.0536 (0.2885)
Conflict in COO: Low	-0.1582 (0.2780)	-0.0211 (0.0260)	-0.0206 (0.0411)	0.1576 (0.4353)	0.1764 (0.3731)	0.0061 (0.0258)	0.2771 (0.2325)
GDP (COA)	-0.6036 (0.6654)	-0.1209* (0.0508)	-0.0779 (0.0737)	-1.202 [†] (0.6942)	-1.222*** (0.1923)	-0.1030** (0.0377)	-1.050 [†] (0.6227)
GDP (COO)	0.1808 (0.3173)	0.0339 (0.0225)	0.0426 (0.0382)	0.4326 (0.4059)	-0.2075 (0.1756)	0.0342 [†] (0.0177)	0.3579 (0.2955)
Civil Liberties Restrictions (COO)	-0.7549** (0.2683)	-0.0464** (0.0167)	-0.0950** (0.0301)	-0.4573 (0.2845)	-0.2616* (0.1239)	-0.0145 (0.0118)	-0.5823*** (0.1650)
<i>Controls</i>							
Refugee Rights	0.3799* (0.1499)	0.0169 (0.0149)	0.0424* (0.0180)	0.4481** (0.1707)	0.2514** (0.0835)	0.0167 (0.0115)	0.0105 (0.1739)
Conflict Duration	0.3752* (0.1673)	0.0320* (0.0126)	0.0452* (0.0211)	0.1909 (0.2598)	0.1462 (0.1776)	0.0130 (0.0129)	-0.0479 (0.0873)
Years Since Last Episode (log)					-0.3360* (0.1421)		
Sample Dyads	Panel 84	Panel 245	Logit rows 84	At risk 74	At risk 241	At risk 234	Logit rows 84
Fixed-effects							
Dyad	Yes	Yes	Yes	Yes		Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics							
Observations	1,834	3,913	1,834	1,266	3,221	3,214	1,834

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***, 0.001, **, 0.01, *, 0.05, †, 0.1

The dependent variable is the ten per cent rate rule throughout: an episode is a dyad-year in which return exceeds a tenth of the stock at risk. Columns 4 to 6 use episode onset, one in the first year of a run, estimated on the dyad-years at risk of starting an episode; interior years of a run are removed from the risk set rather than coded zero. Column 5 is the discrete-time hazard: year fixed effects and log years since the last episode, without dyad fixed effects, because duration is collinear with dyad and year effects together; its dyad count is the number of clusters rather than the number of absorbed effects. Column 7 refits the paper's count model, outcome and offset unchanged, on the rows the conditional logit retains. Standard errors clustered by dyad throughout. Logit, linear-probability and PPML coefficients are on three different scales and should not be compared across columns of different estimators.

Table 23: Sensitivity of the two hypotheses to the episode threshold

Rate threshold	Episode-years	N	Dyads	Conditional logit		Linear probability	
				Forced Return: Severe	War Ending	Forced Return: Severe	War Ending
0.010	853	2,176	101	3.07***	0.74*	0.513***	0.085*
0.025	687	2,097	97	2.90***	1.02**	0.480***	0.122**
0.050	559	2,023	93	2.91***	1.14***	0.465***	0.138**
0.075	468	1,919	87	3.12***	1.28***	0.438***	0.151***
0.100	400	1,834	84	2.75***	1.20**	0.364***	0.133**
0.150	300	1,795	81	2.27***	1.09***	0.263***	0.105**
0.200	236	1,615	72	2.13***	1.09**	0.202***	0.094**
0.250	199	1,530	68	2.16***	0.61 [†]	0.174***	0.062*
0.330	144	1,280	62	2.08**	0.97*	0.112*	0.059*

Each row recodes the outcome at a different rate threshold and re-estimates the paper's specification with dyad and year fixed effects and standard errors clustered by dyad. The observation and dyad counts are for the conditional logit; the linear probability model retains 3,913 dyad-years in 245 dyads at every threshold. The two estimators are on different scales. Significance: [†] 0.10, * 0.05, ** 0.01, *** 0.001.

GRR's return counts are hand-coded from archival reporting, which means that year-to-year changes within a dyad may partly reflect reporting noise rather than genuine shifts in return. Treating the outcome as a count can therefore imply more precision than the underlying sources warrant. As a robustness check, this section recodes the outcome as binary: whether a dyad experienced a substantial return episode in a given year. This approach discards variation in the exact number returned and focuses instead on episodes large enough to be unlikely to result from minor coding differences.

There is no uniquely correct threshold for what counts as a substantial episode, so I report results across several alternatives. Table 20 presents seven coding rules in two families. The *level* rules use common thresholds across all dyads: returns greater than five, ten, or twenty-five percent of the population at risk, or absolute returns greater than twenty-five or fifty thousand people. The *change* rules instead define an episode relative to a dyad's own history: the return rate must exceed the dyad mean by more than one within-dyad standard deviation, or be at least three times its median return rate among years in which any return occurred.

Table 21 re-estimates the paper's main specification using each definition, with the count model in column 1 for comparison. Table 22 fixes the threshold at ten percent and varies the estimator. Table 23 then shows how the estimates change across nine different thresholds.

The association with severe forced return is robust across all of these definitions and estimators. A dyad-year under a severe forced-return policy records a substantial return episode 51 percent of the time, compared with 8 percent otherwise. The coefficient remains statistically significant whether a substantial episode is defined as returns exceeding ten percent of the population at risk or fifty thousand people. The war-ending window is also positive in every specification and statistically significant in most, although the result is less stable than for severe forced return. Its strength varies somewhat with the threshold, and the relationship is weaker when the outcome captures the onset of an episode rather than whether an episode is already under way.

2.4 The 2009 Source Change

[Placeholder. ANALYSIS UNDERWAY.]

3 Robustness Checks

Every column in this section departs from column 1 of Table 2 of the paper and states its departure; everything else is held fixed. The estimation sample is the paper’s own — dyad-years with more than 1,000 refugees in the previous year — and the offset throughout is the corrected stock at risk rather than the raw lagged stock, which differs from it in the dyad-years where recorded return exceeds the lagged figure.

3.1 Alternative Measures of Return

Table 24 varies the dependent variable. Model 1 reproduces the baseline. Model 2 restricts to voluntary returns only, excluding cases coded as forced repatriation. Model 3 uses a centered three-year moving average of total returns, smoothing annual noise and accommodating lagged responses to changing conditions. The baseline results hold across both alternatives. Severe forced return remains positive and significant when forced returns are removed from the outcome, though its magnitude falls by roughly half, from 1.08 to 0.51. This is what the decomposition in the main text implies: part of the association is coerced return counted as return, and the remainder is voluntary return that a coercive policy brings forward. Moderate forced return is no longer significant in Model 2 and is signed negatively (-0.20), consistent with these policies operating through coercion rather than inducing voluntary movement. The post-war window is unaffected by either change and is in fact larger on voluntary returns alone (1.20) than in the baseline (1.01), which is the expected direction: the returns that follow a war ending are disproportionately voluntary.

3.2 Alternative Measures of Conflict and War Endings

The post-war window is built from an extended version of Fearon’s (2017) civil war list, which dates wars by their definitive beginnings and endings rather than by an annual threshold on battle deaths. Fearon’s original coding runs to 2014; the version used here carries thirty-one of those cases forward with revised end dates and adds seventeen new ones, and is then merged with the inter-state wars coded to a refugee origin. A dyad-year counts as post-war only if the origin is also at peace in that year, since origins with overlapping wars can have one war end while another runs.

Two things could be wrong with that. The extension is this project’s own work and could be driving the result, and the definitive-dates approach could be the wrong way to date an ending in the first place. Table 25 addresses both before Table 26 turns to how conflict itself is measured.

3.2.1 Alternative Definitions of a War Ending

Table 24: Alternative Measures of the Dependent Variable

Dep. Var.:	Total Repatriation Total Returns	Voluntary Repatriation Voluntary Returns	Total Repat. (3-yr MA) 3-Year MA
Model:	(1)	(2)	(3)
<i>Variables</i>			
<i>Stay constraint (H2)</i>			
Forced Return: Severe	1.084*** (0.1914)	0.5055** (0.1667)	0.9461*** (0.2144)
Forced Return: Moderate	0.4727* (0.2265)	-0.1970 (0.1425)	0.4857* (0.2009)
Conflict in COA: High	0.5822† (0.3063)	0.3688 (0.3023)	0.3881† (0.2332)
Conflict in COA: Low	0.8090*** (0.2081)	0.2264 (0.1432)	0.6728*** (0.1603)
<i>Preferences (H1)</i>			
War Ending	1.005*** (0.2171)	1.198*** (0.1969)	0.6038*** (0.1736)
Conflict in COO: High	0.1081 (0.2783)	0.1502 (0.2902)	0.2674 (0.3066)
Conflict in COO: Low	0.3022 (0.2248)	0.5132* (0.2257)	0.2736 (0.1985)
GDP (COA)	-1.119† (0.5893)	-0.2935 (0.5799)	-0.9874† (0.5919)
GDP (COO)	0.3890 (0.2842)	0.2702 (0.2772)	0.1589 (0.2567)
Civil Liberties Restrictions (COO)	-0.5345*** (0.1573)	-0.4747** (0.1517)	-0.4879*** (0.1358)
<i>Controls</i>			
Refugee Rights	-0.0094 (0.1695)	0.0693 (0.1665)	-0.0263 (0.1621)
Conflict Duration	-0.0281 (0.0855)	-0.0394 (0.0932)	-0.0982 (0.0927)
<i>Fixed-effects</i>			
Dyad	Yes	Yes	Yes
Year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	2,327	2,281	2,207
Squared Correlation	0.75658	0.86471	0.80687
Pseudo R ²	0.79946	0.79509	0.82691

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***, 0.001, **, 0.01, *, 0.05, †: 0.1*

Standard errors clustered by dyad in parentheses. All models: Poisson PML with log stock-at-risk offset and dyad and year fixed effects. Model 3 uses a centered 3-year moving average of total returns as the dependent variable.

Table 25: Robustness: Alternative Definitions of a War Ending

Model:	Main (1)	Fearon original (2)	Total Repatriation Civil only (3)	UCDP spells (4)	No size floor (5)	+ Ending yr (6)
<i>Variables</i>						
<i>Stay constraint (H2)</i>						
Forced Return: Severe	1.084*** (0.1914)	1.100*** (0.1941)	1.089*** (0.1916)	1.073*** (0.2071)	1.081*** (0.1910)	1.079*** (0.1901)
Forced Return: Moderate	0.4727* (0.2265)	0.4806* (0.2236)	0.4846* (0.2222)	0.4327† (0.2416)	0.4719* (0.2258)	0.4778* (0.2287)
Conflict in COA: High	0.5822† (0.3063)	0.5689† (0.3105)	0.5896† (0.3075)	0.5270† (0.3022)	0.5870† (0.3050)	0.5829† (0.3045)
Conflict in COA: Low	0.8090*** (0.2081)	0.8101*** (0.2090)	0.8053*** (0.2099)	0.8282*** (0.1990)	0.8094*** (0.2081)	0.8100*** (0.2073)
<i>Preferences (H1)</i>						
War Ending	1.005*** (0.2171)	1.028*** (0.2256)	0.9953*** (0.2220)	1.294*** (0.2566)	0.9917*** (0.2179)	1.041*** (0.2086)
War Ending (Year Of)						0.1900 (0.2362)
Conflict in COO: High	0.1081 (0.2783)	0.0812 (0.2786)	0.1093 (0.2811)	0.0844 (0.2751)	0.1095 (0.2794)	0.1132 (0.2785)
Conflict in COO: Low	0.3022 (0.2248)	0.2756 (0.2194)	0.2972 (0.2243)	0.2847 (0.2278)	0.3067 (0.2244)	0.3023 (0.2247)
GDP (COA)	-1.119† (0.5893)	-1.088† (0.5938)	-1.094† (0.5929)	-1.116† (0.6097)	-1.123† (0.5873)	-1.120† (0.5888)
GDP (COO)	0.3890 (0.2842)	0.3588 (0.2861)	0.3885 (0.2845)	0.2834 (0.2910)	0.3824 (0.2836)	0.4103 (0.2802)
Civil Liberties Restrictions (COO)	-0.5345*** (0.1573)	-0.5356** (0.1671)	-0.5139** (0.1684)	-0.6014*** (0.1424)	-0.5343*** (0.1567)	-0.5322*** (0.1576)
<i>Controls</i>						
Refugee Rights	-0.0094 (0.1695)	-0.0191 (0.1721)	-0.0154 (0.1708)	0.0266 (0.1626)	-0.0143 (0.1696)	-0.0100 (0.1704)
Conflict Duration	-0.0281 (0.0855)	-0.0243 (0.0836)	-0.0312 (0.0846)	-0.0754 (0.0892)	-0.0272 (0.0850)	-0.0299 (0.0850)
All controls	Yes	Yes	Yes	Yes	Yes	Yes
Ending list	Extended + inter-state	Fearon original only	Extended civil only	UCDP spells	No 25,000 floor	Extended + inter-state
Endings	41	33	36	76	75	41
<i>Fixed-effects</i>						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	2,327	2,327	2,327	2,327	2,327	2,327
Pseudo R ²	0.79946	0.79929	0.79907	0.79944	0.79924	0.79964

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

All columns are the specification of Table 2 of the paper with the post-war window rebuilt from a different list of war endings; nothing else changes. Column 1 is the list the paper uses. Column 2 keeps only the cases Fearon coded and this project did not revise, so it excludes both the post-2014 extension and the revised end dates. Column 3 keeps the extended civil-war list but drops the inter-state endings. Column 4 dates endings from UCDP high-intensity spells, where an ending is a year in which battle deaths fall below the threshold rather than the end of a named war. Column 5 removes the requirement that the origin have at least 25,000 refugees in the ending year, which is the one filter that selects endings on a quantity related to the outcome. Column 6 adds the ending year itself, so the window is identified from years one to three rather than zero to three. PPML with the log stock at risk as offset, dyad and year fixed effects, SEs clustered by dyad.

The window survives every definition. It is 1.03 on Fearon’s original cases alone, against 1.01 on the list the paper uses, so the extension is not manufacturing the result — which is the check that matters most, since the extension is this project’s own coding. Dropping the inter-state endings leaves it at 1.00, as the case-by-case comparison elsewhere implies it should. Dating endings from UCDP spells instead gives the largest estimate in the table (1.29), and the ending year itself is not distinguishable from zero (0.19), which is consistent with the timing reported in the main text: return follows an ending rather than accompanying it. Severe forced return is between 1.07 and 1.10 throughout.

Column 5 removes the size floor, which is the filter most likely to matter for inference: requiring an origin to have at least 25,000 refugees in the ending year selects endings on a quantity that is related to the outcome. Dropping it takes the list from 41 endings to 75 and leaves the window at 0.99 against 1.01. The estimate is mildly increasing in the floor — raising it to 100,000 gives 1.11 — which is the concentration result again, since post-war return is somewhat larger where the displaced population was larger. The floor is retained because it matches the rule that defines which origins enter GRR at all, and because the descriptive tables elsewhere divide return by the stock at the ending, a quantity that is not meaningful when the denominator is a few hundred people. It is not doing any work in the estimate.

3.2.2 Alternative Measures of Conflict

Having fixed the ending, the second question is how conflict itself is coded. The concern is that the UCDP variable switches between categories year by year, so a single year coded as peace between years of severe conflict may not reflect the kind of resolution that would enable return. Model 1 reproduces the baseline. Model 2 replaces the ordinal conflict indicators with logged battle deaths, a continuous measure of intensity available from 1989. Model 3 dates origin-country conflict at $t - 2$ to allow for a delayed response. Model 4 replaces the origin conflict levels with a de-escalation indicator equal to one when conflict falls from high-intensity to low or none, capturing transitions rather than levels. Model 5 adds an indicator for a peace agreement in the previous year, drawn from the UCDP Peace Agreements Dataset and merged to the origin’s active conflict via UCDP `conflict_id`. Model 6 adds the PITF measure of warfare and violence, combining its Ethnic War and Revolutionary War datasets, which ends in 2017.

The asylum-country push coefficients are stable across all six specifications, and severe forced return is significant at the 0.001 level in every one. Origin-country conflict remains a weak predictor of return throughout, and logged battle deaths in the origin are indistinguishable from zero (0.01). Battle deaths in the asylum country enter positive and significant (0.13), which is the push channel measured continuously rather than in categories. The peace-agreement indicator enters negative and significant (-0.44), reinforcing the paper’s central finding that formal peace does not predict return, and the PITF war indicator enters positive (1.30), on the same pattern as origin conflict elsewhere in the paper.

3.3 Alternative Measures of Repression

A separate concern, distinct from how conflict is measured, is that refugees may decline to return not because of active armed conflict but because of broader state repression that conflict measures do not capture. The Freedom House civil liberties measure (`o_civlib_z`) is included in the main specification; here I supplement it with three additional measures of origin-country repression and political exclusion. Because these measures vary in coverage—and because the EPR ethnic-status measures substantially reduce the estimation sample—I report them in two stages.

Political Terror Scale.

The Political Terror Scale (PTS) measures state-perpetrated physical integrity violations—torture, political imprisonment, extrajudicial killings, and disappearances—on a 1–5 ordinal scale. I use the mean of three sources (Amnesty International, Human Rights Watch, and the U.S. State Department), standardized across the estimation sample (`o_terror_z`).

Table 26: Robustness: Alternative Measures of Conflict

Model:	Main (1)	Battle deaths (2)	Total Repatriation COO t-2 (3)	De-escalation (4)	Peace ag. (5)	PITF war (6)
<i>Variables</i>						
<i>Stay constraint (H2)</i>						
Forced Return: Severe	1.084*** (0.1914)	0.8765*** (0.1864)	1.055*** (0.1987)	1.078*** (0.1935)	1.067*** (0.1840)	1.103*** (0.1977)
Forced Return: Moderate	0.4727* (0.2265)	0.4299† (0.2289)	0.5320* (0.2268)	0.4895* (0.2304)	0.5118* (0.2169)	0.4928* (0.2285)
Conflict in COA: High	0.5822† (0.3063)		0.7512* (0.3075)	0.5161† (0.2990)	0.7151* (0.3029)	0.5713† (0.3176)
Conflict in COA: Low	0.8090*** (0.2081)		0.8548*** (0.1912)	0.7876*** (0.1940)	0.8390*** (0.2068)	0.8223*** (0.2080)
<i>Preferences (H1)</i>						
War Ending	1.005*** (0.2171)	0.9683*** (0.1963)	0.9144*** (0.2040)	1.024*** (0.1926)	0.9988*** (0.2062)	0.9602*** (0.2092)
Conflict in COO: High	0.1081 (0.2783)				0.0804 (0.2717)	0.0686 (0.2812)
Conflict in COO: Low	0.3022 (0.2248)				0.2414 (0.2195)	0.3135 (0.2276)
GDP (COA)	-1.119† (0.5893)	-1.188** (0.4575)	-1.208* (0.6075)	-1.188* (0.5999)	-1.190* (0.5502)	-1.062† (0.5914)
GDP (COO)	0.3890 (0.2842)	0.3900 (0.2771)	0.4921† (0.2790)	0.4133 (0.2732)	0.4332 (0.2786)	0.4048 (0.2904)
Civil Liberties Restrictions (COO)	-0.5345*** (0.1573)	-0.4932** (0.1504)	-0.5065*** (0.1479)	-0.5238*** (0.1488)	-0.5429*** (0.1632)	-0.5592*** (0.1612)
<i>Controls</i>						
Refugee Rights	-0.0094 (0.1695)	0.1875 (0.1833)	0.0766 (0.1730)	0.0066 (0.1667)	0.0272 (0.1675)	-0.0306 (0.1797)
Conflict Duration	-0.0281 (0.0855)	0.0215 (0.0971)	0.0682 (0.0794)	0.0762 (0.0829)	0.0169 (0.0741)	-0.0871 (0.0865)
<i>Alternative conflict measures</i>						
Battle Deaths (COA, log)		0.1331*** (0.0376)				
Battle Deaths (COO, log)		0.0057 (0.0412)				
Conflict in COO: High (t-2)			-0.1791 (0.2162)			
Conflict in COO: Low (t-2)			-0.0608 (0.1153)			
De-escalation (COO)				-0.3021 (0.2288)		
Peace Agreement (COO, t-1)					-0.4381** (0.1530)	
PITF War/Violence (COO)						1.302* (0.5792)
All controls	Yes	Yes	Yes	Yes	Yes	Yes
Years	1980-2023	1989-2023	1980-2023	1980-2023	1980-2023	1980-2017
<i>Fixed-effects</i>						
Dyad	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	2,327	2,134	2,128	2,327	2,214	2,238
Pseudo R ²	0.79946	0.81014	0.80939	0.79919	0.80883	0.80007

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

All models carry the full specification of Table 2 of the paper, including the post-war window, and differ only in how origin- and asylum-country conflict is measured. Columns 2 to 4 replace the ordinal conflict levels; columns 5 and 6 add a measure alongside them. PPML with the log stock at risk as offset, dyad and year fixed effects, SEs clustered by dyad.

Genocide/politicide.

A binary indicator equal to one in any country-year with an ongoing episode of genocide or politicide, drawn from the Political Instability Task Force (PITF) Geno-Politicide event dataset (`o_genocide`). The PITF episode list ends in 2017; observations after 2017 are coded as missing.

Ethnic political status (EPR).

Reported in Section 4.1. The ordinal EPR status variable (`o_eth_status`) takes nine values from ‘self-exclusion’ to ‘monopoly’. Refugee groups are matched to EPR group identifiers via the EPR Refugee Dataset, which links populations of at least 2,000 refugees to EPR groups in neighboring or proximate asylum countries (within 950 km of the origin border). For dyads outside this coverage I code the refugee group manually from World Refugee Survey reports.¹ EPR coverage is unavailable for some refugee groups, reducing the estimation sample.

Table 27 reports the three specifications. Column 1 reproduces the main specification. Column 2 adds PTS and genocide/politicide on the full estimation sample. The asylum-country push coefficients are stable across all three specifications, and Freedom House civil liberties continues to enter negatively. The Political Terror Scale is indistinguishable from zero once civil liberties are controlled for (-0.01), so the repression result operates through the Freedom House measure rather than through physical-integrity violations specifically. Surprisingly, the genocide/politicide indicator enters positive and significant (0.83), opposite to expectation.

The ethnic political status of the refugee group is a further measure of exclusion at home, but it belongs with the other tests of ethnicity as an alternative explanation and is reported in Section 4.1 rather than here.

3.4 Alternative Independent Variable Measures

Table 28 tests sensitivity to how the independent variables are operationalized. Model 1 reproduces the baseline. Model 2 replaces continuous origin and asylum GDP per capita with quartile dummies (Q1 reference) to relax linearity assumptions; other controls retain their continuous form. Model 3 lags all independent variables by one year, addressing the endogeneity concern that asylum-country conflict and return are measured contemporaneously and that the positive coefficient could reflect reverse causality (the “refugee warrior” hypothesis in which refugee populations contribute to host-state conflict rather than the reverse). Model 4 adds UNHCR funding per refugee in the asylum country as a proxy for repatriation funding. The two EPR-TEK indicators for transnational ethnic kin in the asylum country, which an earlier version of this table carried, are reported with the other tests of ethnicity in Section 4.1. Model 5, which precedes it in the table, adds an indicator for an active tripartite repatriation agreement between the origin state, the host state and UNHCR. This is the most formal expression of UNHCR involvement in a return, and complements the per-refugee funding measure in Model 4: funding measures how much UNHCR spends, an agreement measures whether it has formally brokered the return. Agreements are coded for the whole panel but are rare, occurring in only 43 dyad-years, so the coefficient is identified off few switches and should not be read as precisely estimated. It enters positive and significant (0.64), and severe forced return is unchanged beside it (1.05), so a brokered agreement is associated with more return without displacing the coercion result. Given how few switches identify it, this is best read as consistent with the funding result in Model 4 rather than as independent evidence.

Models 6 and 7 restore three covariates that an earlier version of this appendix carried in its baseline and the paper does not. Model 6 adds educational attainment in the origin and the asylum country and the gap between them, on the reasoning that education proxies both the resources a refugee has to re-establish at home and what would be given up by leaving. Model 7 adds the urban share of the asylum country and an indicator for the majority of the refugee population living in camps, which is the most direct measure GRR

¹The same ethnic group is carried forward as long as the refugee population is continuous; the assignment is set to missing when a discontinuity in the population could plausibly indicate a change in ethnic composition. Two case-specific adjustments warrant note: EPR classifies Somalis as ‘irrelevant’ within Somalia on grounds of ethnic homogeneity, and I reclassify this group as high-status; Peuhl/Goula refugees from the Central African Republic appear in the EPR Refugee Dataset but not in the EPR Power Dataset, and are coded as ‘powerless’ based on desk research. The 2021 EPR Refugee values are carried forward from 2020.

Table 27: Robustness: Alternative Repression Measures

Model:	Total Repatriation	
	Main (1)	+ PTS/Genocide (2)
<i>Variables</i>		
<i>Stay constraint (H2)</i>		
Forced Return: Severe	1.084*** (0.1914)	1.049*** (0.2001)
Forced Return: Moderate	0.4727* (0.2265)	0.4213† (0.2410)
Conflict in COA: High	0.5822† (0.3063)	0.5953† (0.3152)
Conflict in COA: Low	0.8090*** (0.2081)	0.7914*** (0.2155)
<i>Preferences (H1)</i>		
War Ending	1.005*** (0.2171)	0.9562*** (0.1987)
Conflict in COO: High	0.1081 (0.2783)	0.0246 (0.2883)
Conflict in COO: Low	0.3022 (0.2248)	0.3576† (0.2152)
GDP (COA)	-1.119† (0.5893)	-0.6990 (0.5910)
GDP (COO)	0.3890 (0.2842)	0.2922 (0.2955)
Civil Liberties Restrictions (COO)	-0.5345*** (0.1573)	-0.5785** (0.1826)
<i>Controls</i>		
Refugee Rights	-0.0094 (0.1695)	0.0178 (0.1819)
Conflict Duration	-0.0281 (0.0855)	-0.0612 (0.0895)
<i>Alternative repression measures</i>		
Political Terror Scale (COO)		-0.0102 (0.1290)
Genocide/Politicide (COO)		0.8349* (0.3391)
<i>Fixed-effects</i>		
Dyad	Yes	Yes
Year	Yes	Yes
<i>Fit statistics</i>		
Observations	2,327	1,994
Squared Correlation	0.75658	0.77177
Pseudo R ²	0.79946	0.80460

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

All models: PPML with log stock-at-risk offset, dyad and year FE; SEs clustered by dyad. Column 1 reproduces column 1 of Table 2 of the paper. Column 2 adds the Political Terror Scale (mean of Amnesty/HRW/State Dept ratings) and a binary indicator for ongoing genocide/politicide (PITF). Column 3 adds EPR ordinal ethnic status (Dominant omitted as reference), reducing the sample to dyads with EPR Refugee Dataset coverage.

carries of the conditions of asylum. These belong here, as alternative measures of the conditions refugees face, rather than in the specification: they are outcomes of the same processes the model is trying to explain, and the urban and camp series are missing for roughly 650 dyad-years.

The main results are stable across all five specifications; severe forced return ranges from 1.08 to 1.23 and is significant at the 0.001 level throughout. Model 3 is the check that matters most for reverse causality, and it does not weaken the result: with every independent variable measured a year earlier, severe forced return is larger than in the baseline (1.21 against 1.08), asylum-country conflict remains significant at both intensities, and the post-war window remains positive (0.62). Origin-country conflict is negatively signed but not significant at either intensity. The UNHCR per-refugee funding coefficient in Model 4 is large and positive (0.85); because UNHCR funding to asylum countries includes resources earmarked for repatriation, this most plausibly reflects active return programs rather than indicating that funding causally drives return.

3.5 Alternative Data Subsets

Table 29 tests whether the results hold across different subsets of the data. Model 1 reproduces the baseline using the full GRR dataset (1980–2023). Model 2 restricts to the WRS period (1980–2008) to assess sensitivity to the data source transition in 2009, when the primary source shifts from the World Refugee Survey to U.S. State Department Country Reports on Human Rights Practices; the WRS period has richer source material on stocks, forced returns, and unregistered populations. The two checks that swap in UNHCR’s return figures and the registered rather than the broad refugee stock have moved to Section 2.1, where they sit with the rest of the comparison between the two sources; they are rungs 3 to 5 of Table 18.

Model 4 excludes dyads associated in the literature with the “refugee warrior” hypothesis, under which refugee populations, camps, or exile communities may become militarized and contribute to conflict spillover in host states (Zolberg 1989; Salehyan and Gleditsch 2006).² Excluding these cases assesses whether the main results are driven by settings in which refugee presence may be more plausibly endogenous to asylum-country conflict. Model 5 excludes dyads with asylum countries in Europe and North America, given that WRS provides less detail on specific nationalities within high-income countries. Model 6 restricts to dyads where refugees originally fled armed conflict rather than non-war forms of persecution. Model 7 restricts to the top three asylum countries per origin by refugee stock, addressing concerns that refugees from some countries are more dispersed across multiple asylum destinations than others; because origins vary in the number of asylum countries they appear in, highly dispersed origins contribute more dyads and receive greater implicit weight when each dyad-year is weighted equally (see Section 3.5.1 below).

The core results are robust across all seven specifications. Severe forced return is significant at the 0.01 level or better in every column, and the post-war window at the 0.001 level in every column, ranging from 0.98 to 1.12. Two columns are worth noting. Models 2 and 3 split the panel at the 2009 source change and are the pair to read together. They do not agree, and the disagreement is worth stating plainly. Severe forced return is 0.73 in the World Refugee Survey period and 1.59 in the State Department period, significant in both, so the coercion result holds on either source and is stronger on the later one. The post-war window does not: it is 1.12 in the Survey period and 0.05, indistinguishable from zero, in the State Department period.

Two things are confounded in that contrast and this design cannot separate them. The later period has fewer war endings with large populations still in exile, so there may simply be less of the phenomenon to detect — the State Department subsample is 901 dyad-years against 1,260 for the Survey period. And the sources differ in what they record: the Survey reports on return movements directly, whereas the State Department reports are organised around human rights conditions and give return figures less consistently, which is why GRR uses UNHCR as a baseline for those years. The preference-component (H1) result should therefore be read as resting mainly on the pre-2009 evidence. The check on the source change proposed in Section 2.4, which would code both sources for the overlapping years, is the one that would separate these two explanations.

Excluding the “refugee warrior” dyads raises the severe-policy coefficient rather than lowering it (1.59), which is the opposite of what the endogeneity concern predicts: the result does not depend on the cases in

²The excluded dyads are: RWA–COD, RWA–TZA, AFG–PAK, KHM–THA, PSE–LBN, PSE–JOR, and NIC–HND.

Table 28: Robustness: Alternative Independent Variables

	Total Repatriation						
Model:	Main (1)	GDP Quartiles (2)	Lagged IVs (3)	+ Funding (4)	+ Tripartite (5)	+ Education (6)	+ Urban/Camp (7)
<i>Variables</i>							
<i>Stay constraint (H2)</i>							
Forced Return: Severe	1.084*** (0.1914)	1.098*** (0.1997)	1.214*** (0.2874)	1.228*** (0.2163)	1.046*** (0.1790)	1.119*** (0.1859)	1.065*** (0.2036)
Forced Return: Moderate	0.4727* (0.2265)	0.4828* (0.1941)	0.2307 (0.2521)	0.5539* (0.2316)	0.4659* (0.2239)	0.5116* (0.2235)	0.4787* (0.2224)
Conflict in COA: High	0.5822† (0.3063)	0.8524** (0.2717)	0.4850* (0.2365)	-0.0576 (0.3420)	0.5975† (0.3089)	0.5147 (0.3269)	0.5898* (0.2810)
Conflict in COA: Low	0.8090*** (0.2081)	0.6772*** (0.1954)	0.5351** (0.1794)	0.9048*** (0.2561)	0.9058*** (0.2413)	0.8535*** (0.2017)	0.7881*** (0.2036)
<i>Preferences (H1)</i>							
War Ending	1.005*** (0.2171)	0.9461*** (0.2219)	0.6170* (0.2713)	0.7617*** (0.2234)	1.021*** (0.2025)	0.9467*** (0.2102)	1.005*** (0.2212)
Conflict in COO: High	0.1081 (0.2783)	0.0532 (0.2738)	-0.4418 (0.2828)	-0.2259 (0.3230)	0.1792 (0.2848)	0.1425 (0.2813)	0.1139 (0.2871)
Conflict in COO: Low	0.3022 (0.2248)	0.2905 (0.2231)	-0.1434 (0.2020)	0.0414 (0.2504)	0.2804 (0.2327)	0.3262 (0.2231)	0.3118 (0.2286)
GDP (COA)	-1.119† (0.5893)		-0.2660 (0.6609)	-1.071 (0.6966)	-1.264* (0.5638)	-1.232* (0.6180)	-1.069† (0.5842)
GDP (COO)	0.3890 (0.2842)		0.0935 (0.2941)	0.6226† (0.3285)	0.3250 (0.2598)	0.4437 (0.2839)	0.3737 (0.2817)
Civil Liberties Restrictions (COO)	-0.5345*** (0.1573)	-0.5738** (0.1751)	-0.2841† (0.1603)	-0.3933* (0.1738)	-0.5441*** (0.1603)	-0.5609*** (0.1679)	-0.5214*** (0.1539)
<i>Controls</i>							
Refugee Rights	-0.0094 (0.1695)	0.0575 (0.1673)	0.1077 (0.1578)	0.0385 (0.1512)	-0.0147 (0.1663)	-0.0186 (0.1836)	0.0117 (0.1702)
Conflict Duration	-0.0281 (0.0855)	-0.0319 (0.0896)	0.1141 (0.0774)	-0.0565 (0.0951)	-0.0248 (0.0897)	-0.0768 (0.0906)	-0.0607 (0.0923)
<i>Alternative and additional measures</i>							
GDP (COA): Q2		-0.6802* (0.2789)					
GDP (COA): Q3		-0.4467 (0.4992)					
GDP (COA): Q4		-1.124 (1.356)					
GDP (COO): Q2		-0.7166** (0.2773)					
GDP (COO): Q3		-0.3631 (0.3537)					
GDP (COO): Q4		0.0318 (0.3956)					
UNHCR Funding p.p. (COA)				0.8483** (0.3005)			
Tripartite Agreement					0.6421** (0.2092)		
Education (COA)						-0.2081 (1.247)	
Education (COO)						-0.2292 (0.7820)	
Urban							-0.2904 (0.2491)
Majority in Camps							-0.1902 (0.5097)
<i>Fixed-effects</i>							
Dyad	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>							
Observations	2,327	2,327	2,164	1,928	2,327	2,202	2,325
Pseudo R ²	0.79946	0.80431	0.79826	0.82505	0.80237	0.80247	0.80051

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Poisson PML, log stock-at-risk offset, dyad + year FEs, SEs clustered by dyad. M1: column 1 of Table 2 of the paper. M2: GDP (COO/COA) quartiles (Q1 ref.). M3: all independent variables at t-1, including the war-ending window. M4: + UNHCR funding p.p. (z-log). M5: + an indicator for an active tripartite repatriation agreement between origin, host and UNHCR, which is coded panel-wide but occurs in only 43 dyad-years. M6: + educational attainment in both countries and the origin-asylum gap. M7: + urban share and an indicator for the majority of the refugee population living in camps; the urban and camp series are missing for roughly 650 dyad-years, so M7 runs on a smaller sample.

which refugee presence is most plausibly a cause of host-state conflict.

Table 29: Robustness: Alternative Datasets, Periods, Samples, and Offsets

Dep. Var.:				Total Repatriation			
Model:	Main	WRS	USHR	Excl.Warrior	Excl.ENA	War.Orig	Top.3.COA
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Variables</i>							
<i>Stay constraint (H2)</i>							
Forced Return: Severe	1.084*** (0.1914)	0.7329** (0.2273)	1.586*** (0.2686)	1.595*** (0.3069)	1.039*** (0.1937)	1.066*** (0.1844)	1.054*** (0.1866)
Forced Return: Moderate	0.4727* (0.2265)	0.3431† (0.1885)	0.2507 (0.3282)	0.3408 (0.2121)	0.4871* (0.2248)	0.4681* (0.2283)	0.5028* (0.2232)
Conflict in COA: High	0.5822† (0.3063)	0.9557*** (0.2760)	-0.9404* (0.4629)	0.7024** (0.2271)	0.6370* (0.2948)	0.5384† (0.3183)	0.6356* (0.3074)
Conflict in COA: Low	0.8090*** (0.2081)	0.6311** (0.2331)	0.6434* (0.3074)	0.5107* (0.2159)	0.7943*** (0.2098)	0.8401*** (0.2052)	0.8116*** (0.2228)
<i>Preferences (H1)</i>							
War Ending	1.005*** (0.2171)	1.119*** (0.2534)	0.0500 (0.5178)	0.7508*** (0.2275)	0.9792*** (0.2258)	1.029*** (0.2166)	1.070*** (0.2203)
Conflict in COO: High	0.1081 (0.2783)	0.1351 (0.3065)	-0.0756 (0.4995)	-0.0732 (0.3195)	0.0673 (0.2844)	0.1663 (0.2674)	0.1174 (0.2887)
Conflict in COO: Low	0.3022 (0.2248)	0.4031† (0.2447)	-0.2861 (0.3633)	-0.0218 (0.2205)	0.2839 (0.2268)	0.3290 (0.2216)	0.3182 (0.2282)
GDP (COA)	-1.119† (0.5893)	-0.0338 (0.6626)	-5.076** (1.739)	-1.603*** (0.4424)	-1.145† (0.6044)	-1.112† (0.5925)	-1.232* (0.6089)
GDP (COO)	0.3890 (0.2842)	0.3528 (0.3167)	2.580*** (0.5272)	0.4607† (0.2699)	0.3042 (0.2925)	0.4304 (0.2839)	0.3365 (0.2949)
Civil Liberties Restrictions (COO)	-0.5345*** (0.1573)	-0.4655** (0.1703)	-0.3266 (0.3649)	-0.4875** (0.1591)	-0.5886*** (0.1649)	-0.5638*** (0.1602)	-0.5225*** (0.1574)
<i>Controls</i>							
Refugee Rights	-0.0094 (0.1695)	-0.0481 (0.2128)	0.0128 (0.2806)	0.0658 (0.1460)	0.0252 (0.1657)	-0.0172 (0.1916)	0.0258 (0.1780)
Conflict Duration	-0.0281 (0.0855)	0.0221 (0.1087)	0.2175 (0.1647)	0.0198 (0.0906)	-0.0361 (0.0881)	-0.0207 (0.0887)	-0.0285 (0.0885)
All controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Source	Both	WRS	USHR	Both	Both	Both	Both
Years	1980-2023	1980-2008	2009-2023	1980-2023	1980-2023	1980-2023	1980-2023
<i>Fixed-effects</i>							
Dyad	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>							
Observations	2,327	1,260	901	2,242	1,877	2,099	2,007
Pseudo R ²	0.79946	0.82553	0.82331	0.79223	0.79668	0.80194	0.80085

Clustered (Dyad) standard-errors in parentheses

Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1

All models: Poisson PML with log stock-at-risk offset, dyad and year fixed effects. Standard errors clustered by dyad. Model 3 restricts to the years coded from State Department reports. Model 4 excludes dyads linked in the literature to the refugee warrior hypothesis. Model 5 excludes dyads with asylum countries in Europe and North America. Model 6 restricts to dyads where refugees originally fled armed conflict (o_pers_nonwar == 0). Model 7 restricts to the top 3 asylum countries per origin-year by refugee stock. The checks that substitute UNHCR returns or the registered stock are reported in Table 18.

3.5.1 Distribution of Asylum Countries

Origins vary in the number of asylum destinations they appear in, because the sample construction rule includes asylum countries for each origin until the cumulative displaced population reaches 75% of the origin’s total. Figure 4 shows the distribution: most origins are concentrated in a small number of hosts, while a subset are dispersed across many (e.g., Ukraine appears in ten asylum countries). Table 30 lists the origins that appear in more than three asylum countries.

Because more-dispersed origins contribute more dyads, they receive greater implicit weight under equal dyad-year weighting, which could shift the estimand if peripheral hosts differ systematically from major ones. To assess sensitivity to this feature, I restrict the sample to the top three asylum countries per origin-year (preserving the full data for most origins), reported as Model “Top.3” in Table 29. The main findings are robust to this restriction.

Table 30: Origins with more than three asylum countries in the analytic sample

Origin (ISO3)	No. asylum countries	Asylum countries (ISO3)
UKR	10	CZE, DEU, ESP, GBR, ITA, NLD, POL, RUS, SVK, USA
YEM	10	CAN, DEU, DJI, EGY, JOR, MYS, NLD, SDN, SOM, USA
COD	7	BDI, COG, RWA, TZA, UGA, ZAF, ZMB
GEO	7	AUT, DEU, FRA, GBR, ITA, UKR, USA
IRN	7	DEU, GBR, IRQ, NLD, SWE, TUR, USA
BGD	6	CAN, FRA, IND, ITA, USA, ZAF
CIV	6	AGO, FRA, GHA, GIN, ITA, LBR
GIN	6	AGO, BEL, DEU, FRA, ITA, USA
PAK	6	AFG, CAN, DEU, GBR, ITA, USA
SRB	6	AUT, CHE, DEU, FRA, GBR, ITA
BIH	5	DEU, HRV, SRB, SWE, USA
BLR	5	DEU, FRA, POL, SWE, USA
COG	5	COD, FRA, GAB, USA, ZAF
EGY	5	AUS, CAN, DEU, ITA, USA
LBN	5	CAN, DEU, SWE, SYR, USA
MAR	5	AUT, DEU, ESP, FRA, ITA
ROU	5	CAN, DEU, HUN, SWE, USA
TJK	5	AFG, KGZ, RUS, TKM, UZB
ALB	4	DEU, FRA, GBR, USA
CHL	4	CAN, MEX, SWE, USA
CMR	4	DEU, NGA, TCD, USA
CUB	4	BRA, MEX, URY, USA
GHA	4	DEU, GBR, ITA, TGO
IRQ	4	DEU, IRN, JOR, SYR
KEN	4	CAN, ETH, USA, ZAF
NGA	4	CMR, DEU, ITA, NER
PER	4	CAN, ESP, ITA, USA
POL	4	CAN, DEU, SWE, USA
RUS	4	AUT, DEU, FRA, USA
SDN	4	ETH, SSD, TCD, UGA
SEN	4	GMB, GNB, ITA, MRT
TUR	4	CHE, DEU, FRA, IRQ
VEN	4	BRA, COL, ESP, PER

3.6 Alternative Specifications

Table 31 varies the model specification. Model 1 reproduces the baseline Poisson model. Model 2 uses OLS with log returns as the dependent variable. Model 3 clusters standard errors two ways, by origin country and asylum country, to account for correlation across dyads sharing a common origin or destination rather than only within dyads. Model 4 estimates a negative binomial model, which relaxes the Poisson equidispersion

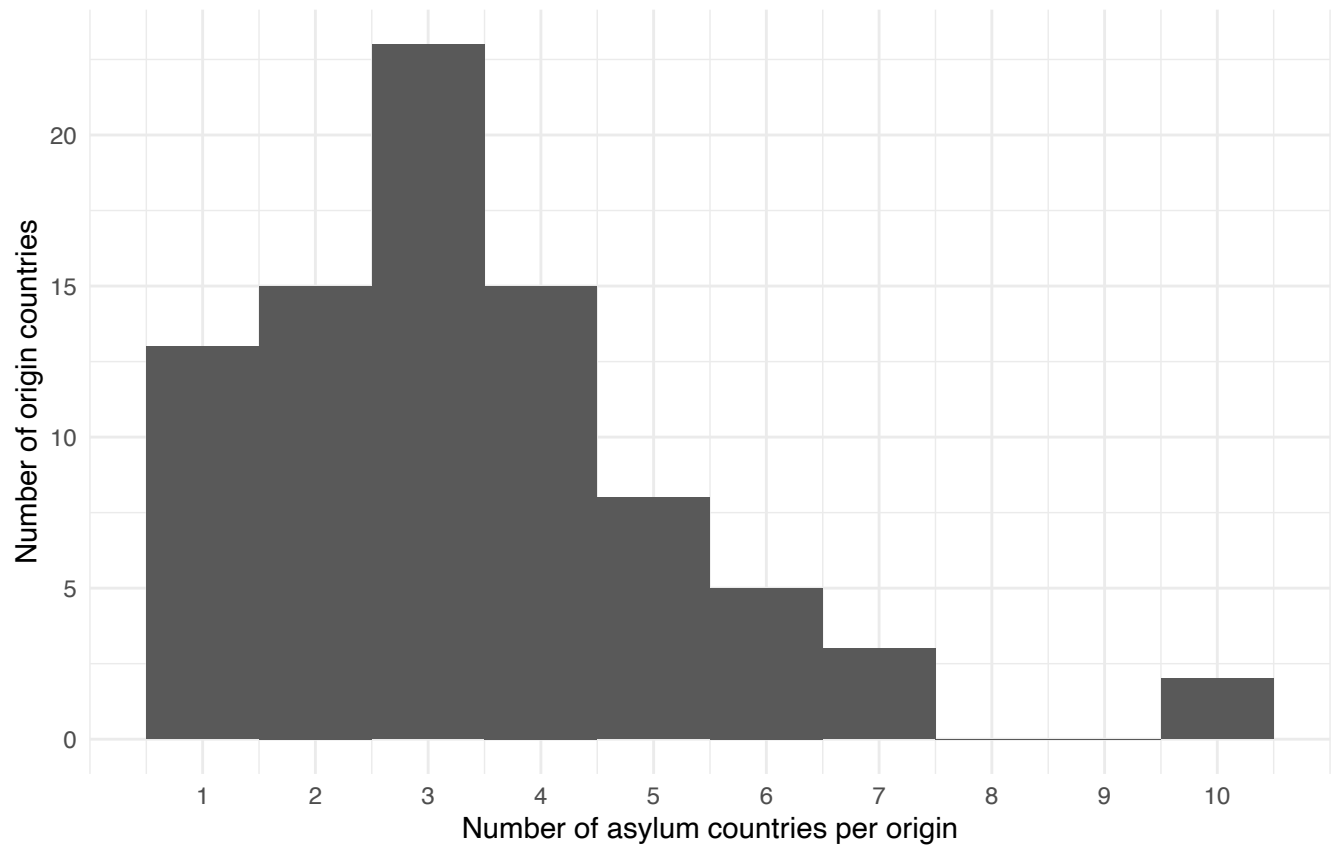


Figure 4: Histogram of the number of unique asylum countries per origin

assumption. Model 5 introduces origin-by-year fixed effects, which absorb all time-varying origin-country confounders (including origin-country conflict, ethnic status, and conflict years) and identify asylum-side variables from variation across hosts within the same origin-year.

The results hold across all five specifications. Severe forced return is positive and significant throughout, and asylum-country conflict retains its sign everywhere, with reduced precision under OLS and negative binomial (Models 2 and 4). Coefficients are not comparable in magnitude across estimators: OLS is on the log scale and the negative binomial has a different link, so the 4.57 and 2.73 reported there should be read for sign and significance rather than as rate ratios. Two-way clustering leaves the estimates unchanged to two decimal places and only slightly alters precision, indicating that correlation across dyads sharing an origin or a host is not inflating the baseline standard errors.

The origin-by-year fixed effects model (Model 5) recovers the asylum-side baseline, confirming that the main asylum-side patterns are not driven by unobserved time-varying origin-country shocks. These fixed effects necessarily absorb every origin-year variable, including the post-war window, origin conflict, origin income, and civil liberties, so the column identifies the stay-constraint channels alone and is silent on H1. Severe forced return is larger there (1.34) than in the baseline.

3.6.1 Pooled Specification

The within-dyad PPML estimates identify effects from year-to-year variation conditional on dyad fixed effects. I complement these with a pooled PPML specification that retains year fixed effects but omits dyad fixed effects, allowing identification from both within- and between-dyad variation. Log dyadic distance, absorbed by dyad fixed effects in the within specification, is added here as a standard gravity control. A further motivation for the pooled specification is sample coverage: two-way fixed-effects PPML restricts identification to dyads with sufficient within-cluster variation in the outcome, dropping a substantial share of the panel. The pooled specification draws on the full dataset, allowing me to assess whether the principal associations hold across the broader population of dyads.

The principal push channels hold under pooled identification: severe forced return is 1.09 in the full pooled specification against 1.08 within dyads, and the post-war window is larger (1.31 against 1.01). The main difference is that, when variation across cases is included, both income terms become considerably more precisely estimated and origin income switches sign, entering negative. This is the expected consequence of dropping the dyad fixed effects: between-dyad differences in income are correlated with a great deal else, which is why the paper reports the within-dyad estimates as its main results and this specification as a check on coverage rather than as an alternative to them.

3.7 Influence of the Largest Cases

The empirical distribution of refugee return is heavily skewed: the top five percent of dyad-years account for 88.5 percent of all returns. Because PPML identification is weighted by event counts, this concentration raises the question of whether the main estimates are carried by a small number of canonical episodes. Table 33 addresses this in two ways: by re-estimating the baseline after dropping high-mass cases, and by re-estimating on the top ten dyads only.

By and large, the patterns hold, and the coercion result strengthens rather than weakens. Severe forced return is larger in every drop-cases column than in the full sample—1.44 dropping the five largest dyads, 1.53 dropping the top one percent of dyad-years, and 1.45 dropping the top five percent, against 1.08 in the baseline—and it remains significant at the 0.001 level throughout. Within the top ten dyads alone it is smaller (0.86) and only marginally significant, on 275 dyad-years.

The post-war window behaves differently, and the difference is informative. It survives every exclusion but attenuates as the largest cases are removed, from 1.01 in the full sample to 0.84 and 0.92 in the first two columns and 0.42 when the top five percent of dyad-years are dropped. Low-intensity asylum-country conflict does not survive at all once high-mass cases are excluded, falling from 0.81 to statistically indistinguishable from zero, while entering strongly within the top ten dyads (1.24). The picture is therefore that coercion predicts return throughout the distribution, whereas the return that follows peace, and the return that

Table 31: Robustness: Alternative Specifications

Dep. Var.:	Total Repatriation Main	log(trepat+1) OLS	Two-way.CL	Total Repatriation Neg.Binom	Origin.x.Year
Model:	(1) Poisson	(2) OLS	(3) Poisson	(4) Neg. Bin.	(5) Poisson
<i>Variables</i>					
<i>Stay constraint (H2)</i>					
Forced Return: Severe	1.084*** (0.1914)	4.575*** (0.4150)	1.084*** (0.1284)	2.728*** (0.5082)	1.338*** (0.3375)
Forced Return: Moderate	0.4727* (0.2265)	1.633*** (0.2903)	0.4727† (0.2479)	1.223* (0.5588)	1.364*** (0.1628)
Conflict in COA: High	0.5822† (0.3063)	0.5423 (0.4686)	0.5822* (0.2651)	0.7692 (0.5065)	0.6685 (0.5082)
Conflict in COA: Low	0.8090*** (0.2081)	0.3911 (0.2974)	0.8090*** (0.1888)	0.2315 (0.4050)	0.9703*** (0.1766)
<i>Preferences (H1)</i>					
War Ending	1.005*** (0.2171)	0.8103* (0.3450)	1.005*** (0.2701)	0.8693* (0.4155)	
Conflict in COO: High	0.1081 (0.2783)	-0.6553 (0.4260)	0.1081 (0.2743)	0.5653 (0.5189)	-0.5803 (0.7794)
Conflict in COO: Low	0.3022 (0.2248)	-0.2280 (0.2663)	0.3022 (0.2176)	-0.1331 (0.3061)	0.0528 (0.4133)
GDP (COA)	-1.119† (0.5893)	-0.8760 (0.5728)	-1.119† (0.6589)	-0.6039 (1.043)	-2.142† (1.114)
GDP (COO)	0.3890 (0.2842)	0.3117 (0.2617)	0.3890 (0.2978)	0.2230 (0.5692)	
Civil Liberties Restrictions (COO)	-0.5345*** (0.1573)	-0.4916** (0.1757)	-0.5345*** (0.1545)	-0.4886 (0.3297)	
<i>Controls</i>					
Refugee Rights	-0.0094 (0.1695)	0.2152 (0.2017)	-0.0094 (0.1873)	0.3105 (0.2927)	0.0959 (0.1621)
Conflict Duration	-0.0281 (0.0855)	0.2777* (0.1322)	-0.0281 (0.0931)	0.4669* (0.2085)	0.0901 (0.1841)
Log Stock at Risk		0.7881*** (0.1037)			
Controls	Yes	Yes	Yes	Yes	Yes
Estimator	Poisson	OLS	Poisson	Neg. Binom.	Poisson
Clustering	Dyad	Dyad	Origin + Asylum	Dyad	Dyad
DV	trepat	log(trepat+1)	trepat	trepat	trepat
Fixed-effects					
Dyad	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	
coo_iso-Year					Yes
<i>Fit statistics</i>					
Observations	2,327	3,913	2,327	2,327	1,397
Pseudo R ²	0.79946	0.15281	0.79946	0.03790	0.92991

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

All models include dyad fixed effects. Standard errors in parentheses. Model 2 uses OLS with log returns as DV. Model 3 clusters by origin and asylum country. Model 4 uses negative binomial. Model 5 adds origin-by-year fixed effects; origin-side coefficients absorbed by these FE are automatically dropped.

Table 32: Pooled PPML: Identification Across and Within Dyads

Dependent Variable: Model:	Total Repatriation		
	(1)	(2)	(3)
<i>Stay constraint (H2)</i>			
Forced Return: Severe	1.011*** (0.2222)	1.088*** (0.1836)	1.547*** (0.2306)
Forced Return: Moderate	0.4167† (0.2231)	0.4396* (0.1940)	-0.0420 (0.2761)
Conflict in COA: High	1.040*** (0.2993)	0.5571† (0.3004)	0.4739 (0.3487)
Conflict in COA: Low	0.8459*** (0.2051)	0.7154*** (0.1724)	0.3466 (0.2286)
<i>Preferences (H1)</i>			
War Ending		1.311*** (0.2196)	1.239*** (0.3166)
Conflict in COO: High	0.1202 (0.3085)	0.2198 (0.2131)	0.1251 (0.2786)
Conflict in COO: Low	0.5080† (0.2926)	0.2451 (0.2162)	0.2484 (0.2580)
GDP (COA)		-0.8440*** (0.1757)	-0.8153*** (0.2164)
GDP (COO)		-0.4982*** (0.1415)	-0.4106* (0.1695)
Civil Liberties Restrictions (COO)	-0.2908* (0.1429)	-0.4830*** (0.0923)	-0.2488* (0.1147)
Repatriation Assistance			0.4155*** (0.1191)
<i>Controls</i>			
Refugee Rights		0.0632 (0.0710)	0.0834 (0.0698)
Conflict Duration		0.1290 (0.0784)	0.0739 (0.0880)
Log Distance	0.1091 (0.1551)	0.0931 (0.1223)	0.1284 (0.1479)
Fixed-effects			
Year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	4,302	3,880	2,286
Squared Correlation	0.46084	0.66053	0.69158
Pseudo R ²	0.65710	0.74349	0.71329

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

accompanies low-intensity conflict in the host, are concentrated in the major episodes. This is consistent with the exposure argument in the main text: the largest dyads are where the post-war windows with substantial populations at risk occur.

Table 33: Robustness: Concentration of Returns

Dependent Variable:	Total Repatriation				
Model:	Full (1)	Drop top 5 dyads (2)	Drop top 1% yrs (3)	Drop top 5% yrs (4)	Top 10 dyads (5)
<i>Stay constraint (H2)</i>					
Forced Return: Severe	1.084*** (0.1914)	1.438*** (0.2470)	1.529*** (0.1854)	1.446*** (0.3406)	0.8551* (0.4151)
Forced Return: Moderate	0.4727* (0.2265)	0.1372 (0.1854)	0.4172* (0.1744)	0.5698** (0.2140)	0.6822 (0.4316)
Conflict in COA: High	0.5822† (0.3063)	0.5196* (0.2553)	0.5021* (0.2386)	0.3513 (0.2478)	0.4874 (0.6493)
Conflict in COA: Low	0.8090*** (0.2081)	0.2708 (0.2563)	0.2700 (0.2510)	0.0107 (0.1738)	1.245*** (0.3691)
<i>Preferences (H1)</i>					
War Ending	1.005*** (0.2171)	0.8424** (0.2576)	0.9201*** (0.2222)	0.4155* (0.1851)	0.8669* (0.3962)
Conflict in COO: High	0.1081 (0.2783)	0.0803 (0.3083)	0.0599 (0.2824)	-0.5251* (0.2651)	-0.7082*** (0.1758)
Conflict in COO: Low	0.3022 (0.2248)	0.1654 (0.2343)	0.2807 (0.2050)	-0.2465 (0.1663)	-0.3354 (0.2651)
GDP (COA)	-1.119† (0.5893)	-0.5185 (0.5867)	-0.6247 (0.4929)	0.1105 (0.5732)	-3.488* (1.394)
GDP (COO)	0.3890 (0.2842)	0.5820† (0.3004)	0.3671 (0.2744)	0.5972** (0.2223)	1.630† (0.8552)
Civil Liberties Restrictions (COO)	-0.5345*** (0.1573)	-0.4130* (0.1811)	-0.5033** (0.1802)	-0.2889 (0.1825)	-0.8905** (0.2766)
<i>Controls</i>					
Refugee Rights	-0.0094 (0.1695)	0.0662 (0.1493)	0.1278 (0.1459)	0.0242 (0.1277)	0.3855* (0.1795)
Conflict Duration	-0.0281 (0.0855)	0.0423 (0.0962)	0.0153 (0.0878)	0.0713 (0.0943)	0.3619 (0.3147)
<i>Fixed-effects</i>					
Dyad	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	2,327	2,199	2,289	2,081	275
Squared Correlation	0.75658	0.61893	0.59267	0.33232	0.88729
Pseudo R ²	0.79946	0.67281	0.64780	0.44166	0.87270

Clustered (Dyad) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

3.8 Aggregate Attribution

The aggregate attribution in the main text asks a different question from the coefficient estimates above. Rather than estimating the association between each factor and return, it measures how much predicted return disappears when that factor is removed where it is observed. These shares therefore depend both on the estimated coefficients and on the distribution of exposure across dyads. I report uncertainty from two sources: coefficient draws that hold the estimation sample fixed, and a dyad-cluster bootstrap that resamples dyads and refits the model.

Table 34: Attributable share of all observed return, with two 95 per cent intervals. The coefficient-draw interval takes 2,000 draws from the estimated sampling distribution of the coefficients and recomputes the shares, holding the estimation sample fixed; this is the procedure behind the footnote in the main text. The dyad-cluster bootstrap resamples dyads with replacement and refits the baseline on each resample, so it also carries uncertainty about which dyads the panel contains; 1,000 of 1,000 resamples converged. The shares must not be added: some dyad-years carry two of the three factors.

Factor	Share of all return	Coefficient draws	Dyad-cluster bootstrap
Severe forced return	25.8%	[12.4%, 39.3%]	[9.4%, 47.5%]
Post-war window (1–3 yrs, at peace)	12.6%	[6.1%, 20.6%]	[3.9%, 22.9%]
Asylum conflict: high	6.3%	[-0.2%, 17.7%]	[-4.9%, 17.3%]

Table 35: The aggregate attribution with the Afghan dyads removed. The baseline is refitted on each subsample. The coefficient on severe forced return is reported beside the shares to show that the two quantities behave differently.

Sample	Dyad-years	Severe forced return	Post-war window	Coefficient on severe
Full estimation sample	2,327	25.8%	12.6%	1.08***
Excluding Afghanistan–Iran	2,304	13.4%	16.7%	0.95***
Excluding Afghanistan–Pakistan	2,304	31.0%	9.4%	1.32***
Excluding both Afghan dyads	2,281	14.5%	12.9%	1.17***

Severe forced return accounts for 25.8 per cent of observed return, compared with 12.6 per cent for the post-war window. Holding the panel fixed, the ratio of these shares is 2.04, with a 95 per cent interval of [0.89, 4.56]. The coercion share exceeds the post-war share in 95 per cent of coefficient draws. This is the uncertainty reported in the main text.

The dyad-cluster bootstrap is less precise. Resampling dyads gives a ratio interval of [0.49, 10.13], with coercion exceeding the post-war window in 77 per cent of resamples. The ordering is therefore well supported conditional on the observed panel, but less certain once variation in which dyads enter the panel is incorporated.

Table 35 shows where this additional uncertainty comes from. Excluding both Afghan dyads raises the coefficient on severe forced return from 1.08*** to 1.17***, but reduces its attributable share from 25.8% to 14.5%. Excluding Afghanistan–Iran alone reverses the ordering of the two shares (13.4% versus 16.7%).

This distinction influences interpretation. The coefficient on severe forced return is not driven by the Afghan cases. The aggregate share is more sensitive to them because Afghanistan–Iran and Afghanistan–Pakistan are among the largest return episodes in the data. They should not be excluded from a decomposition of global return, but the result should be read as a statement about the composition of observed global return rather than as a general property of refugee dyads.

3.8.1 Pooled Specification for the Aggregate Attribution

The baseline estimates use dyad fixed effects, which are appropriate for estimating how changes within a dyad are associated with return. They are less well suited to the aggregate counterfactual. Fixed-effects Poisson drops dyads with no within-dyad variation in return, and dyad fixed effects also absorb between-dyad differences in exposure that are part of the aggregate quantity of interest.

I therefore also calculate the attribution using a pooled specification without dyad or year fixed effects, adding distance, contiguity, and the log stock at risk as controls. This is not an alternative identification strategy for the main coefficients. Its purpose is to calculate the aggregate decomposition on a broader sample and without absorbing the between-dyad component of exposure. It differs from Table 32, which retains year fixed effects and is reported as a robustness check on the coefficients.

Table 36: What each specification covers, against the 5,131-dyad-year estimation frame of 285 dyads. The two cover almost identical shares of recorded return, because the dyads the fixed effects drop are those that never record any; they differ in refugee stock-years, because those dyads hold refugees who do not return.

Specification	Dyad-years	Dyads	Share of frame return	Share of frame stock-years
Baseline: dyad and year fixed effects	2,327	110	77.7%	58.7%
Pooled, no fixed effects	3,880	255	77.8%	73.3%

Table 37: Attributable shares under the pooled specification, with the baseline shares beside them and the two 95 per cent intervals computed exactly as in Table 34. Both intervals belong to the pooled column; 1,000 of 1,000 resamples converged. The shares must not be added, since some dyad-years carry two of the three factors.

Factor	Baseline share	Pooled share	Coefficient draws	Dyad-cluster bootstrap
Severe forced return	25.8%	24.7%	[15.2%, 33.8%]	[9.1%, 47.5%]
Post-war window (1–3 yrs, at peace)	12.6%	14.3%	[6.7%, 25.0%]	[4.3%, 25.2%]
Asylum conflict: high	6.3%	5.9%	[-1.5%, 15.6%]	[-1.1%, 15.4%]

The aggregate shares are very similar across the two specifications. Severe forced return accounts for 24.7 per cent of predicted return in the pooled model, compared with 25.8 per cent under dyad fixed effects. The corresponding figures are 14.3 and 12.6 per cent for the post-war window, and 5.9 and 6.3 per cent for high-intensity asylum conflict. The decomposition is therefore not sensitive to the fixed-effects sample restriction.

The pooled coefficients also reinforce the interpretation of the aggregate result. The post-war coefficient is 1.26 (SE 0.26), compared with 1.01 (SE 0.19) for severe forced return. The larger aggregate share for coercion therefore does not reflect a larger proportional effect. It reflects greater exposure: severe forced return occurs in dyad-years containing substantially larger refugee populations.

Uncertainty is again greater when dyads are resampled. Holding the panel fixed, the ratio of the coercion and post-war shares has a 95 per cent interval of [0.78, 4.30], with coercion larger in 90 per cent of draws. The dyad-cluster bootstrap widens the interval to [0.45, 9.10], with coercion larger in 70 per cent of resamples. The pooled specification therefore leaves the substantive attribution almost unchanged, while confirming that the precise ordering of the two aggregate shares is sensitive to the composition of the panel.

3.9 Leave-One-Out Analyses

Figure 5 presents leave-one-out analyses that iteratively exclude each origin country, asylum country, origin region, and asylum region and re-estimate the main model. Coefficients are stable across most exclusions; Table 38 reports the cases where excluding a region produces a sign change or an absolute coefficient shift exceeding one.

Severe forced return does not appear in that table at all: no regional exclusion reverses its sign or moves it by more than one. Table 39 makes the same point at the level of individual dyads. The largest single-dyad shift is 0.32—dropping Afghanistan–Pakistan, which raises the coefficient from 1.08 to 1.41—and every re-estimate remains significant at the 0.001 level. The central result is therefore not carried by any one origin, host, or region.

The flagged shifts fall on coefficients that are at most marginally significant in the main specification. High-intensity asylum-country conflict, significant only at the ten percent level in the baseline (0.58), turns negative when Sub-Saharan Africa is excluded as either an origin or an asylum region, although the re-estimated coefficients are not themselves significant; at the dyad level it is most sensitive to Rwanda–DR Congo and

the two Afghan dyads. Moderate forced return behaves similarly when Asia and MENA are dropped. Origin-country conflict changes sign at both intensities under several exclusions, which is unsurprising given that it is indistinguishable from zero to begin with. None of these alter the substantive interpretation, but they do indicate that the asylum-conflict channel is the least robust of the push results—which is consistent with what the main text concludes about it on other grounds.

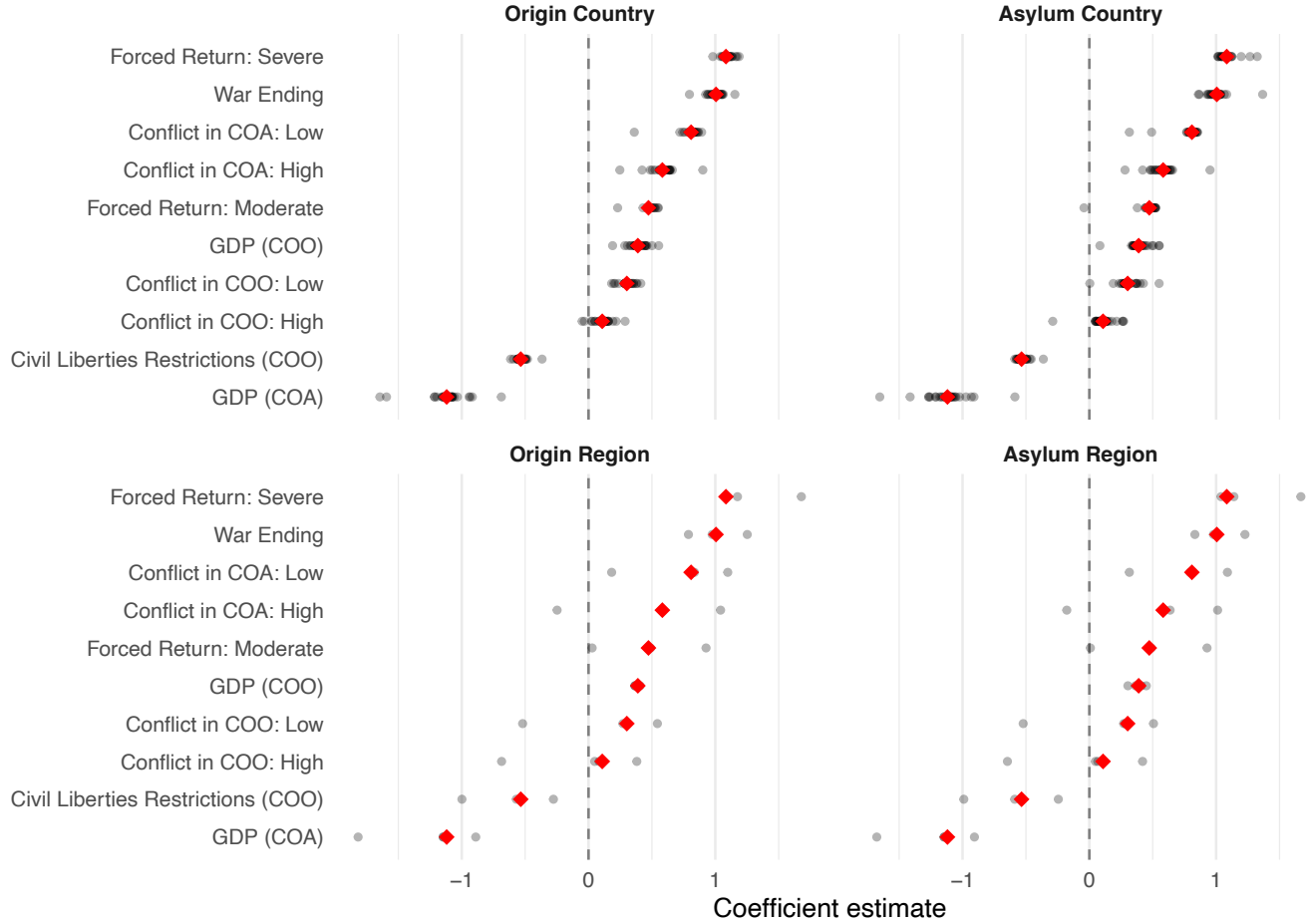


Figure 5: Coefficient stability across leave-one-out analyses

Table 38: Influential Regions: Coefficients with Sign Changes or Large Shifts

Variable	Region Dropped	Drop Type	Main Est.	New Est.	Change
Conflict in COA: High	Sub-Saharan Africa	Origin Region	0.582+	-0.248	0.830
Conflict in COA: High	Sub-Saharan Africa	Asylum Region	0.582+	-0.178	0.760
Conflict in COO: High	Sub-Saharan Africa	Origin Region	0.108	-0.685	0.793
Conflict in COO: High	Sub-Saharan Africa	Asylum Region	0.108	-0.646	0.754
Conflict in COO: Low	Sub-Saharan Africa	Asylum Region	0.302	-0.522+	0.824
Conflict in COO: Low	Sub-Saharan Africa	Origin Region	0.302	-0.520+	0.822

Note: Table shows regions whose exclusion causes sign changes or absolute coefficient shifts above 1. Stars indicate significance of each coefficient: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Bold rows indicate sign reversals where the new coefficient remains statistically significant at $p < 0.05$; sign flips on coefficients that fall back to non-significance after exclusion are shown but not bolded.

4 Alternative Explanations

4.1 Social and Ethnic Ties Between Origin and Asylum

The alternative account this section addresses is that return is driven by social and ethnic proximity between origin and asylum rather than by conditions in either country. Refugees who share a language, a religion, or an ethnic kin group with the host population may be more likely to stay. These measures are almost entirely time-invariant, so they can speak to which dyads refugees return from, but not to when return happens.

I merge three external datasets to the GRR dyads. Common official language, common spoken language, colonial relationship, and ever-same-country come from CEPII GeoDist (Mayer and Zignago 2011). Religious similarity is built from the COW World Religion Project (Maoz and Henderson 2013), following Disdier and Mayer’s (2007) construction: the probability that one person drawn at random from each country adheres to the same tradition, which is the inner product of the two adherence-share vectors. Two versions are used, one over sixteen broad traditions and one over thirty-one denominational categories, the latter separating Sunni from Shia and Catholic from Protestant. Transnational ethnic kinship comes from EPR-TEK (Vogt et al. 2015): whether any ethnic group spans the origin and the host, and what share of each population it accounts for.

Because the measures are fixed within a dyad, the dyad fixed effects in the main specification absorb them entirely. The tests below use origin-by-year fixed effects instead, which identify the tie coefficients from variation across the hosts of one origin in one year: among refugees who left the same country at the same moment, were those who went somewhere culturally closer less likely to return? Two limitations follow. The design requires at least two hosts of the same origin observed in the same year, which reduces the sample to roughly 1,780 dyad-years from 2,327. And the measures are collinear with one another, so each enters on its own, with the summary index as the closest thing to a joint test.

Ties do predict return. Of the 8 measures, 4 are significant at the five percent level. A shared spoken language is associated with less return (-0.52^*), as are a larger kin share of the origin population (-0.26^*) and the summary index of all ties taken together (-0.30^*). These run in the direction the account predicts: refugees who went somewhere culturally closer are less likely to go home. Religious similarity runs the other way and is associated with more return (0.25^*), although the denominational version is not significant.

Severe forced return is unchanged in every column, so cultural proximity predicts return alongside coercion rather than instead of it.

4.1.1 Ethnic Political Status

A separate question concerns the political standing of the refugees’ own ethnic group: whether its position at home predicts return, and whether the position of its co-ethnics in the host country does. A group that gains power at home might be more likely to go back, and one whose kin hold power in the host might be more likely to stay. I test both with the EPR data described above.

Refugees from politically dominant groups return at higher rates than those from excluded ones. The standing of co-ethnics in the host has no bearing on return: neither EPR-TEK indicator is distinguishable from zero (-0.30 for kin in power, 0.20 for kin excluded).

Table 39: Most influential dyads: top five single-dyad drops by absolute coefficient change, per variable.

Variable	Dyad	Main Est.	New Est.	Change
Conflict in COA: High	AFG-PAK	0.582+	0.952***	0.370
Conflict in COA: High	RWA-COD	0.582+	0.300	0.282
Conflict in COA: High	AFG-IRN	0.582+	0.333	0.249
Conflict in COA: High	SOM-ETH	0.582+	0.493	0.089
Conflict in COA: High	MOZ-MWI	0.582+	0.520+	0.063
Conflict in COA: Low	AFG-IRN	0.809***	0.282	0.527
Conflict in COA: Low	AFG-PAK	0.809***	0.492*	0.317
Conflict in COA: Low	KHM-THA	0.809***	0.892***	0.084
Conflict in COA: Low	MMR-THA	0.809***	0.731***	0.078
Conflict in COA: Low	RWA-COD	0.809***	0.763***	0.046
Forced Return: Severe	AFG-PAK	1.084***	1.324***	0.240
Forced Return: Severe	RWA-COD	1.084***	1.287***	0.203
Forced Return: Severe	AFG-IRN	1.084***	0.955***	0.129
Forced Return: Severe	BDI-TZA	1.084***	1.186***	0.102
Forced Return: Severe	MMR-BGD	1.084***	1.020***	0.064
Forced Return: Moderate	AFG-IRN	0.473*	-0.031	0.504
Forced Return: Moderate	AFG-PAK	0.473*	0.379*	0.094
Forced Return: Moderate	BDI-TZA	0.473*	0.528*	0.056
Forced Return: Moderate	ETH-SOM	0.473*	0.521*	0.048
Forced Return: Moderate	RWA-COD	0.473*	0.516*	0.043
GDP (COA)	AFG-IRN	-1.119+	-0.570	0.549
GDP (COA)	KHM-THA	-1.119+	-1.650***	0.531
GDP (COA)	UGA-COD	-1.119+	-1.384*	0.265
GDP (COA)	UGA-SDN	-1.119+	-1.336*	0.217
GDP (COA)	MMR-BGD	-1.119+	-0.931	0.188
War Ending	AFG-IRN	1.005***	1.223***	0.218
War Ending	MOZ-MWI	1.005***	0.860***	0.145
War Ending	AFG-PAK	1.005***	0.868***	0.138
War Ending	IRQ-IRN-IRN	1.005***	1.080***	0.074
War Ending	LBR-GIN	1.005***	0.932***	0.073
Civil Liberties Restrictions (COO)	AFG-IRN	-0.535***	-0.421**	0.114
Civil Liberties Restrictions (COO)	AFG-PAK	-0.535***	-0.458**	0.076
Civil Liberties Restrictions (COO)	ETH-ERI-SDN	-0.535***	-0.601***	0.067
Civil Liberties Restrictions (COO)	MMR-THA	-0.535***	-0.476**	0.059
Civil Liberties Restrictions (COO)	IRQ-IRN-IRN	-0.535***	-0.476**	0.058
Conflict in COO: High	RWA-COD	0.108	-0.136	0.244
Conflict in COO: High	AFG-PAK	0.108	0.264	0.156
Conflict in COO: High	ETH-SOM	0.108	0.262	0.154
Conflict in COO: High	AGO-ZMB	0.108	0.196	0.088
Conflict in COO: High	SDN-SSD-UGA	0.108	0.192	0.084
Conflict in COO: Low	RWA-COD	0.302	0.101	0.201
Conflict in COO: Low	AFG-IRN	0.302	0.493*	0.190
Conflict in COO: Low	ETH-SOM	0.302	0.426+	0.124
Conflict in COO: Low	AFG-PAK	0.302	0.189	0.113
Conflict in COO: Low	MOZ-MWI	0.302	0.372	0.069
GDP (COO)	KHM-THA	0.389	0.192	0.197
GDP (COO)	AFG-PAK	0.389	0.551+	0.162
GDP (COO)	MMR-BGD	0.389	0.550*	0.161
GDP (COO)	SOM-ETH	0.389	0.495+	0.106
GDP (COO)	MMR-THA	0.389	0.286	0.103

Note: Table reports, for each key variable, the five dyads whose exclusion produces the largest absolute change in the coefficient. Stars indicate significance of each coefficient: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001. Bold rows indicate sign reversals where the new coefficient remains statistically significant at p<0.05.

Table 40: Ethnic political status and refugee return

Model:	Total Repatriation	
	Origin status (EPR) (1)	Kin in host (EPR-TEK) (2)
<i>Variables</i>		
<i>Stay constraint (H2)</i>		
Forced Return: Severe	1.323*** (0.2285)	1.109*** (0.1967)
Forced Return: Moderate	0.5053* (0.2254)	0.5234* (0.2421)
Conflict in COA: High	0.4728 (0.3632)	0.6221* (0.2990)
Conflict in COA: Low	1.003*** (0.2034)	0.8401*** (0.2157)
<i>Preferences (H1)</i>		
War Ending	0.6756*** (0.1900)	1.002*** (0.2167)
Conflict in COO: High	0.2447 (0.2743)	0.0921 (0.2835)
Conflict in COO: Low	0.5739** (0.2214)	0.2898 (0.2272)
GDP (COA)	-2.216*** (0.5127)	-1.125† (0.5854)
GDP (COO)	0.2110 (0.2293)	0.3821 (0.2810)
Civil Liberties Restrictions (COO)	-0.4714** (0.1507)	-0.5414*** (0.1629)
<i>Controls</i>		
Refugee Rights	0.0743 (0.1925)	-0.0014 (0.1714)
Conflict Duration	-0.1860† (0.0949)	-0.0227 (0.0868)
Ethnic Status: Discriminated (COO)	-0.9315* (0.4351)	
Ethnic Status: Junior Partner (COO)	-0.6585 (0.4613)	
<i>Ethnic political status</i>		
Ethnic Status: Monopoly (COO)	-1.510** (0.5303)	
Ethnic Status: Powerless (COO)	-0.6464 (0.4737)	
Ethnic Status: Self-Excluded (COO)	-2.632*** (0.6664)	
Ethnic Status: Senior Partner (COO)	-0.5271 (0.4068)	
Ethnic Status: State Collapse (COO)	-1.501*** (0.3988)	
Co-ethnic Kin in Power (COA)		-0.2981 (0.4500)
Co-ethnic Kin Excluded (COA)		0.2020 (0.3520)
All controls	Yes	Yes
<i>Fixed-effects</i>		
Dyad	Yes	Yes
Year	Yes	Yes
<i>Fit statistics</i>		
Observations	1,718	2,327
Pseudo R ²	0.81544	0.80062

Clustered (Dyad) standard-errors in parentheses

*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, †: 0.1*

Column 1 adds the ordinal EPR status of the displaced group in the origin, with Dominant omitted as the reference; EPR coverage is incomplete, which is why the sample is smaller. Column 2 adds the EPR-TEK indicators for whether the group's co-ethnics in the asylum country hold power or are excluded. Both carry the full specification of Table 2 of the paper. PPML with the log stock at risk as offset, dyad and year fixed effects, SEs clustered by dyad.

5 Extensions and Supporting Analyses

5.1 Forced-Return Policies: Predictors, Law, and UNHCR Involvement

The forced-return measure is hand-coded from the archival sources described above. This section examines it in three stages. It first asks when and where forced-return policies occur — which host governments impose them, and which refugee populations they are imposed on — by comparing dyad-years with and without coercion under progressively more demanding designs. It then compares the GRR measure of observed practice with an independently coded measure of statutory refugee protection, which provides an external check on the hand coding. It finally examines UNHCR involvement in the same dyad-years, measured using repatriation funding, recorded repatriation programmes, and tripartite agreements between origin states, asylum states, and UNHCR. Together these analyses describe the conditions under which coercion arises, validate the measure against independent sources, and set out the relationship between international involvement, coercion, and return.

5.1.1 When Do Governments Force Refugees to Return?

The main analysis treats forced-return policy as a constraint on refugees' ability to remain in asylum. That raises an upstream question: when do governments choose to impose that constraint? One possibility is that coercion reflects characteristics of the host government itself. More authoritarian governments, governments with weaker refugee protections, or governments under greater economic or security pressure may be more willing to force refugees out. A second possibility is that governments target particular kinds of refugee populations, for example groups that are socially distant, geographically close, concentrated in camps, or perceived as militarized. A third is that policy responds to the trajectory of the displacement crisis: violence in the origin may raise security concerns and the prospect of further arrivals, while governments may become less willing to accommodate refugees as exile persists.

I examine these possibilities by moving from broad comparisons to progressively narrower ones. I first ask which characteristics distinguish dyad-years with forced-return policies from those without them. I then compare different refugee populations within the same asylum country and year, holding the host government and the conditions it faces fixed. The corresponding origin-by-year comparison asks why refugees from the same country face coercion in one asylum state but not another. Finally, dyad and year fixed effects compare the same origin–asylum population to itself over time. None of these comparisons identifies a causal effect of the covariates. Their value is that they show which associations survive once increasingly large sets of differences across hosts, origins, and refugee populations are removed.

The analysis uses the full GRR policy panel rather than the smaller return-regression sample. I retain dyad-years with a coded forced-return measure and more than 1,000 refugees in the preceding year, excluding 1989 because no *World Refugee Survey* was published. The resulting sample contains 4,448 dyad-years across 272 dyads. Severe forced-return policy is uncommon, appearing in 150 dyad-years (3.4%); some form of forced-return policy appears in 441 (9.9%). Continuous covariates are standardized on this sample. Coverage varies for some external measures, so each model reports its own N .

Table 41: Bivariate correlates of forced-return policy. Each cell is a separate logit of the policy indicator on that correlate alone, with no controls and no fixed effects; standard errors in parentheses. Continuous measures are standardised on the policy sample, so coefficients are per standard deviation. These are descriptive associations and are not to be read as effects.

Correlate	Severe policy	Any policy	Dyad-years
Scale of the refugee population			
Refugee stock in this dyad (log, lagged)	1.35*** (0.10)	0.91*** (0.05)	4,448
Total refugees hosted by the asylum country (log)	0.83*** (0.10)	0.53*** (0.06)	4,448
Camp residence (ordinal)	0.48*** (0.08)	0.39*** (0.05)	4,425
Refugees reported militarized	1.53*** (0.30)	1.60*** (0.21)	4,360
Host government and host pressure			
GDP p.c. in asylum (log)	-0.53*** (0.09)	-0.47*** (0.05)	4,412
V-Dem polyarchy (host)	-0.41*** (0.09)	-0.43*** (0.05)	4,448
Freedom House political rights (host)	0.36*** (0.09)	0.42*** (0.05)	4,111
DWRAP refugee-rights index (host)	-0.12 (0.08)	-0.24*** (0.05)	4,285
Conflict in asylum: high	-0.42 (0.37)	0.45** (0.16)	4,448
Conflict in asylum: low	1.03*** (0.17)	0.70*** (0.11)	4,448
Origin-country conditions			
Conflict in origin: high	1.33*** (0.18)	1.02*** (0.12)	4,448
Conflict in origin: low	0.32 [†] (0.18)	0.28* (0.11)	4,448
Battle deaths in origin (log)	0.63*** (0.08)	0.43*** (0.05)	4,113
GDP p.c. in origin (log)	-1.05*** (0.10)	-0.59*** (0.06)	4,179
Civil-liberties restrictions in origin	0.72*** (0.11)	0.56*** (0.06)	4,403
Cumulative years of origin conflict (log)	0.67*** (0.08)	0.45*** (0.05)	4,448
Dyadic relationship and duration			
Distance origin-asylum (log)	-0.36*** (0.07)	-0.42*** (0.05)	4,399
Origin and asylum contiguous	1.67*** (0.24)	1.47*** (0.13)	4,399
Composite cultural-tie index	0.39*** (0.08)	0.44*** (0.05)	4,274
Years since the dyad entered the panel	-0.12 (0.09)	-0.13* (0.05)	4,448

Taken one at a time, almost every explanation looks plausible. 16 of the 20 correlates in Table 41 are significantly associated with severe policy at the five percent level. Coercion is more common in large refugee populations and in countries hosting more refugees overall; it is also associated with camps, poorer and less democratic hosts, active conflict and higher battle deaths in the origin, poorer and more repressive origins, proximity, and reported militarization. These associations cannot distinguish among the competing explanations because the cases bundle these characteristics together. Large refugee populations are disproportionately found near poor and violent origins, often in poorer neighboring hosts. The more informative question is therefore which relationships remain once those characteristics are put in competition.

Table 42: Four comparisons of increasing severity. Logit coefficients, standard errors clustered by dyad in parentheses. Column 1 and 2 are pooled with year effects. Column 3 compares different origin groups facing the same asylum government in the same year, so every host-level characteristic is absorbed by construction. Column 4 compares hosts of the same origin in the same year, so every origin-level characteristic is absorbed. Columns 5 and 6 compare a dyad to itself over time, so time-invariant dyadic measures drop mechanically. A dash means the design absorbs the variable; *sep.* means the coefficient is a report of a cell with no counter-examples rather than an estimate. All models converged.

Variable	(1) Pooled, severe	(2) Pooled, any	(3) Same host-year, severe	(4) Same origin-year, severe	(5) Same dyad, severe	(6) Same dyad, any
Scale of the refugee population						
Refugee stock in this dyad (log, lagged)	1.71*** (0.32)	0.71*** (0.17)	4.23*** (1.01)	2.20 [†] (1.32)	1.19** (0.40)	0.21 (0.22)
Total refugees hosted by the asylum country (log)	-0.05 (0.24)	0.15 (0.21)	–	-0.74 (1.19)	-0.29 (0.46)	0.38* (0.19)
Camp residence (ordinal)	0.34 (0.24)	0.19 (0.18)	0.63 (1.39)	-0.86 (0.68)	0.67 (0.69)	0.20 (0.28)
Host government and host pressure						
GDP p.c. in asylum (log)	0.18 (0.34)	0.63* (0.28)	–	3.52* (1.42)	-1.26 (1.63)	-1.80* (0.78)
V-Dem polyarchy (host)	0.94** (0.36)	0.46* (0.21)	–	-1.03 (1.26)	0.55 (0.77)	0.23 (0.55)
DWRAP refugee-rights index (host)	0.40* (0.19)	-0.02 (0.16)	–	1.04* (0.48)	0.64* (0.32)	0.14 (0.31)
Conflict in asylum: high	-0.84 (0.56)	0.23 (0.47)	–	-1.39 (1.94)	-0.18 (0.83)	0.00 (0.45)
Conflict in asylum: low	0.48 (0.35)	0.32 (0.36)	–	-0.57 (1.08)	0.53 (0.51)	-0.01 (0.36)
Origin-country conditions						
Conflict in origin: high	0.04 (0.45)	0.11 (0.31)	4.00 (2.46)	–	0.03 (0.86)	-0.04 (0.39)
Conflict in origin: low	0.03 (0.48)	0.09 (0.30)	0.91 (1.90)	–	0.30 (0.46)	0.24 (0.35)
GDP p.c. in origin (log)	-0.85** (0.30)	-0.22 (0.18)	-2.12* (0.90)	–	-0.91 (0.99)	-0.21 (0.50)
Civil-liberties restrictions in origin	-0.17 (0.24)	0.11 (0.22)	-4.70*** (1.34)	–	0.56 (0.39)	0.52* (0.21)
Cumulative years of origin conflict (log)	0.10 (0.24)	0.14 (0.13)	1.19 (1.23)	–	0.10 (0.37)	0.15 (0.21)
Dyadic relationship and duration						
Distance origin–asylum (log)	-0.09 (0.26)	-0.26 (0.18)	-7.16** (2.68)	-0.43 (0.57)	–	–
Origin and asylum contiguous	-0.01 (0.58)	1.21 [†] (0.73)	-4.57 [†] (2.48)	6.94* (2.88)	–	–
Composite cultural-tie index	0.19 (0.21)	0.20 (0.15)	-0.78 (0.95)	-1.84 (1.36)	–	–
Years since the dyad entered the panel	0.15 (0.36)	0.13 (0.23)	-1.84 [†] (1.07)	-3.71* (1.58)	–	–
Specification						
Fixed effects	Year	Year	Asylum × year	Origin × year	Dyad, year	Dyad, year
Dyad-years	3,622	3,828	225	285	917	1,977

One result is unusually consistent: the size of the particular refugee population. In the fully adjusted pooled model, a ten-fold increase in lagged refugee stock is associated with 8.10 times the odds of severe coercion. The same relationship appears when the comparison is restricted to different origin groups facing the same host government, and it remains when the same origin–asylum dyad is compared to itself over time. Refugee stock is the only variable in Table 42 that is positive and distinguishable from zero in the pooled, same-host, and within-dyad severe-policy specifications. The same-origin comparison is less precisely estimated because only 285 dyad-years identify the severe-policy coefficient, but the estimate is again positive and similar in magnitude (2.20[†], $p = 0.095$). For any forced-return policy, the same-origin estimate is clearly distinguishable from zero (1.24**, $p = 0.003$).

This is not simply a story about countries carrying a large refugee burden. When the asylum country’s total refugee population is entered by itself, it predicts severe coercion (0.86* per standard deviation). Once the size of the particular origin group is included, that association falls to -0.24 ($p = 0.26$), while the dyad-specific stock remains strong (1.58***). Governments therefore appear to respond more to the scale of the population against which policy is being made than to the number of refugees in the country as a whole.

The political variables do not yield an equally stable account. More democratic hosts are less coercive in the raw data, but the polyarchy coefficient reverses sign in the adjusted pooled model (0.94**) and is not distinguishable from zero in the same-origin or within-dyad severe-policy comparisons. This makes it difficult to interpret regime type as a simple explanation for which refugee population a government targets or when it escalates. The broad DWRAP refugee-rights index is more anomalous still. Its raw association is negative, but it enters positively after adjustment (0.40*) and remains positive in some of the more restrictive specifications. I do not interpret that reversal substantively. DWRAP combines several dimensions of national refugee law, while the outcome here is a dyad-specific practice. The next subsection therefore turns to the narrower statutory provision most directly relevant to forced return, the right to non-refoulement, which produces the expected negative relationship with coercive practice.

Origin-country violence presents a similar contrast between the raw pattern and the adjusted one. Severe coercion occurs in 1.8% of dyad-years with no recorded battle deaths in the origin and rises monotonically to 12.2% when deaths exceed ten thousand. Yet much of that gradient is accounted for by the refugee

population itself. The battle-death coefficient falls from 0.69** to 0.21 when stock is added, and it is not distinguishable from zero in the fully adjusted pooled, host-and-year, or within-dyad specifications. There is some evidence that hosts distinguish the war producing a particular refugee population from violence around them more generally: with host and year effects, battle deaths in the refugee group's own origin predict coercion while fighting in the host's other neighboring countries does not. But that relationship also disappears once stock and origin poverty are taken into account. The evidence therefore does not suggest that battlefield violence independently triggers severe policy.

Militarization also looks more important in the raw data than it does in the more restrictive comparisons. Severe coercion appears in 13.2% of dyad-years in which refugees are reported militarized, compared with 3.2% otherwise. The association weakens after adjustment and is not distinguishable from zero for severe policy in the same-origin or within-dyad specifications, either contemporaneously or at a one-year lag. Militarization continues to predict the broader category of any forced-return policy in some specifications, so it should not be treated as irrelevant. But there is little evidence that it explains escalation to severe coercion. This result also warrants caution because the militarization and forced-return measures are sometimes coded from the same annual archival narrative, making contemporaneous associations especially difficult to interpret.

Social and cultural ties provide little evidence for another intuitive account, that governments face greater political costs when expelling socially close populations. With year effects alone, 6 of 16 estimable tie coefficients are significant, and 6 of these are positive: culturally proximate groups are, if anything, more likely to face coercion. Much of this association is geographic. Once distance and contiguity are taken into account, and then the remaining observables are added, only 3 of 32 coefficients remain significant. In the origin-by-year and asylum-by-year comparisons, only 4 of 31 estimable coefficients do, none on the severe-policy margin. No measure of social ties behaves consistently across specifications. These results do not show that ties are irrelevant, but they provide no support for the idea that social proximity systematically protects a refugee population from coercion.

Table 43: Bilateral ties with forced-return POLICY as the outcome, not return. Each cell is a separate logit with one tie measure entered on its own alongside the stated conditioning set, standard errors clustered by dyad. The measures are almost time-invariant, so there are no dyad fixed effects; column 5 identifies the tie from variation across the hosts of one origin in one year and column 6 from variation across the origin groups in one host in one year. Column 3 adds distance and contiguity, column 4 adds the full pooled observable set. *sep.* marks a cell with no counter-examples rather than an estimate. Conclusions from Section 4.1, which asks which hosts refugees return from, do not transfer here.

Tie measure	(1) Year FE	(2) + region	(3) + geography	(4) + observables	(5) Origin $\times$ year	(6) Asylum $\times$ year
Severe forced-return policy						
Common official language	0.49 (0.54)	0.25 (0.58)	0.30 (0.64)	0.35 (0.39)	1.15 (1.60)	5.25 (4.93)
Common spoken language	0.22 (0.56)	0.07 (0.57)	0.10 (0.66)	0.19 (0.38)	0.49 (1.41)	-0.27 (1.26)
Religious similarity, broad tradition	0.36 (0.30)	0.19 (0.25)	0.14 (0.28)	0.27 (0.22)	-2.10 (1.30)	-1.45 (1.07)
Religious similarity, denominational	-0.06 (0.12)	-0.46* (0.22)	-0.65** (0.22)	-0.38 (0.28)	-2.10 (1.40)	-3.41 [†] (1.77)
Any transnational ethnic kin	1.24* (0.61)	0.70 (0.69)	0.49 (0.72)	0.49 (0.66)	-1.54 (2.19)	sep.
Kin share of origin population	0.21* (0.10)	0.09 (0.14)	0.02 (0.15)	0.02 (0.14)	-0.33 (0.55)	-1.08 (0.75)
Colonial or common-state history	0.65 (0.54)	0.69 (0.52)	0.53 (0.58)	1.03* (0.45)	2.07 (1.77)	-4.41 (3.35)
Summary index of all ties	0.35 (0.26)	0.19 (0.28)	0.11 (0.35)	0.23 (0.22)	-1.45 (1.26)	-0.53 (1.00)
Any forced-return policy						
Common official language	0.40 (0.32)	0.20 (0.32)	0.17 (0.34)	0.20 (0.30)	0.56 (0.81)	3.15*** (0.79)
Common spoken language	0.07 (0.31)	-0.11 (0.31)	-0.20 (0.34)	-0.12 (0.29)	-0.23 (0.81)	0.97 (0.99)
Religious similarity, broad tradition	0.42* (0.18)	0.27 (0.19)	0.15 (0.18)	0.33 [†] (0.18)	0.76* (0.35)	0.07 (0.54)
Religious similarity, denominational	0.13 (0.13)	-0.16 (0.18)	-0.39 [†] (0.20)	-0.31 (0.24)	-0.47 (0.45)	-1.29 (0.79)
Any transnational ethnic kin	1.33*** (0.35)	1.08* (0.45)	0.71 [†] (0.41)	0.94* (0.37)	0.76 (0.74)	2.00** (0.66)
Kin share of origin population	0.32** (0.11)	0.25* (0.12)	0.10 (0.16)	0.13 (0.15)	0.09 (0.42)	1.31*** (0.34)
Colonial or common-state history	0.25 (0.34)	0.21 (0.33)	-0.15 (0.42)	-0.28 (0.41)	-0.31 (0.84)	-0.24 (1.26)
Summary index of all ties	0.42** (0.14)	0.30* (0.14)	0.10 (0.16)	0.22 (0.16)	0.38 (0.58)	0.95 [†] (0.53)

Table 44: Two clocks. Each cell is the probability that a new severe forced-return policy begins, with onsets over dyad-years at risk in parentheses. The risk set is dyad-years whose previous CALENDAR year is observed in the sample and carried no severe policy, so 1988 and 1990 are never treated as consecutive. Age is measured from the dyad’s first year of observed refugee presence in the full panel.

Years since the dyad entered the panel	All sizes	Under 100,000 refugees	100,000 refugees or more
0–1	0.00% (0/20)	0.00% (0/11)	0.00% (0/9)
2–3	3.98% (9/226)	3.64% (6/165)	4.92% (3/61)
4–5	1.53% (4/262)	1.04% (2/193)	2.90% (2/69)
6–10	1.36% (9/663)	0.83% (4/481)	2.75% (5/182)
11–15	1.16% (8/687)	0.60% (3/497)	2.63% (5/190)
16–20	1.16% (7/604)	0.00% (0/464)	5.00% (7/140)
21+	0.79% (11/1400)	0.18% (2/1106)	3.06% (9/294)

The timing of severe policy reinforces the importance of scale, but it does not reveal a fixed population threshold or a fixed point in time at which governments become coercive. Collapsing consecutive severe-policy years into episodes yields 62 episodes across 42 dyads and 34 host countries. Among episodes whose beginnings are observed rather than left-censored at 1980, the median onset occurs 6 years after the refugee situation enters the panel. The median population at onset is approximately 160,000, and 61% of episodes begin with at least 100,000 refugees in the host country.

The six-year median should not be read as evidence of “host fatigue.” Table 44 shows that the apparent decline in onset with crisis age comes almost entirely from smaller refugee populations. Among dyad-years with at least 100,000 refugees, the relationship between crisis age and the onset of severe policy is essentially flat (-0.04 , $p = 0.87$). Among smaller populations it is sharply negative (-1.27^{***}). Governments therefore do not appear simply to become less tolerant after refugees have remained for a particular number of years. Large populations remain exposed to roughly similar onset risk whether the displacement is relatively new or long-running; the declining risk with age is concentrated among populations that have become small.

Most severe coercion also occurs before the crisis in the origin has ended. 67% of severe-policy dyad-years with observed conflict status occur during ongoing origin conflict, and 63% of severe-policy episodes begin while fighting continues. Yet here too, population size accounts for much of the relationship. Adding lagged refugee stock reduces the high-intensity conflict coefficient from 1.76 to 0.62, an attenuation of approximately 65%; with the remaining controls added, the conflict coefficient falls to 0.04 and is not distinguishable from zero. In the simpler model containing conflict and stock, a ten-fold increase in refugee population is associated with about 4.9 times the odds of severe policy. This is a descriptive decomposition, not a mediation analysis, because the conflict itself helps produce the displacement whose size is being controlled. But it clarifies the pattern: coercion is concentrated during war largely because that is when hosts are confronting the largest displaced populations.

The combination of scale and ongoing conflict suggests a broader interpretation. Governments imposing severe policies are often facing a large population while the crisis that produced it remains unresolved. Continued fighting gives them little reason to expect the population to fall quickly and leaves open the possibility of further displacement. The analysis does not measure governments’ expectations about future arrivals, so this mechanism cannot be tested directly here. It is nevertheless more consistent with the observed pattern than an account in which coercion is mechanically triggered by battlefield violence, the passage of time, or stable characteristics of either the host government or the refugee group.

Table 45: Two competing security explanations, under progressively more demanding comparisons. Logit coefficients on the row variable, standard errors clustered by dyad. Battle deaths are UCDP counts and begin in 1989, so those models run on the smaller sample the N columns report. The militarization flag is spliced from two non-overlapping World Refugee Survey coding series and is sometimes read from the same archival narrative as the forced-return code itself, which is why the contemporaneous and lagged within-dyad estimates are shown side by side. *sep.* marks a cell with no counter-examples rather than an estimate.

Specification	Severe policy	Any policy	N (severe)	N (any)
Origin battle deaths (log, per SD)				
Raw, year FE	0.69** (0.25)	0.47*** (0.13)	3,985	4,113
+ log refugee stock	0.21 (0.19)	0.16 (0.10)	3,985	4,113
+ stock and origin GDP p.c.	0.10 (0.18)	0.11 (0.10)	3,792	3,916
+ distance and contiguity	0.11 (0.18)	0.10 (0.11)	3,745	3,868
Full pooled covariates	0.45 (0.51)	-0.36 (0.42)	3,493	3,613
Same host-year	1.72* (0.73)	0.24 (0.24)	221	733
Host and year FE	0.30 (0.25)	0.09 (0.13)	2,000	2,946
Dyad and year FE	0.42 (0.33)	0.11 (0.15)	893	1,959
Violence in the host's other neighbours, host and year FE	-0.06 (0.23)	0.05 (0.14)	2,055	3,006
Refugees reported militarized				
Raw, year FE	1.41* (0.55)	1.51*** (0.40)	4,121	4,345
Full pooled covariates	0.81 [†] (0.47)	1.14** (0.40)	3,576	3,778
Same host-year	sep.	5.34*** (1.58)	229	751
Host and year FE	0.72 (0.47)	1.74*** (0.48)	2,136	3,145
Same origin-year	0.58 (1.22)	1.61 [†] (0.86)	287	802
Dyad and year FE, contemporaneous	-0.05 (0.50)	1.27* (0.53)	1,022	2,198
Dyad and year FE, one-year lag	-0.72 (0.45)	0.25 (0.41)	946	2,062

Taken together, the results do not imply that governments adopt a severe policy whenever a refugee population crosses a particular threshold, nor that population size is the only correlate of coercion. What distinguishes size is its consistency. It is the one predictor that retains the same positive relationship with severe policy in the pooled data, across refugee populations facing the same host, and within the same dyad over time; the same-origin estimate points in the same direction but is estimated from much thinner variation. The broader pattern is therefore one of coercion concentrated in large, unresolved displacement crises. Governments appear most likely to restrict the option of remaining when they are already accommodating a substantial refugee population and the crisis producing that population has not yet run its course.

5.1.2 De Jure Law and De Facto Practice

DWRAP provides an independent comparison with the GRR forced-return measure. It codes refugee policy from statute for 90 host countries in the sample, covering 95% of the panel. Its non-refoulement measure distinguishes statutes that permit refoulement, are silent or ambiguous, or grant an explicit right to non-refoulement. Unlike GRR, it measures formal law rather than observed state practice and is coded from independent sources.

Table 46: Hand-coded forced-return practice by the externally coded statutory position on refoulement. Restricted to dyad-years in which both are coded.

Statutory position	Dyad-years	Hosts	Any policy	Severe policy
Explicit provision permitting refoulement	509	34	15.3%	3.9%
Silent or ambiguous	698	29	21.3%	7.0%
Explicit right to non-refoulement	3,044	66	6.8%	2.5%

Table 47: Logit of hand-coded forced return on the statutory right to non-refoulement, against the two remaining positions. Standard errors clustered by dyad. The dyad-fixed-effects columns are identified from statutory changes within a host over time.

Specification	Explicit right to non-refoulement	Dyad-years
Any, host + year FE	-1.15*** (0.34)	3,321
Any, dyad + year FE	-1.09* (0.48)	2,128
Severe, host + year FE	-1.29* (0.61)	2,268
Severe, dyad + year FE	-1.66* (0.83)	970

The two measures generally align. Forced return occurs in 6.8% of dyad-years in which the host grants an explicit right to non-refoulement, compared with 21.3% where the statute is silent or ambiguous. The statutory right is also negatively associated with coercion in each specification in Table 47, including models with dyad and year fixed effects. The correspondence between independently coded law and observed practice provides an external check on the GRR measure.

The relationship is not fully monotonic. Hosts whose statutes explicitly permit refoulement record less coercion than those whose statutes are silent or ambiguous. This category is small and concentrated in relatively few hosts, so Table 47 focuses on the contrast between an explicit right to non-refoulement and the other statutory positions.

Formal protection nevertheless does not eliminate coercion. Among dyad-years in which the host grants an explicit right to non-refoulement, 206 record a forced-return policy and 77 record a severe policy, spanning 30 host countries. De jure protection is therefore associated with less coercion, but violations remain common enough to distinguish statutory commitments from de facto practice.

5.1.3 UNHCR Involvement

I use three measures of UNHCR involvement. Repatriation funding per refugee is drawn from UNHCR financial reporting and is coded for 3,015 dyad-years. A second measure records whether UNHCR operated a repatriation programme in the dyad-year, using the *World Refugee Survey*; it is coded for 4,238 dyad-years and extends back to 1980. The third records an active tripartite repatriation agreement between the origin state, asylum state, and UNHCR, and occurs in 43 dyad-years.

5.1.3.1 Comparing the Three Measures

Table 48: Agreement between the three measures of UNHCR involvement. Each row is restricted to the dyad-years in which both of its measures are coded, which is why the samples differ. The last two columns give the share of dyad-years recording the second measure among those that do and do not record the first.

Pair	Both coded	Both present	Correlation	Second given first	Second given not
Funding and Programme	2,449	186	0.31	24.3%	3.8%
Funding and Tripartite agreement	2,599	26	0.10	3.3%	0.7%
Programme and Tripartite agreement	4,205	32	0.22	7.5%	0.3%

The measures overlap only partially. A repatriation programme is recorded in 24.3% of funded dyad-years, compared with 3.8% of unfunded dyad-years, but their correlation is only 0.31. Among the 2,449 dyad-years in which all three measures are observed, 65.8% record none and only 24 record all three. Funding without either a programme or an agreement accounts for 23.6% of these observations.

The three variables therefore capture related but distinct forms of involvement. Funding records financial resources, the programme flag records an operational presence, and a tripartite agreement records formal institutional involvement. They should not be treated as interchangeable measures.

5.1.3.2 When UNHCR Is Involved

Table 49: UNHCR involvement by decade, as the share of coded dyad-years recording each measure. The funding cell for the 2000s is empty because the series is uncoded for every dyad-year from 2000 to 2010, and the 2020s cell rests on the 1980–2020 remainder only; the coded denominators are printed so that the rates are read against them.

Period	Dyad-years	Funding (coded)	Programme (coded)	Tripartite
1980s	335	24.6%	334	10.6%
1990s	872	34.1%	868	18.0%
2000s	1,244	–	–	12.2%
2010s	1,366	29.9%	1,242	5.9%
2020s	631	30.3%	155	4.9%

Table 50: UNHCR involvement by whether the origin country is still at war. Shares are of coded dyad-years within each row.

Status	Dyad-years	Funding	Programme	Tripartite	Severe policy
Origin at peace	2,734	21.4%	8.3%	0.7%	1.8%
Origin at war	1,714	43.1%	13.4%	1.3%	5.8%

UNHCR involvement varies substantially over time. Repatriation programmes are most common in the 1990s, appearing in 18.0% of coded dyad-years, compared with 5.9% in the 2010s and 4.9% in the 2020s. Tripartite agreements are also concentrated in the earlier period, reaching 4.5% in the 1990s and becoming rare thereafter.

These trends must be read against changes in source coverage. The programme series declines gradually across the source transition in 2009, including within Sub-Saharan Africa, suggesting that its decline is not solely a coding break. The tripartite series is less comparable over time: agreements are recorded in 2.0% of dyad-years before 2009 and 0.0% afterwards, but the post-2008 State Department reports do not

systematically document agreements. The tripartite measure is therefore used below only within the years covered by its original source. The funding series has a separate coverage gap from 2000 to 2010 and after 2020.

Table 50 also shows that involvement is not restricted to post-conflict settings. Repatriation funding is recorded in 43.1% of dyad-years in which conflict is ongoing in the origin, compared with 21.4% where the origin is at peace. The corresponding programme rates are 13.4% and 8.3%. Severe forced return is likewise more common while the origin remains at war (5.8% versus 1.8%). UNHCR involvement and coercive return therefore frequently occur before conflict in the origin has ended.

5.1.3.3 Where UNHCR Is Involved

Table 51: UNHCR involvement and forced return by the region of the asylum country. Shares are of coded dyad-years within each region.

Asylum region	Dyad-years	Dyads	Funding	Programme	Tripartite	Severe policy
Europe & North America	1,831	145	15.9%	1.3%	0.0%	0.9%
Sub-Saharan Africa	1,500	75	45.2%	18.8%	1.9%	4.1%
Asia & MENA	941	37	32.9%	14.9%	1.4%	7.1%
Latin America & Caribbean	176	15	21.7%	10.0%	1.1%	2.3%

Table 52: The ten dyads with the most years of a recorded UNHCR repatriation programme, with their funding, agreement and coercion records beside them.

Dyad	Coded years	Programme years	Funded years	Agreement years	Severe-policy years
AFG-PAK	43	30	21	1	6
RWA-COD	43	28	17	2	1
LKA-IND	38	20	10	2	4
BDI-TZA	43	18	4	2	16
AFG-IRN	43	17	21	4	27
SOM-KEN	33	14	16	0	5
MMR-BGD	35	13	7	2	8
LBR-GIN	27	12	7	0	0
AGO-COD	32	10	12	0	1
AGO-ZMB	30	10	11	0	0

UNHCR involvement is concentrated in the Global South, particularly in Sub-Saharan Africa. A repatriation programme is recorded in 18.8% of coded dyad-years in Sub-Saharan Africa and 14.9% in Asia and the Middle East, compared with 1.3% in Europe and North America. Funding follows the same pattern (45.2% versus 15.9%). Severe forced-return policies are also more common in the regions where UNHCR involvement is greatest, occurring in 4.1% of Sub-Saharan African host dyad-years and 0.9% of European and North American host dyad-years.

The dyads with the most programme-years show the same overlap. They include Afghans in Pakistan and Iran, Rwandans in the Democratic Republic of the Congo, Sri Lankan Tamils in India, Burundians in Tanzania, Somalis in Kenya, and Rohingya in Bangladesh. For example, Afghanistan–Iran records 17 programme-years, 4 years under a tripartite agreement, and 27 years of severe forced return. Burundi–Tanzania records 18 programme-years and 16 severe-policy years. UNHCR involvement is therefore common in, rather than absent from, many of the major coercive-return cases.

5.1.3.4 Involvement and Coercion

Table 53: Share of dyad-years recording a forced-return policy, by whether UNHCR is involved. Restricted to dyad-years in which both the policy and the involvement measure are coded.

Measure	Dyad-years present	Any: absent	Any: present	Severe: absent	Severe: present
Repatriation funding recorded	796	8.2%	16.5%	2.1%	6.8%
UNHCR repatriation programme	429	7.8%	27.7%	2.1%	14.0%
Tripartite agreement in force	43	9.6%	39.5%	3.2%	25.6%

Table 54: Each UNHCR measure added to a model of severe forced-return policy. Logit with year fixed effects and dyad-clustered standard errors, on the refugee stock, host income, host autocracy, statutory refugee rights, camp residence, asylum- and origin-country conflict, distance, contiguity, origin income, origin civil liberties and cumulative conflict years; the full set of controls is included and not shown.

Measure	Coefficient	Dyad-years
Repatriation funding recorded	0.33 (0.57)	1,944
UNHCR repatriation programme	1.30** (0.41)	3,386
Tripartite agreement in force	1.26* (0.62)	3,545

Forced-return policies are more common in dyad-years with UNHCR involvement on all three measures. Severe policies occur in 2.1% of dyad-years without a repatriation programme and 14.0% of those with one. Among dyad-years covered by a tripartite agreement, the corresponding rate is 25.6%.

The programme indicator remains positively associated with severe coercion after the controls in Table 54 (1.30**), while the tripartite coefficient is less precisely estimated (1.26*). Funding itself is not independently associated with coercion once population size and the other covariates are included (0.33).

These estimates are descriptive rather than causal. UNHCR is likely to become involved precisely where repatriation is already being organised, so its presence cannot be interpreted as producing coercion. The narrower result is that formal international involvement does not imply the absence of coercive return.

5.1.4 UNHCR Assistance and Realized Return

The preceding analyses use UNHCR involvement to examine its relationship with coercion. I next use the same measures to evaluate H3, which concerns refugees' ability to finance return. Because the data contain no dyad-year measure of refugees' own resources, the observable evidence is limited to external assistance.

The three measures have different periods of coverage. Repatriation funding is observed for 1980–1999 and 2011–2020, the programme indicator for 1980–2008, and the tripartite-agreement register through 2002. Models using each measure are therefore estimated over the years in which its source can distinguish presence from absence.

A remaining difficulty is selection into assistance. UNHCR involvement is more common where conflict in the origin continues and where coercion is also more common, so a positive association between assistance and return is not by itself evidence for H3. The analyses below therefore examine the amount rather than simply the presence of funding, the timing of return around tripartite agreements, and whether the association with assistance differs across the conditions identified by the theory.

Table 55: Each measure of UNHCR involvement added to the paper’s baseline, with recorded return as the outcome. PPML with dyad and year fixed effects, the log stock at risk as an offset and dyad-clustered standard errors; every column carries the full main specification. Each row is estimated on the years its own source covers, so the samples differ by design; the two indented rows show what ignoring that bound would produce and are diagnostics for the restriction rather than estimates of the measure. The last column reports the coefficient on a severe forced-return policy in the same model, which is 1.08 in Table 2 of the paper.

Measure	Coefficient (SE)	p	Dyad-years	Dyads	Severe forced return
Funding per refugee (standardised log)	0.31** (0.12)	0.007	1,378	100	1.29*** (0.31)
UNHCR repatriation programme, 1980–2008	0.67** (0.24)	0.005	1,225	90	0.56* (0.28)
2009–2023, after the source change	-0.97* (0.49)	0.047	867	77	1.99*** (0.33)
Tripartite agreement in force, 1980–2002	0.55 [†] (0.28)	0.051	873	82	0.84** (0.27)
full panel, post-2002 zeros included	0.64** (0.21)	0.002	2,327	110	1.05*** (0.18)
Jointly, common sample: funding	0.35* (0.14)	0.012	635	68	1.17*** (0.31)
programme	0.58* (0.24)	0.014	635	68	
tripartite agreement	0.42 (0.28)	0.124	635	68	

Table 55 reports each measure over the years covered by its source. All three coefficients are positive. Funding per refugee has a coefficient of 0.31** ($p = 0.007$), while the programme indicator is 0.67** ($p = 0.005$) over 1980–2008. The tripartite coefficient is 0.55[†] within the period covered by its register and is estimated from 21 dyads.

On the common sample of 635 dyad-years and 68 dyads, funding and the programme indicator remain positive when all three measures enter together (0.35* and 0.58*), while the tripartite coefficient is not distinguishable from zero (0.42, $p = 0.12$).

The coefficient on severe forced return remains large in each specification. Its variation across rows mainly reflects the different periods covered by the three sources: adding the assistance measure to the corresponding sample changes the severe forced-return coefficient by at most 0.18. The assistance results therefore do not account for the association between coercion and return reported in the main specification.

5.1.4.1 Funding Presence and Amount

Table 56: The amount of repatriation funding against its presence. Amount is the standardised log of funding per refugee; the binary records whether any funding is recorded. All rows are the paper’s baseline with the stated terms added, and the offset already expresses return as a rate on the stock at risk, so the last two rows add that stock a second time as an explicit regressor.

Measure	Coefficient (SE)	p	Dyad-years	Dyads
Amount, all coded dyad-years	0.31** (0.12)	0.007	1,378	100
Amount, funded dyad-years only	0.82*** (0.21)	<0.001	557	73
Any funding recorded (binary), alone	0.14 (0.30)	0.641	1,378	100
Binary and amount together: binary	-1.06* (0.47)	0.023	1,378	100
amount	0.64*** (0.16)	<0.001	1,378	100
Amount, log stock at risk added as a regressor	0.31** (0.11)	0.006	1,378	100
log stock at risk	-0.10 (0.16)	0.539	1,378	100

The funding results differ between the extensive and intensive margins. A binary indicator for any recorded repatriation funding is not associated with higher return (0.14, $p = 0.64$). Among funded dyad-years, however, the amount of funding is positively associated with return (0.82***). When the binary indicator and funding amount enter together, their coefficients are -1.06* and 0.64***, respectively.

The distinction is substantively relevant because funded dyad-years vary considerably in the resources provided. The median is \$20 per refugee, with an interquartile range from \$6 to \$51. The positive association with return is therefore concentrated in the amount of assistance rather than the simple presence of a funding allocation.

This pattern is more consistent with a resource constraint than with an account based only on selection into receiving assistance, although it does not eliminate selection in the amount allocated. It is also not explained by refugee-population size. Adding the log stock at risk as a separate regressor leaves the funding coefficient at 0.31**, while the stock coefficient is -0.10.

5.1.4.2 Coverage of the Programme Measure

The programme indicator is restricted to 1980–2008 because its source changes after that year. The *World Refugee Survey* reports directly on UNHCR operations, whereas the State Department reports used from 2009 focus on state conduct and mention UNHCR programmes less systematically. A zero after 2008 therefore cannot be interpreted in the same way as a zero beforehand.

The estimates are consistent with this measurement break. Over 1980–2008, the programme coefficient is 0.67** ($p = 0.005$); over 2009–2023 it is -0.97* ($p = 0.047$). Pooling the two periods gives 0.24 ($p = 0.42$). I therefore use the programme measure only through 2008 in the analyses that follow.

5.1.4.3 Timing of Tripartite Agreements

Table 57: Recorded return as a share of the stock at risk, by year relative to the signature of a tripartite repatriation agreement. Built from the 24 signings in the ExCom register that match a dyad in the estimation sample. Pooled rates, not a fitted model: each row uses the dyad-years observed at that distance from a signature, and the composition changes across rows.

Years from signature	Return rate	Dyad-years	Dyads
-4	2.4%	12	11
-3	3.7%	18	17
-2	2.8%	17	14
-1	5.1%	23	19
0	8.2%	22	19
+1	18.1%	22	19
+2	17.4%	22	19
+3	13.4%	20	18
+4	13.9%	16	14

Table 57 describes return around the signature of a tripartite agreement. In the four years before signature, pooled return rates range from 2.4% to 3.7%, reaching 5.1% in the year before signature. The rate is 8.2% in the signature year and 18.1% and 17.4% in the following two years. The descriptive series therefore shows little increase before signature and a substantial increase around and after it.

The model-based comparison is less precise. Over the period covered by the agreement register, the coefficient for the three years up to and including signature is -0.21 ($p = 0.57$), while the coefficient for the following two years is 0.46 ($p = 0.12$). Neither reaches conventional significance, and the analysis rests on only 24 agreement signings across 21 dyads. The timing evidence is therefore suggestive rather than conclusive.

5.1.4.4 Where Assistance Predicts Return

Table 58: Assistance interacted with the two conditions the theory says should change what it does. The first three rows use the funding amount and the post-war window on the funding sample; the last three use the programme flag and a severe forced-return policy on the 1980–2008 programme sample. Main effects are conditional on the interacted term being zero and are not comparable with Table 55.

Measure	Coefficient (SE)	p	Dyad-years	Dyads
War ending, no funding recorded	0.53 (0.38)	0.160	1,378	100
Funding amount, outside the post-war window	0.28* (0.12)	0.022	1,378	100
interaction	0.39† (0.22)	0.079	1,378	100
Severe forced return, no programme	1.54*** (0.34)	<0.001	1,225	90
UNHCR programme, no severe policy	1.23*** (0.20)	<0.001	1,225	90
interaction	-1.82*** (0.24)	<0.001	1,225	90

The interactions in Table 58 examine whether the association with assistance varies across the conditions identified by the theory. Funding has a positive interaction with the post-war window ($0.39^\dagger$, $p = 0.079$). The main post-war coefficient is correspondingly smaller and less precise (0.53), indicating that the association between a war ending and return is stronger where repatriation resources are greater.

The programme interaction points in the opposite direction under severe coercion. A UNHCR programme is positively associated with return where no severe forced-return policy is present (1.23^{***} , $p < 0.001$), while its interaction with severe forced return is -1.82^{***} ($p < 0.001$). Assistance therefore has a stronger association with return in the absence of severe coercion.

This interaction should be interpreted cautiously. Return already equals 11.7% of the stock at risk in severe-coercion dyad-years, compared with 7.1% elsewhere. The negative interaction may therefore partly reflect less remaining scope for a programme to increase return where coercion has already raised it.

5.1.4.5 Summary

Taken together, the three measures provide qualified support for H3. Funding per refugee and the programme indicator are positively associated with return over the periods in which they are observed, including when both enter on their common sample. The tripartite measure is positive but imprecisely estimated and is based on relatively few dyads. The funding results are concentrated in the amount provided rather than the simple presence of funding, and the timing of tripartite agreements provides no clear evidence that elevated return precedes their signature.

These results do not identify a causal effect of UNHCR assistance. The allocation of assistance is endogenous to the circumstances in which return is being organised, and the available data cannot remove that selection. The evidence is therefore narrower: return is systematically higher where greater repatriation resources are available, particularly in settings where coercion is absent and after conflict in the origin has ended. This is consistent with the mobility constraint in H3, but it does not establish that additional assistance itself causes return.

5.2 Conditional Effects of the Post-War Window

Panel B of the figure in the results section reports the return-rate ratio associated with a war ending at each value of four permissive conditions, and the interaction that tests whether the condition changes the effect of peace rather than the level of return. This section reports the four models that panel is drawn from. It was a table in the main body until the figure replaced it; the figure carries the two conditional estimates and the interaction, and what the table adds is the condition's own association with return outside the post-war window, the number of post-war dyad-years on each side of each split, and the fit statistics.

Each column interacts the post-war window with one binary condition, cut at the sample median in the direction predicted to favour return; funding is cut at the median among funded dyad-years. The first row names the omitted category, so the second row is the association with a war ending when the condition is absent, the fourth is the interaction, and the fifth is their sum — the association when the condition is present, with a standard error that carries the covariance of the two terms. Every column carries the full main-specification right-hand side, including the continuous variable each binary moderator is cut from.

Table 59: Interactions between the post-war window and permissive conditions

Term	(1) Poor asylum	(2) Rich origin	(3) Large funding	(4) Free origin
Window and condition				
Condition absent means	Rich asylum	Poor origin	No large funding	Repressive origin
War ending, condition absent	0.03 (0.24)	1.17*** (0.25)	0.57 (0.42)	0.85*** (0.20)
Condition alone, no war ending	0.20 (0.34)	0.81** (0.30)	0.69** (0.24)	-0.67 [†] (0.40)
War ending x condition	1.41*** (0.25)	-0.60 (0.41)	0.66 (0.56)	0.60 (0.40)
War ending, condition present	1.44*** (0.21)	0.57 [†] (0.32)	1.23** (0.48)	1.45*** (0.43)
Support and fit				
Post-war dyad-years, condition absent	62	90	53	93
Post-war dyad-years, condition present	76	48	42	45
All main-specification variables	Yes	Yes	Yes	Yes
Dyad and year fixed effects	Yes	Yes	Yes	Yes
Dyad-years	2,327	2,327	1,378	2,327
Pseudo R2	0.808	0.802	0.783	0.801

Read down a column rather than across the table. Asylum-country income is the one condition whose interaction is distinguishable from zero: a war ending is worth essentially nothing in the wealthier half of hosts and a great deal in the poorer half. The other three interactions are not significant, and in the case of origin income the point estimate runs against the direction the theory predicts. Two of the conditions do predict return on their own, outside any window — a richer origin and a large repatriation programme — which is a different claim from conditioning the effect of peace and is why the two rows are reported separately.

5.3 Predicted Return Rates Quoted in the Results

The results section quotes model-predicted rates of return as a guide to the size of the associations it reports. This section collects them, states how each is constructed, and reports the interval on the contrast so that the precision of each comparison is on the page. These are marginal predictions from the fitted PPML models, not event-study estimates or identified dynamic effects.

Both tables come from the event-time specification described in Section 5.3.1 below: the eleven window bins, with the far-pre term omitted, so every bin including the year before the event is measured against a dyad-year more than five years from its event rather than against $t - 1$.

Table 60 follows the dyad-years actually standing at each distance from an event and predicts them twice: once as observed, and once with every event term from the event year onward set to zero. The counterfactual column is therefore the same dyad-years without the event, and the difference between the columns is the estimated association at that distance. Because the two predictions differ by one set of terms in the linear predictor, the pre-event predictions coincide by construction; this design cannot show a pre-trend, and the joint lead tests reported in Section 5.3.1 are what bear on the run-up.

Table 60: Predicted return around three events

	Event occurs	Event does not occur	Ratio	95% interval on the ratio	Dyad-years
A severe forced-return policy begins					
Event year	9.2%	4.0%	2.31	[1.17, 4.24]	42
One year after	19.1%	5.5%	3.48	[1.71, 6.66]	40
Two years after	8.5%	5.2%	1.63	[0.76, 3.28]	34
A war ends in the origin					
Event year	8.9%	9.5%	0.94	[0.58, 1.56]	67
One year after	27.3%	14.5%	1.88	[1.12, 3.45]	42
Two years after	25.8%	9.4%	2.73	[1.48, 5.03]	48
Conflict begins in the asylum country					
Event year	9.0%	10.5%	0.86	[0.31, 1.99]	112
One year after	7.9%	10.2%	0.77	[0.32, 1.74]	103
Two years after	8.9%	8.8%	1.02	[0.49, 1.76]	87

Table 61 instead holds the event fixed and moves a standing condition. It takes the whole estimation frame, switches on the post-war window term for the stated year, and sets one covariate to the twenty-fifth and seventy-fifth percentiles of its own distribution with everything else as observed. The model is additive in each condition, so the ratio between the two columns is the same at every distance by construction; what changes across rows is the level, not the contrast. The funding rows rest on the smaller set of dyad-years for which funding is coded and are not strictly comparable in level with the other three.

Table 61: Predicted return after a war ending, by standing condition

	25th percentile	75th percentile	Ratio	95% interval on the ratio
Income in the asylum country: 25th vs 75th percentile				
One year after	22.2%	6.6%	3.34	[0.82, 15.62]
Two years after	32.1%	9.6%	3.34	[0.82, 15.62]
Income in the origin: 25th vs 75th percentile				
One year after	12.1%	21.1%	0.57	[0.22, 1.40]
Two years after	17.5%	30.6%	0.57	[0.22, 1.40]
Civil liberties restrictions in the origin: 25th vs 75th				
One year after	25.2%	11.7%	2.15	[1.45, 3.20]
Two years after	36.5%	17.0%	2.15	[1.45, 3.20]
Repatriation funding per refugee: 25th vs 75th percentile				
One year after	5.8%	11.3%	0.52	[0.34, 0.80]
Two years after	10.0%	19.5%	0.52	[0.34, 0.80]

Why the ratio and not the level. A predicted level compounds the uncertainty of every coefficient in the model at once, and the ninety-five per cent interval on any single rate in these tables spans roughly an order

of magnitude; the interval on the wartime baseline alone runs from about 2 to about 72 per cent. A ratio formed within each simulation draw is far better determined, because the dyad, year and offset contributions and every coefficient not being varied cancel. The levels are therefore quoted in the text as an indication of scale, and the ratio column is what carries the inference. This is also why the figure in the results section plots ratios rather than levels.

5.3.1 Construction

Predictions are formed on the linear predictor. For each dyad-year the dyad, year and offset contribution is recovered by subtracting $\mathbf{X}\hat{\beta}$ from the fitted linear predictor and is then held fixed, so that only the terms being varied move. Intervals come from drawing 400 coefficient vectors from the estimated sampling distribution and recomputing the quantity for each draw; the ratio is formed within each draw.³ Uncertainty in the fixed effects is excluded, so the intervals understate total uncertainty.

Each event is dated to its nearest onset, and a dyad that is ever treated but whose onset cannot be dated is dropped rather than counted as untreated. Each panel also drops the contemporaneous control for its own treatment, which would otherwise absorb the event. The war-ending clock uses the corrected window, in which a post-war year must be a year of peace, so no post-event bin contains a year in which the origin is fighting again.

³The draw count is FH_NSIM in 0_scripts/exploratory_v2.Rmd, and the tables in this section are produced by 0_scripts/predicted_rates_table.R.

6 Literature Review

Table 62: List of studies for Figure 1 of the paper

Journal	Authors	Year	Title	Population
Columbia University Press (book)	Adelman, Howard; Barkan, Elazar	2011	No Return, No Refuge: Rites and Rights in Minority Repatriation	Multiple (minorities)
Working Paper (EBRD)	Adema, Joop; Aksoy, Cevat Giray; Giesing, Yvonne; Poutvaara, Panu	2026	Refugee Return	Ukrainian
IMR	Al Husein, Nawras; Wagner, Natascha	2023	Determinants of Intended Return Migration among Refugees: A Comparison of Syrian Refugees in Germany and Turkey	Syrian
Working Paper (World Bank-UNHCR Joint Data Center)	Alrababah, Ala; Casalis, Marine	2024	Understanding Refugee Return: Key Findings, Gaps, and Future Research	Multiple
BJPS	Alrababah, Ala; Masterson, Daniel; Casalis, Marine; Hangartner, Dominik; Weinstein, Jeremy	2023	The Dynamics of Refugee Return: Syrian Refugees and Their Migration Intentions	Syrian
Peace Economics, Peace Science and Public Policy	Arias, María Alejandra; Ibáñez, Ana María; Querubín, Pablo	2014	The Desire to Return during Civil War: Evidence for Internally Displaced Populations in Colombia	Colombian
International Migration Review	Barnett, Michael	2001	Humanitarianism with a Sovereign Face: UNHCR in the Global Undertow	Multiple
Forced Migration Review	Barnett, Michael	2001	UNHCR and the Ethics of Repatriation	Multiple
Cornell University Press (book)	Barnett, Michael; Finnemore, Martha	2004	Rules for the World: International Organizations in Global Politics	Multiple
J Development Economics	Beaman, Lori; Onder, Harun; Onder, Stefanie	2022	When Do Refugees Return Home? Evidence from Syrian Displacement in Mashreq	Syrian
Working Paper	Beber, Bernd; Roessler, Philip; Scacco, Alexandra	2021	Coping with Partition: Wealth, Security, and Migration in Post-Separation Sudan	South Sudanese
International Migration	Black, Richard; Gent, Saskia	2006	Sustainable Return in Post-conflict Contexts	Multiple
Berghahn Books (edited volume)	Black, Richard; Koser, Khalid	1999	The End of the Refugee Cycle?: Refugee Repatriation and Reconstruction	Multiple
JCR	Bove, Vincenzo; Di Salvatore, Jessica; Elia, Leandro	2025	What it Takes to Return: UN Peacekeeping and the Safe Return of Displaced People	South Sudanese
Cambridge University Press (book)	Bradley, Megan	2013	Refugee Repatriation: Justice, Responsibility and Redress	Multiple
World Development	Camarena, Kara Ross	2022	Repatriation during Conflict: A Signaling Analysis	Multiple
AJPS	Camarena, Kara Ross; Hägerdal, Nils	2020	When Do Displaced Persons Return? Postwar Migration among Christians in Mount Lebanon	Lebanese (Christians)

Table 62: List of studies for Figure 1 of the paper (*continued*)

Journal	Authors	Year	Title	Population
Refugee Survey Quarterly	Chimni, B. S.	1999	From Resettlement to Involuntary Repatriation: Towards a Critical History of Durable Solutions to Refugee Problems	Multiple
Journal on Migration and Human Security JEMS	Crisp, Jeff; Long, Katy	2016	Safe and Voluntary Refugee Repatriation: From Principle to Practice	Multiple
	Di Saint Pierre, Francesca; Martinovic, Borja; De Vroome, Thomas	2015	Return Wishes of Refugees in the Netherlands: The Role of Integration, Host National Identification and Perceived Discrimination	Multiple (Afghan, Iranian, Iraqi, Somali)
International Migration	Dinger, Osman Bahadır; Sahin-Mencütek, Zeynep	2025	Refugee Return, Reintegration, and Citizenship Practices in Post-Conflict Syria	Syrian
International Migration	Eastmond, Marita	2006	Transnational Returns and Reconstruction in Post-war Bosnia and Herzegovina	Bosnian
BJPS	Gerver, Mollie	2018	Refugee Repatriation and the Problem of Consent	South Sudanese
Political Studies	Gerver, Mollie; Simon, Miranda; Ghosn, Faten	2025	Refugee Resettlement and Preferences	Syrian
APSR	Ghosn, Faten; Chu, Tiffany S.; Simon, Miranda; Braithwaite, Alex; Frith, Michael; Jandali, Joanna	2021	The Journey Home: Violence, Anchoring, and Refugee Decisions to Return	Syrian
J Refugee Studies	Hardgrove, Abby	2009	Liberian Refugee Families in Ghana: The Implications of Family Demands and Capabilities for Return to Liberia	Liberian
African Studies Review	Harrell-Bond, Barbara E.	1989	Repatriation: Under What Conditions Is It the Most Desirable Solution for Refugees? An Agenda for Research	Multiple
Third World Quarterly [TBC]	İçduygu, Ahmet; Nimer, Maissam	2020	The Politics of Return: Exploring the Future of Syrian Refugees in Jordan, Lebanon and Turkey	Syrian
Journal of Immigrant and Refugee Studies	Kaya, Serdar; Orchard, Phil	2020	Prospects of Return: The Case of Syrian Refugees in Germany	Syrian
Journal of Immigrant and Refugee Studies	Kayaoglu, Ayselin; Şahin-Mencütek, Zeynep; Erdoğan, M. Murat	2021	Return Aspirations of Syrian Refugees in Turkey	Syrian
J Refugee Studies	Klinthäll, Martin	2007	Refugee Return Migration: Return Migration from Sweden to Chile, Iran and Poland 1973-1996	Chilean, Iranian, Polish
J Refugee Studies	Koser, Khalid	1997	Information and Repatriation: The Case of Mozambican Refugees in Malawi	Mozambican
Oxford University Press (book)	Long, Katy	2013	The Point of No Return: Refugees, Rights, and Repatriation	Multiple
UNHCR PDES Working Paper	Long, Katy	2010	Back to Where You Once Belonged: A Historical Review of UNHCR Policy and Practice on Refugee Repatriation	Multiple
J Refugee Studies	MacDonald, Anna; Porter, Holly	2020	The Politics of Return: Understanding Trajectories of Displacement and the Complex Dynamics of 'Return' in Central and East Africa	Multiple (Central/East African)
J Refugee Studies	Nandi, Sarah	2025	"We were together and we had our own family in each other": Refusing Repatriation and Forging Gendered Belonging as Hijra Refugees in Kolkata	Bangladeshi (Hijra)

Table 62: List of studies for Figure 1 of the paper (*continued*)

Journal	Authors	Year	Title	Population
JEMS	Omata, Naohiko	2013	The Complexity of Refugees' Return Decision-Making in a Protracted Exile: Beyond the Home-Coming Model and Durable Solutions	Liberian
J Refugee Studies	Sahin-Mencütek, Zeynep	2021	Governing Practices and Strategic Narratives for Syrian Refugee Returns	Syrian
International Migration Review	Schwartz, Stephanie	2025	Return-Without-Refoulement [TBC: confirm exact title]	Burundian
J Refugee Studies	Stefanovic, Djordje; Loizides, Neophytos; Parsons, Samantha	2015	Home is Where the Heart Is? Forced Migration and Voluntary Return in Turkey's Kurdish Regions	Kurdish
International Migration	Stefansson, Anders H.	2006	Homes in the Making: Property Restitution, Refugee Return, and Senses of Belonging in a Post-war Bosnian Town	Bosnian
Book chapter (US Committee for Refugees)	Stein, Barry N.; Cuny, Frederick C.	1994	Refugee Repatriation during Conflict: Protection and Post-Return Assistance	Multiple
Conflict Management and Peace Science	Toft, Monica Duffy	2007	The Myth of the Borderless World: Refugees and Repatriation Policy	Multiple
Geopolitics	Um, Khatharya	2023	Refugee Return, Reintegration, and Sustainable Futurity: Politics, Pitfalls and Possibilities of Repatriation in Post/Conflict Situations	Cambodian; Burmese
Journal of Behavioral and Experimental Economics	Vakhitov, Volodymyr; Zaika, Nataliia; Kandul, Serhiy	2025	Return Intentions of Ukrainian Refugees: The Role of National Identity and Pride	Ukrainian
Comparative Migration Studies	Van Houte, Marieke; Siegel, Melissa; Davids, Tine	2016	Deconstructing the Meanings of and Motivations for Return: An Afghan Case Study	Afghan
Working Paper	Weber, Sigrid; Hartman, Alexandra	2022	Property Rights and Post-conflict Recovery: Theory and Evidence from IDP Return Movements in Iraq	Iraqi (Yazidi and Sunni Muslim)
Book chapter (Routledge)	Whitaker, Beth Elise	2003	Changing Priorities in Refugee Protection: The Rwandan Repatriation from Tanzania	Rwandan
J Refugee Studies	Zakirova, Komila;	2021	The Road Back Home is Never Long: Refugee Return Migration	Multiple
J Refugee Studies	Buzurukov, Bilol	2021	Refugees and Their Return Home: Unsettling Matters	Multiple
J Refugee Studies	Zetter, Roger	2021		
J Refugee Studies	Žíla, Ondřej	2026	'Partial Returns': Displacement, Mobility, and Translocal Connections in Post-war Bosnia and Herzegovina	Bosnian